

Chapter 58. Weaning from Mechanical Ventilation

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Weaning from Mechanical Ventilation: Introduction

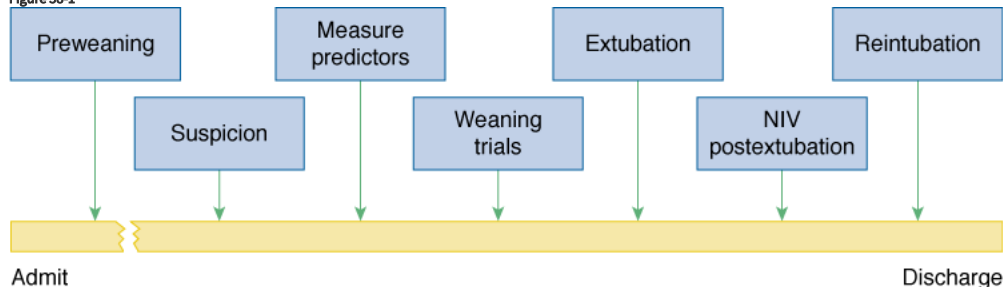
Thirty years ago, the weaning of patients from the ventilator was relegated to nurses and respiratory therapists. It aroused little interest among physicians. It certainly wasn't thought worthy of serious scientific inquiry. All this has changed. No other area of critical care has undergone so great a transformation. But the illumination has also cast shadows. In particular, discussion of weaning is now bedeviled by imprecise language. This can be seen as just deserts insofar as few clinicians use the term *weaning* in the strict literal sense—a gradual reduction in the level of ventilator support. Instead, most patients today are taken off the ventilator cold turkey. It would be fine if the confused language stopped there. But this is only one small example of how fundamental scientific misunderstanding has arisen from imprecise word choices.

Under the cloak of imprecise language, much muddled thinking, flawed logic, and misinterpretation has crept into the field. These language problems are not just pedantic quirks. Instead, they impede the rigor of research in this area, as well as interpretation of the findings. Communication is also hindered by the lumping together of many distinct components of this complex process. To enhance clarity, we divide weaning into seven stages.

Seven Stages of Weaning

We divide weaning into seven stages to draw attention to areas that receive minimal attention (Fig. 58-1). Stage 1 is preweaning, when no attempt at weaning is desirable. For example, when a patient is receiving 80% oxygen and positive end-expiratory pressure (PEEP) of 15 cm H₂O, performing any disconnect from the ventilator (for measurement of weaning predictors) is inappropriate and may even be dangerous. Every ventilated patient begins at stage 1, and some patients never get beyond that stage. For example, in a prospective study of 249 ventilated patients,¹ sixty-five patients (26%) died during mechanical ventilation without any attempt at weaning. In another report of 357 patients entered into a trial of weaning techniques,² 12.9% never reached the stage of any active weaning attempt. We identify this preweaning stage to emphasize the importance of the transition between it and the next stage of a patient's clinical course.

Figure 58-1



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Seven stages of weaning. Stage 1 is preweaning, a stage that many patients never get beyond. Stage 2 is the period of diagnostic triggering, the time when a physician begins to think that the patient might be ready come off the ventilator. Stage 3 is the time of measuring and interpreting weaning predictors. Stage 4 is the time of decreasing ventilator support (abruptly or gradually). Stage 5 is either extubation (of a weaning-success patient) or reinstitution of mechanical ventilation (in a weaning-failure patient). Stage 6 is use of noninvasive ventilation after extubation. Stage 7 is reintubation. Failure to appreciate stage 2 probably leads to the greatest delays in weaning.

Stage 2 is the period during which the clinician contemplates the possibility that the patient is ready for weaning. This statement may seem obvious to the point of banality. But the point at which this thought first enters the mind of a physician managing a complex patient is not so straightforward. In such a patient, the key act is for a physician to think that the patient just might come off the ventilator successfully. Except for self-extubations, this decision is not made by the patient. The idea has to begin in the doctor's brain. Several large studies have documented that many patients are ventilated for a week or more and the ventilator is then successfully discontinued on the first day that weaning predictor tests are measured.^{3,4} A physician has to ask himself or herself whether the patient might have tolerated extubation a day or so earlier. Failure to recognize this second stage may be the greatest obstacle to expeditious weaning.

Psychologists have undertaken extensive research into how probabilistic reasoning is employed in making decisions, investigating how people perceive, process, and evaluate the probabilities of uncertain events.⁵⁻⁷ Studies have clarified how interactions between the demands of a task and the limitations of a thinker negatively impact cognitive processes. Research on inferences that involve a sequence of steps has revealed that people make wrong decisions because they are more confident in their judgments (such as deciding that a patient is not ready for a T-tube trial) than is validly justified by the data on which the decisions are

In his recently published book, *Thinking, Fast and Slow*, the Nobel laureate, Daniel Kahneman, brings together the voluminous research conducted by his group and other investigators on human decision making over the past five decades.⁷ Kahneman presents human thinking as involving two independent systems. System One is amazingly fast, intuitive, and effort-free—but prone to error. System Two is the slow process of forming judgments based on conscious thinking and deductive reasoning. To activate System Two requires mental effort and hard work. Consequently, people turn to System One for most decisions. One of the discoveries cited by the Nobel committee was Kahneman's demonstration that overreliance on simple "seat-of-the-pants" decisions (characteristic of System One) can lead to large systematic errors, which have serious and persistent consequences.^{6,7} An example of a System One error in ventilator weaning is a clinician's intuition that a patient is not ready for a T-tube trial. Another System One error pertinent to weaning, and highlighted by Kahneman's research, is a clinician's failure to pay close attention to prior probability (Bayesian reasoning)—a failure that leads to major errors in decision making.⁶ When taking care of a ventilator-supported patient, physicians should be mindful of these cognitive processes and employ compensatory tactics such as the use of screening tests to spot a patient's readiness for weaning. By alerting an unsuspecting physician to a patient's readiness to tolerate unassisted ventilation—hours or days before he or she would otherwise order a spontaneous breathing trial—weaning-predictor tests circumvent the cognitive errors inherent in clinical decision making.⁸

Stage 3 is the time of obtaining physiologic measurements that serve as predictors, and interpreting the data appropriately in the context of each patient's unique clinical condition. The critical word here is *interpretation*. It is imperative to be clear about why these predictor tests are being performed, the influence of a patient's preexisting condition on the interpretation of the results, what action to take based on the results, and when it is prudent to adjust the thresholds for taking action. These points may seem self-evident, but the literature is replete with evidence of cloudy thinking on each point.

Stage 4 is to decrease ventilator support. Support is either removed abruptly and completely (T-tube trial) or gradually decreased over hours or days. Stage 5 is extubation of a patient who tolerated stage 4 or reinstitution of mechanical ventilation in a patient who failed the weaning trial. Stage 6 is continued ventilator support after extubation using noninvasive ventilation; this stage applies to only a minority of patients. Stage 7 is reintubation, usually accompanied by the reinstitution of mechanical ventilation.

Pathophysiology of Weaning Failure

Over the last 20 years, understanding of the mechanisms that cause patients to fail their first attempt to recommence spontaneous breathing has increased considerably. Advances in this aspect of weaning research have been enormously greater than change in clinical management. Greater understanding of pathophysiology has led to new approaches to the timing of the weaning process, prediction of outcome, and techniques used for weaning. Delineation of pathophysiologic principles led to the undertaking of clinical trials. The trials, in contrast, have contributed little to our understanding of the pathophysiology of weaning failure.

Research on pathophysiology has been limited to failure of attempts at spontaneous breathing when a still-intubated patient is first disconnected from the ventilator. Virtually no pathophysiologic research has been conducted in patients who develop acute respiratory failure in the hours immediately after extubation. Likewise, very little research has been conducted in patients who fail repeated weaning attempts, and, as a consequence, may be transferred to centers that specialize in the delivery of mechanical ventilation in the post-intensive care setting.

When intubated patients are disconnected from the ventilator and left to breathe on their own, about a fifth are unable to sustain spontaneous ventilation. If the trial is extended, these weaning failure patients will develop hypercapnia unless severe hypoxemia first intervenes. The pathophysiologic mechanisms that cause weaning failure can be divided into those occurring at the level of control of breathing, mechanics of the lung and chest wall, the respiratory muscles, the cardiovascular system, and gas-exchange properties of the lung.

Control of Breathing

The physiologic processes that fall under the heading of control of breathing primarily include afferent and efferent signals, and the processing of these signals in the brainstem. Clinical research on control of breathing has primarily focused on the overall level of respiratory motor output, termed respiratory drive. In human research, it is not realistic to obtain electrode recordings from the respiratory centers, and it is extremely difficult to measure phrenic nerve traffic. Consequently, a number of indirect methods have been used to assess respiratory drive. Measurement of the ventilatory response to hypercapnia or hypoxia is used in ambulatory patients, but difficult to apply in weaning failure patients. Electromyographic (EMG) recordings from the diaphragm reflect phrenic nerve traffic, but are difficult to standardize among patients.

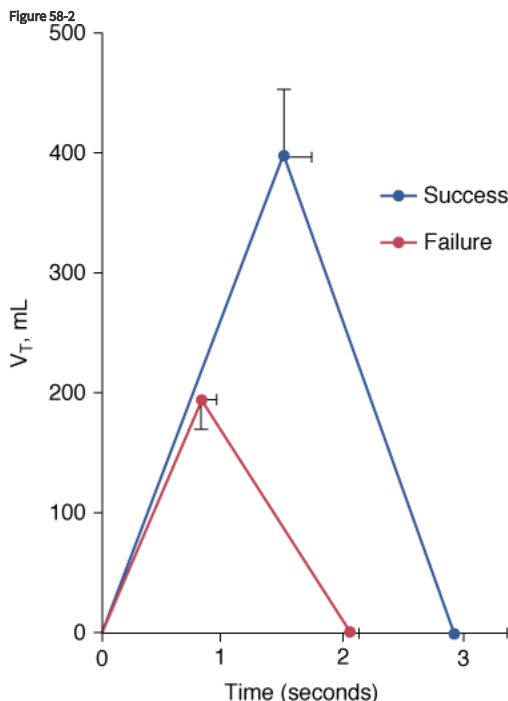
Most data on respiratory drive in weaning failure patients has been obtained with two techniques: the airway occlusion method ($P_{0.1}$) and mean inspiratory flow (V_T/T_I) of breathing pattern analysis. In a spontaneously breathing patient, it is not possible to measure $P_{0.1}$ on every breath or even at frequent intervals. If $P_{0.1}$ is measured repeatedly, the act of measurement will alter respiratory drive.⁹ As such, continuous measurements of respiratory drive over the evolution of weaning failure have been limited to V_T/T_I . V_T/T_I suffers from the limitation that oral airflow is far removed from the brainstem. Any intervening impediment, such as abnormal lung mechanics, can cause a decrease in V_T/T_I . Thus, there is always the possibility that V_T/T_I is providing an underestimate of respiratory drive. An advantage, however, of breathing pattern analysis over other methods of measuring respiratory drive is that it also provides information on respiratory timing.

Tobin et al¹⁰ studied seventeen patients who underwent a T-tube trial of weaning. Seven patients developed severe distress, and arterial carbon dioxide tension (Pa_{CO_2}) rose from 42 to 56 mm Hg, and pH fell from 7.43 to 7.35. Between the beginning and end of the trial, which lasted 40 ± 11 minutes, the patients developed an increase in V_T/T_I : 265 ± 27 to 328 ± 32 mL/s. These findings were surprising. At that time, it was expected that acute hypoventilation—the physiologic terminology for

Loading [Contrib]/a11y/accessibility-menu.js accompanied by a decrease in drive. Yet, not one patient had a value of V_T/T_I below the 95% confidence limits of normal

subjects. Although V_T/T_I was not depressed in the weaning failure patients, it was not higher than in the weaning success patients. Subsequent studies, using $P_{0.1}$, revealed that respiratory drive is higher in weaning failure patients than in weaning success patients.^{11–15}

The patients also showed marked changes in respiratory timing (Fig. 58-2). Upon resumption of spontaneous breathing, the weaning-failure patients exhibited immediate and marked shortening of inspiratory time (T_I): 0.81 ± 0.11 versus 1.41 ± 0.27 seconds in weaning-success patients. Within the respiratory centers, expiratory time (T_E) is strongly coupled to T_I . Consequently, T_E was also shorter in the weaning-failure patients than in the weaning-success patients: 1.24 ± 0.27 versus 2.48 ± 0.47 seconds. The combined changes in T_I and T_E led to a marked increase in respiratory frequency (f): 32.3 ± 2.3 versus 20.9 ± 2.8 breaths/min. Because the rate of inspiratory flow (V_T/T_I) was equivalent in the two patient groups, the short T_I resulted in a lower tidal volume (V_T) in the weaning-failure patients: 194 ± 23 versus 398 ± 56 mL. The decrease in V_T was balanced by the increase in f , and thus minute ventilation ($\dot{V}_E$) was equivalent in the two groups. A decrease in V_T without an increase in $\dot{V}_E$ must result in higher overall dead space ventilation (V_D/V_T). Indeed, the combined changes in V_T and f accounted for 81% of the increase in P_{aCO_2} observed in the weaning-failure patients. From the above discussion, it is evident that the fundamental abnormality in control of breathing in weaning failure is a shortening of T_I .



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The mean respiratory cycle during spontaneous breathing in seven weaning-failure and ten weaning-success patients. The early termination of inspiratory time in the weaning-failure patients leads to a decrease in tidal volume (V_T). The decrease in inspiratory time, coupled with a decrease in expiratory time, results in a faster respiratory frequency. Bars represent 1 standard error (SE). (Used, with permission, from Tobin et al.¹⁰)

Several groups of investigators have shown that the combination of increased f and low V_T is a characteristic abnormality in weaning-failure patients.

Vassilakopoulos et al.¹⁶ studied thirty patients at two points in time. Measurements were first obtained shortly after the patients failed a T-tube trial. Measurements were repeated approximately 9 days later, shortly before the patients were successfully extubated. The investigators found that an index of rapid shallow breathing, frequency-to-tidal-volume ratio (f/V_T), was lower in weaning failure than in weaning-success patients: 62 ± 21 versus 98 ± 38 . They obtained additional detailed measurements of lung mechanics and respiratory muscle function, and found that only two variables, tension-time index and f/V_T , were significant determinants of weaning failure.

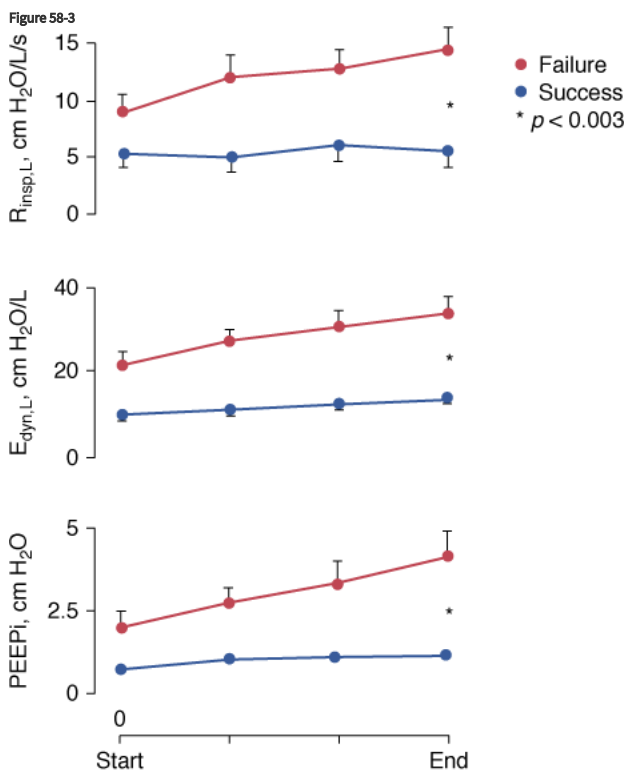
Research indicates that most weaning-failure patients develop an increase in respiratory drive as they experience progressive ventilatory failure. Clinical experience, however, suggests that at least some patients have depressed respiratory drive. Jubran and Tobin¹⁵ observed that two of seventeen (11.8%) weaning-failure patients developed P_{aCO_2} values of greater than 70 mm Hg during a T-tube trial, and yet detailed measurements of their lung mechanics and respiratory muscle function were within the range of the weaning-success patients. These limited data suggest that perhaps 10% of patients who develop hypercapnia during a failed weaning trial may do so primarily because of respiratory center depression.

Respiratory Mechanics

Physiologic variables for quantifying lung mechanics can be grouped under three major headings: resistance, elastance, and gas trapping. The most detailed data on weaned from mechanical ventilation comes from a study by Jubran and Tobin of thirty-one patients with chronic obstructive

pulmonary disease (COPD) undergoing a weaning trial.¹⁷ Over the course of a trial of spontaneous breathing lasting 45 ± 8 minutes, seventeen patients developed acute distress and an increase in (from 45 to 58 mm Hg), requiring the reinstitution of mechanical ventilation. The remaining fourteen patients tolerated the trial and were successfully extubated; these served as a control group.

At the start of the trial, inspiratory lung resistance was equivalent in the weaning-failure and weaning-success patients: 9.0 ± 1.7 versus 5.3 ± 1.1 cm H₂O/L/s (Fig. 58-3).¹⁷ By the end of the trial, resistance increased to 14.8 ± 2.0 cm H₂O/L/s in the failure patients, but it did not change in the success patients. Four factors may account for the increase in resistance: (a) an increase in inspiratory flow (unlikely because the increase in flow was no higher in failure patients than in success patients); (b) a decrease in lung volume (unlikely because most patients also develop gas trapping); (c) accumulation of secretions (unlikely because all patients had been suctioned before the trial, and secretions were no different in the two groups); and (d) bronchoconstriction. Bronchoconstriction appears to be the most likely explanation in that patients with COPD have heightened airway reactivity—although it is not clear why this should be greater in weaning-failure than in weaning-success patients.



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Inspiratory resistance of the lung ($R_{insp,L}$), dynamic lung elastance ($E_{dyn,L}$), and intrinsic positive end-expiratory pressure (PEEPi) in seventeen weaning-failure patients and fourteen weaning-success patients. Data displayed were obtained during the second and last minute of a T-tube trial, and at one-third and two-thirds of the trial duration. Between the onset and end of the trial, the failure group developed increases in $R_{insp,L}$ ($p < 0.009$), $E_{dyn,L}$ ($p < 0.0001$), and PEEPi ($p < 0.0001$) and the success group developed increases in $E_{dyn,L}$ ($p < 0.006$) and PEEPi ($p < 0.02$). Over the course of the trial, the failure group had higher values of $R_{insp,L}$ ($p < 0.003$), $E_{dyn,L}$ ($p < 0.006$), and PEEPi ($p < 0.009$) than the success group. (Used, with permission, from Jubran and Tobin.¹⁷)

Dynamic lung elastance was higher in weaning-failure patients than in weaning-success patients at the start of the trial: 21.2 ± 3.4 versus 9.9 ± 1.7 cm H₂O/L (see Fig. 58-3).¹⁷ At the end of the trial, elastance increased to 34.1 ± 4.0 cm H₂O/L in the failure patients and to 14.0 ± 2.0 cm H₂O/L in the success patients. The elevated elastance at the start of the trial was probably secondary to frequency dependence of elastance. Three factors may account for the increase in elastance over the course of the trial: (a) dynamic hyperinflation (this possibility is supported by a twofold increase in intrinsic PEEP [PEEPi] by the end of the trial); (b) development of subclinical pulmonary edema secondary to increased left ventricular afterload; and (c) microatelectasis (a possibility supported by the marked decrease in V_T).

PEEPi, an indirect measure of gas trapping, was higher in the failure patients than in the success patients at the onset of the trial: 2.0 ± 0.5 versus 0.7 ± 0.1 cm H₂O. By the end of the trial, PEEPi increased to 4.1 ± 0.8 cm H₂O in the failure patients and to 1.1 ± 0.2 cm H₂O in the success patients (see Fig. 58-3).¹⁷

Jubran and Tobin did not partition total PEEPi into the component resulting from expiratory muscle contraction and that resulting from an increase in end-expiratory lung volume. This information was subsequently obtained by the same group of investigators,¹⁸ who partitioned total PEEPi into that resulting from expiratory muscle contraction (abdominal muscles, expiratory rib cage muscles, or both) by calculating the rise in gastric pressure (Pga) between the onset of expiratory flow and the point of rapid decline in esophageal pressure (Pes), and the remaining portion, suggesting an increase in end-expiratory lung volume. After correcting for expiratory muscle contribution, the remaining portion of total PEEPi increased between the start and end of the weaning trial in seven of the ten patients.¹⁸ These data suggest that many weaning-failure patients develop dynamic hyperinflation. Expiratory flow limitation¹⁹ and tachypnea, through a decrease

in time available of exhalation,¹⁰ are the most likely determinants of dynamic hyperinflation. It should be recognized that it has not been possible to obtain direct measurements of end-expiratory lung volume in patients experiencing acute respiratory failure, and the use of esophageal pressure to estimate this entity is based on many assumptions.²⁰

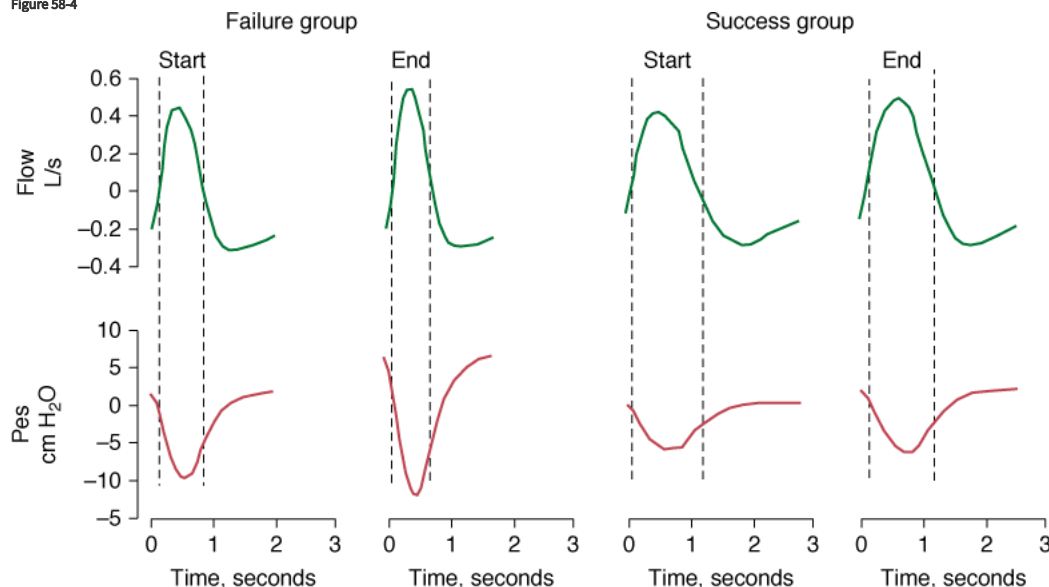
Other investigators have also reported a worsening of lung mechanics in weaning-failure patients. An innovative approach was employed by Vassilakopoulos et al.¹⁶ They first studied patients at the end of a T-tube trial. Then they reinstituted ventilation in the assist-control mode, sedated the patients, and hyperventilated them to abolish spontaneous respiratory muscle activity. Next they adjusted the ventilator settings to simulate a patient's pattern of spontaneous breathing, and measured lung mechanics under passive conditions. The investigators studied the patients at two points in time: shortly after they first failed a T-tube trial, and approximately 9 days later, shortly before they were successfully extubated. Between the time of weaning failure and weaning success, airway resistance decreased from 9.6 ± 3.4 to 7.9 ± 3.3 cm H₂O/L/s, static PEEPi decreased from 6.1 ± 2.5 to 3.8 ± 2.7 cm H₂O, and static respiratory compliance did not change. The investigators also disconnected the patients from the ventilator (after first delivering some breaths simulating spontaneous breathing), and allowed them to exhale freely until zero expiratory flow was reached. This point was taken as the elastic equilibrium volume of the respiratory system, and the increase in functional residual capacity secondary to gas trapping (PEEPi) was taken as the difference between inspired and expired volume. This volume was 327 ± 180 mL during weaning failure, and it fell to 213 ± 175 at the time of weaning success.

The observation that weaning-failure patients display more severely deranged lung mechanics than do weaning-success patients raises the question of whether the derangements might be detectable ever before patients reattempt spontaneous breathing (i.e., while patients are still receiving full ventilator support). Jubran and Tobin²¹ studied lung and chest wall mechanics of patients before the onset of a weaning trial. Inspiratory resistance was about fourteen times higher than that in healthy subjects but it was equivalent in the failure and success patients: 13.9 and 13.0 cm H₂O/L/s, respectively. Static elastance of the lung and the chest wall were similar in the two groups.²¹ Dynamic elastance of the lung was higher in the failure patients than in the success patients, 28 ± 3 versus 17.8 ± 2 cm H₂O/L, and this was the only measurement of passive mechanics that differentiated the two groups. Dynamic PEEPi during passive ventilation was equivalent in the groups. This picture contrasts with the more severely deranged mechanics in failure patients than in success patients during a weaning trial.¹⁷ The difference indicates that something in the act of spontaneous breathing, rather than an intrinsic abnormality in respiratory mechanics, is responsible for the marked difference between failure and success patients during a weaning trial.

Patient Effort

The deterioration in lung mechanics during a failed weaning trial leads to increased work of breathing. The increased respiratory work is made manifest by greater swings in esophageal pressure (Fig. 58-4). In the study of Jubran and Tobin, pressure-time product was not different in the weaning-failure and weaning-success patients at the onset of a spontaneous breathing trial: 255 ± 59 and 158 ± 23 cm H₂O*s/min (normal: 94 ± 12).¹⁷ At the end of the trial, pressure-time product increased more in the failure patients than in the success patients: 388 ± 68 versus 205 ± 25 cm H₂O*s/min. The increase in effort in the failure patients resulted from worsening of all elements of respiratory mechanics. Partitioning of the increase in pressure-time product at end of the trial revealed that the fraction caused by PEEPi increased by 111%, that caused by the non-PEEPi elastic component increased by 33%, and the fraction caused by the resistive component increased by 42%.¹⁷

Figure 58-4

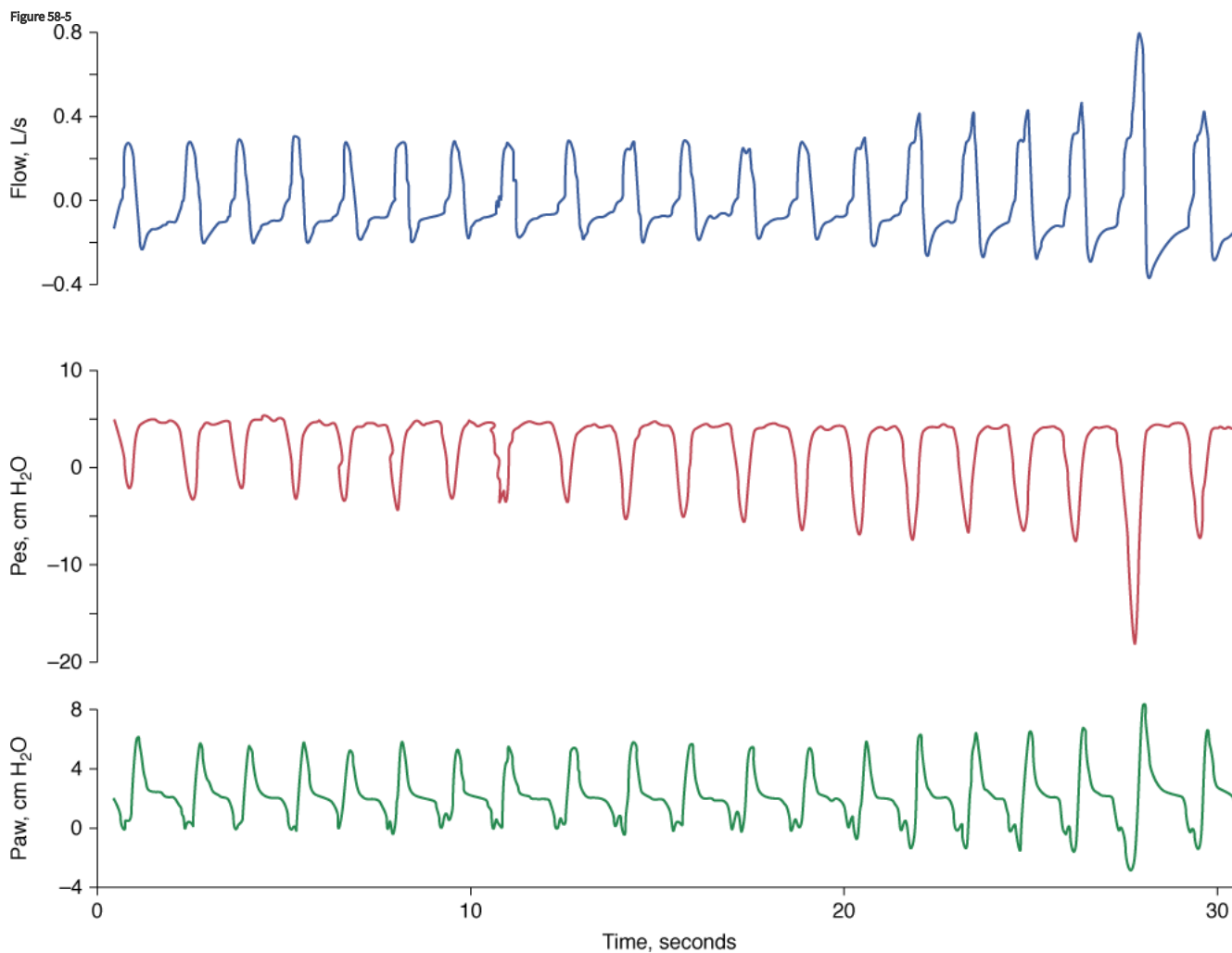


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Ensemble average plots of flow and esophageal pressure (*Pes*) at the start and end of a T-tube trial in seventeen weaning-failure patients and fourteen weaning-success patients. At the start of the trial, the inspiratory excursion in *Pes* was greater in the failure patients, and it increased further by the end of the trial. To generate the plots, the data were divided into twenty-five equal time intervals over a single respiratory cycle for each of the 5 breaths for each patient in the

two groups. For a given patient, the 5 breaths from the start of the trial were then superimposed and aligned with respect to time, and the average at each time point was calculated. The group mean tracings were then generated by ensemble averaging of the individual mean from each patient. The same procedure was performed for breaths at the end of the trial. (Used, with permission, from Jubran and Tobin.¹⁷)

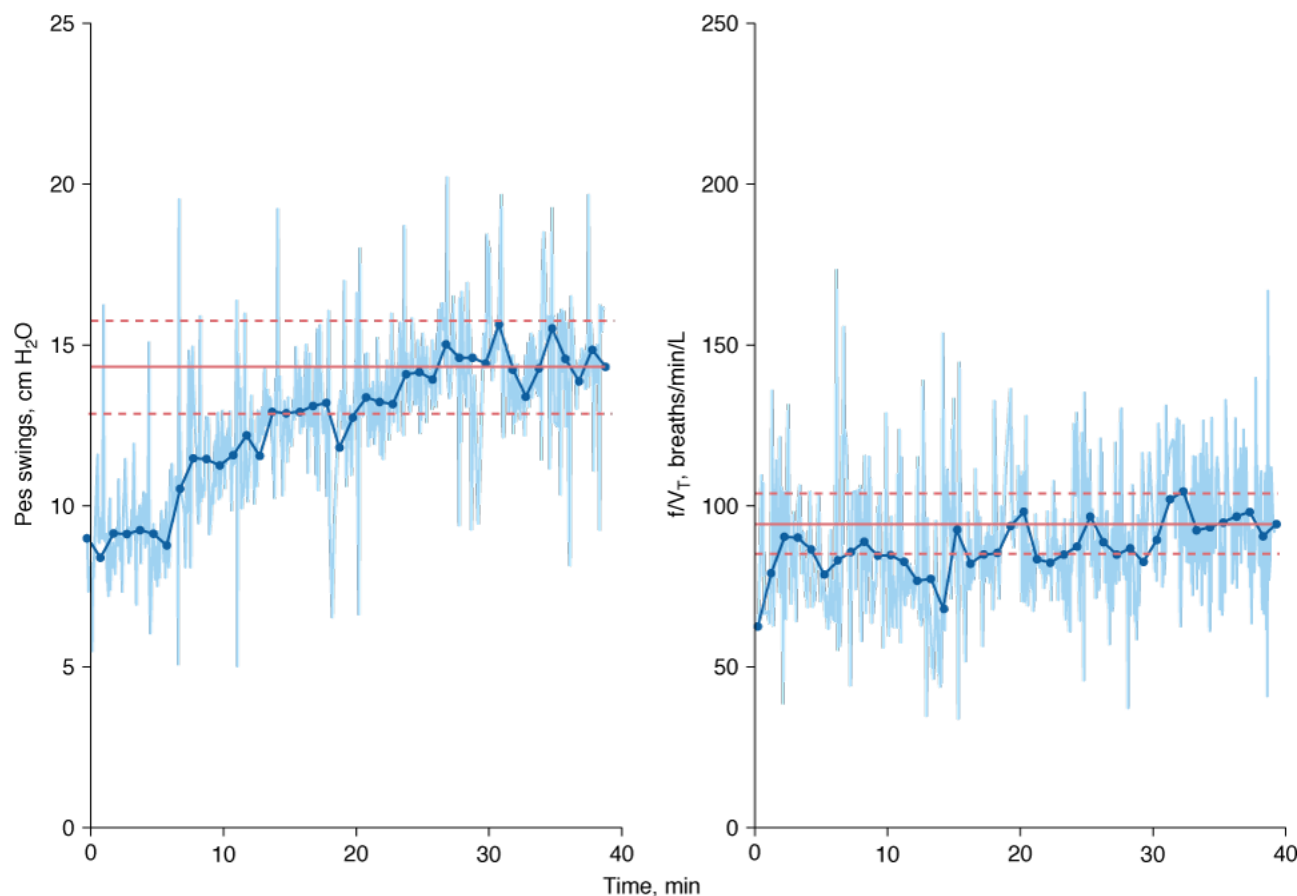
In sixty patients being weaned from mechanical ventilation, Jubran et al²² characterized the changes in esophageal-pressure swings over the course of a trial of spontaneous breathing (Fig. 58-5). The median time to reach $\pm 10\%$ of the average value of esophageal pressure during the last minute of the trial was 7.5 (interquartile range [IQR]: 4.2 to 14.8) minutes in the weaning-failure patients and 5 (IQR: 2 to 8.5) minutes in the weaning-success patients (Fig. 58-6). In contrast, f/V_T , a measure of rapid shallow breathing, reached $\pm 10\%$ of its final average value at 2 (IQR: 1 to 2) minutes in both the weaning-success and weaning-failure patients. The more gradual and progressive increase in esophageal pressure over time may have resulted from a slow increase in partial pressure of carbon dioxide (P_{CO_2}), as commonly occurs in weaning-failure patients.^{10,17} The relative constancy in f/V_T may be related to the increases in both inspiratory resistive and inspiratory elastic loads that occur in weaning-failure patients. These loads have opposing effects on breathing pattern. A resistive load slows respiratory frequency while preserving tidal volume, thereby producing a decrease in f/V_T .^{23,24} In contrast, an elastic load increases respiratory frequency accompanied by a decrease in tidal volume, both of which will produce an increase in f/V_T .^{25,26}



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Progressive increase in inspiratory effort in a weaning-failure patient. Flow (*top panel*), esophageal pressure (Pes, *middle panel*), and airway pressure (Paw, *bottom panel*) in a patient who developed severe respiratory distress while receiving continuous positive airway pressure of 5 cm H₂O. The patient had developed respiratory failure (requiring mechanical ventilation) after developing a pulmonary embolus subsequent to undergoing lobectomy for lung cancer. The swings in esophageal pressure became progressively more negative over the first 30 seconds of the trial.

Figure 58-6



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Time-series plot of swings in esophageal pressure (*Pes*; *left panel*) and frequency-to-tidal-volume ratio (f/V_T ; *right panel*) during a trial of spontaneous breathing in a weaning-failure patient. Dots represent 1-minute averages. The solid horizontal line indicates the average value of *Pes* swings and f/V_T during the final minute of the trial. The dashed lines indicate $\pm 10\%$ of the final minute values for *Pes* swings and f/V_T . The time taken to reach $\pm 10\%$ of the final value was 14 minutes for *Pes* swings and 2 minutes for f/V_T . (Used, with permission, from Jubran et al.²²)

Respiratory Muscles

Research into the mechanisms whereby abnormalities of the respiratory muscles might contribute to weaning failure has focused on inspiratory muscle strength and respiratory muscle fatigue. Muscle strength has been assessed by measuring the pressure generated during a maximal inspiratory effort against an occluded airway.²⁷ Early investigators reported that maximal inspiratory pressure ($P_{I\max}$) was lower in weaning-failure patients than in weaning-success patients, but later investigators reported no difference between the two groups.^{14,28–31} The pattern of reporting suggests the possibility of test-referral bias, whereby patients with the lowest values of $P_{I\max}$ were deliberately excluded from the subsequently conducted studies.³² Another consideration is the well-recognized difficulty of ensuring reliable measurements of $P_{I\max}$, given its total dependence on patient motivation and cooperation—an even greater challenge in critically ill patients.²⁷ Studies using phrenic nerve stimulation, and specifically the technique of twitch interpolation, revealed that patients being weaned from mechanical ventilation typically make submaximal efforts when $P_{I\max}$ is being measured. Moreover, Laghi et al.²⁹ found that six of nine weaning-failure patients had twitch transdiaphragmatic pressure (P_{di}) values below 10 cm H₂O. Healthy subjects have twitch P_{di} values of 35 to 39 cm H₂O, and stable patients with COPD have values of 17 to 20 cm H₂O. Contrary to previous thinking, these data indicate that weaning-failure patients may have considerable muscle weakness.

For years, researchers and clinicians have believed that most if not all weaning-failure patients develop respiratory muscle fatigue by the time a failed weaning trial is stopped. This belief has been largely based on observations made by Cohen et al.³³ These investigators studied twelve patients who exhibited difficulties during weaning. Seven patients developed a shift in the power spectrum of the EMG signal recorded from the diaphragm, a finding judged to signify muscle fatigue. Six of the seven patients also exhibited paradoxical motion of the abdomen (inward displacement of the abdomen during inspiration) and four exhibited respiratory alternans (phasic alternation between the contribution of the rib cage and abdominal compartments to V_T). The changes in rib cage–abdominal motion were not observed in the five patients who did not develop EMG changes. The investigators concluded that respiratory muscle fatigue was a common cause of weaning failure, and that its presence could be detected by finding paradoxical motion of the abdomen.

Subsequent detailed recordings of rib cage–abdominal motion revealed that when paradoxical motion of the abdomen occurs in weaning-failure patients, it occurs
i Loading [Contrib]/a11y/accessibility-menu.js the ventilator and displays no progression over time.³⁴ When quantified objectively, the extent of abdominal paradox was no

greater in weaning-failure patients than in weaning-success patients. In studies of healthy volunteers, fatigue was found to be neither necessary nor sufficient to induce abnormal rib cage–abdominal motion.³⁵ These data indicated that rib cage–abdominal motion could not be used for detecting respiratory muscle fatigue. The studies, however, did not exclude the possibility that fatigue is common in weaning-failure patients.

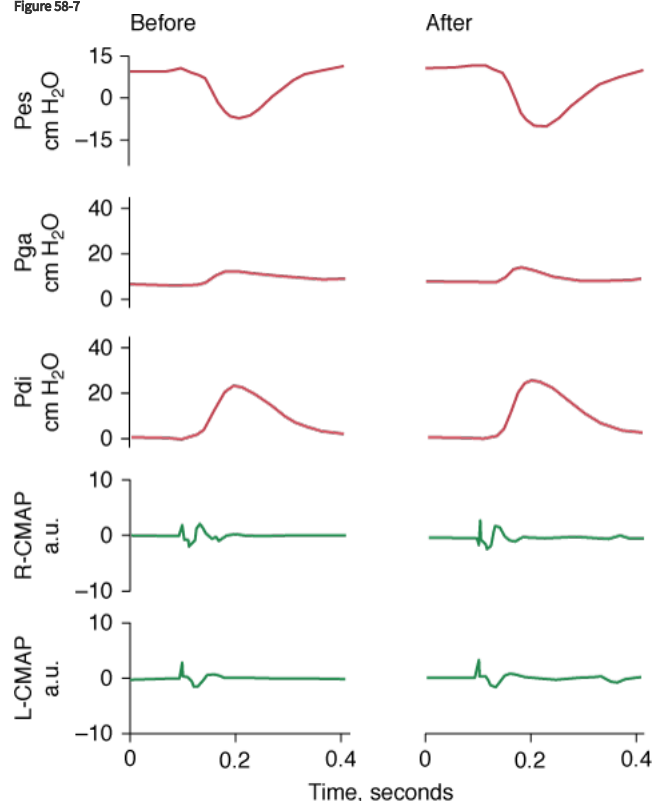
Investigators also evaluated a more complex measure of fatigability, tension–time index. Tension–time index is the product of two fractions:

$$(\text{mean pressure per breath}/P_{I\max}) \times (T_I/T_{TOT})$$

Studies in healthy volunteers show that respiratory muscle fatigue becomes inevitable when subjects breathe against an inspiratory load that causes tension–time index to rise above a threshold of 0.15. In a number of studies,^{16,17,36} many more weaning-failure patients than weaning-success patients were found to exhibit tension–time index values above 0.15. As such, weaning-failure patients experience workloads that are sufficient to induce respiratory muscle fatigue.

The EMG power spectrum and tension–time index provide only indirect evidence of fatigue, and do not provide direct proof of its occurrence. In neurophysiologic terms, fatigue means that a muscle is generating less force in response to a given neural stimulus than it had generated in the past. The most direct method for detecting fatigue in patients is to stimulate the phrenic nerves in the neck and measure the resulting change in Pdi. The challenge with use of phrenic nerve stimulation in critically ill patients is to ensure that successive twitches are all generated at the same end-expiratory lung volume, a constant degree of neural depolarization is achieved by the stimulator, and twitch potentiation (the increase in pressure that occurs with a recent forceful contraction) is avoided. Laghi et al²⁹ measured twitch Pdi using phrenic stimulation in eleven weaning-failure patients and eight weaning-success patients before and after a T-tube trial. Twitch Pdi was 8.9 ± 2.2 cm H₂O before the trial and 9.4 ± 2.4 cm H₂O after the trial in the weaning-failure patients (Fig. 58-7). The respective values in the weaning-success patients were 10.3 ± 1.5 and 11.2 ± 1.8 cm H₂O. No patient in either group exhibited a fall in twitch Pdi. The failure to develop fatigue was surprising because seven of the eight weaning-failure patients had a tension–time index above 0.15.

Figure 58-7

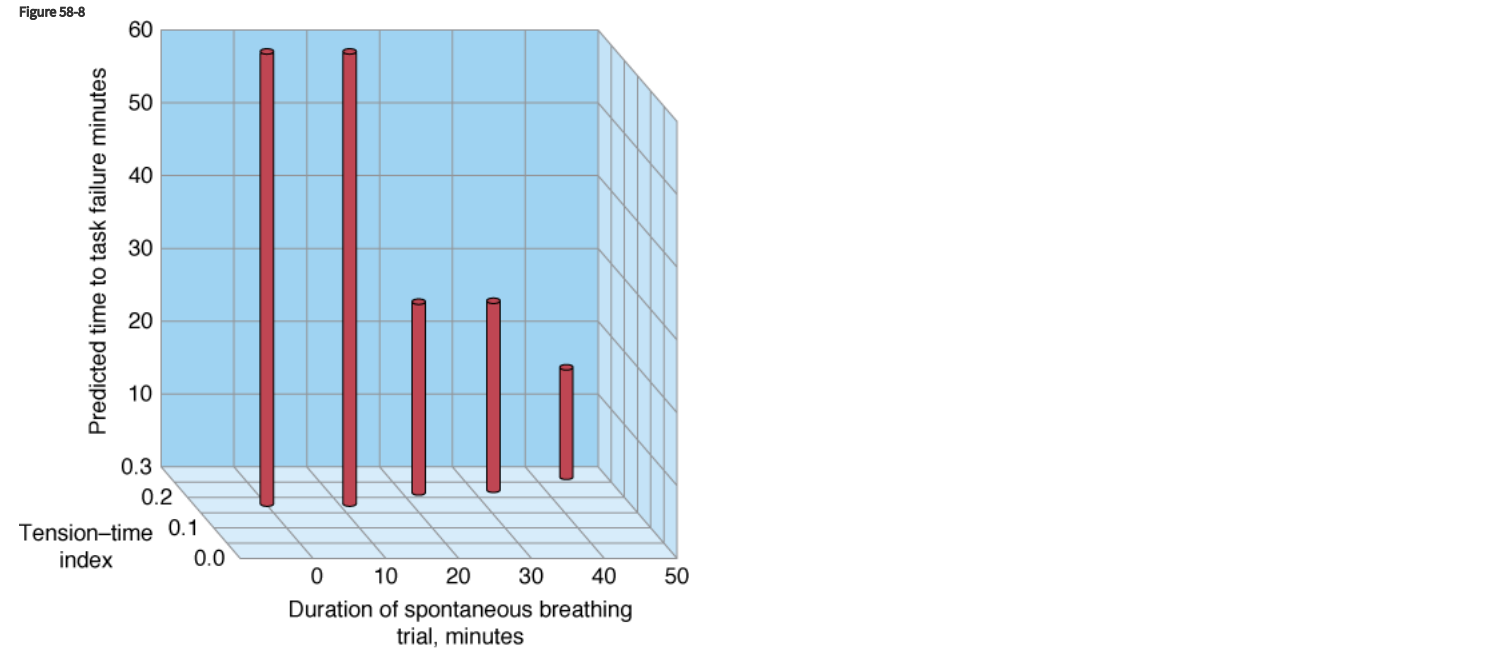


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Esophageal pressure (*Pes*), gastric pressure (*Pga*), transdiaphragmatic pressure (*Pdi*), and compound motor action potentials (*CAMP*) of the right and left hemidiaphragms after phrenic nerve stimulation before (*left*) and after (*right*) a T-tube trial in a weaning-failure patient. The end-expiratory value of *Pes* and the amplitude of the right and left *CAMP*s were the same before and after the trial, indicating that the stimulations were delivered at the same lung volume and that the stimulations achieved the same extent of diaphragmatic recruitment. The amplitude of twitch *Pdi* elicited by phrenic nerve stimulation was the same before and after weaning. (Used, with permission, from Laghi et al.²⁹)

The most likely reason that patients did not develop fatigue is because physicians reinstituted mechanical ventilation before there was enough time for its development. The relationship between tension–time index and the length of time that a load can be sustained until task failure follows an inverse-power function. Bellemare and Grassino³⁷ expressed the relationship as: time to task failure = $0.1 (\text{tension–time index})^{-3.6}$. Figure 58-8 shows the increase in tension–time index over the course of the weaning trial²⁹ and predicted time to task failure.³⁷ At the point that the physician reinstituted mechanical ventilation, patients were on 13 minutes of spontaneous breathing before developing task failure. In other words, clinical manifestations of severe

respiratory distress were evident for a substantial time before the patients were predicted to develop fatigue. In an intensive care setting, these clinical signs will lead attendants to reinstitute mechanical ventilation before fatigue has time to develop.

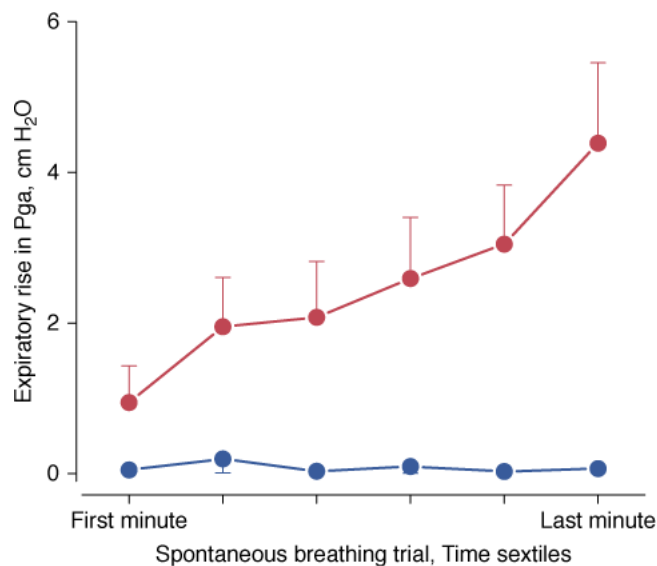


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Interrelationship between the duration of a spontaneous breathing trial, tension-time index of the diaphragm, and predicted time to task failure in nine patients who failed a trial of weaning from mechanical ventilation. The patients breathed spontaneously for an average of 44 minutes before a physician terminated the trial. At the start of the trial, the tension-time index was 0.17, and the formula of Bellemare and Grassino³⁷ (see text for details) predicted that patients could sustain spontaneous breathing for another 59 minutes before developing task failure. As the trial progressed, the tension-time index increased and the predicted time to development of task failure decreased. At the end of the trial, the tension-time index reached 0.26. That patients were predicted to sustain spontaneous breathing for another 13 minutes before developing task failure clarifies why patients did not develop a decrease in diaphragmatic twitch pressure. In other words, physicians interrupted the trial on the basis of clinical manifestations of respiratory distress, before patients had sufficient time to develop contractile fatigue. (Used, with permission, from Laghi and Tobin.²⁷).

In a study of nineteen patients being weaned from mechanical ventilation,¹⁸ all but one of the eleven weaning-failure patients exhibited expiratory muscle activity (the exception being a patient with paraplegia). Expiratory muscle activity—as quantified by the expiratory rise in Pga—was absent in all but three of eight weaning-success patients, and its magnitude was trivial in the remainder (Fig. 58-9). At the onset of the trial, the expiratory rise in Pga was equivalent in the failure and success groups, 0.9 ± 0.5 and 0.1 ± 0.1 cm H₂O, respectively. At the end of the trial, the expiratory rise in Pga increased to 4.4 ± 1.1 cm H₂O in the failure group ($p = 0.0005$), whereas it did not change, 0.1 ± 0.1 cm H₂O, in the success group. Compared with the success group, the failure group exhibited larger increases in expiratory rise in Pga ($p = 0.004$). In the failure group, expiratory muscle activity accounted for $53 \pm 4\%$ of total PEEPi throughout the weaning trial.





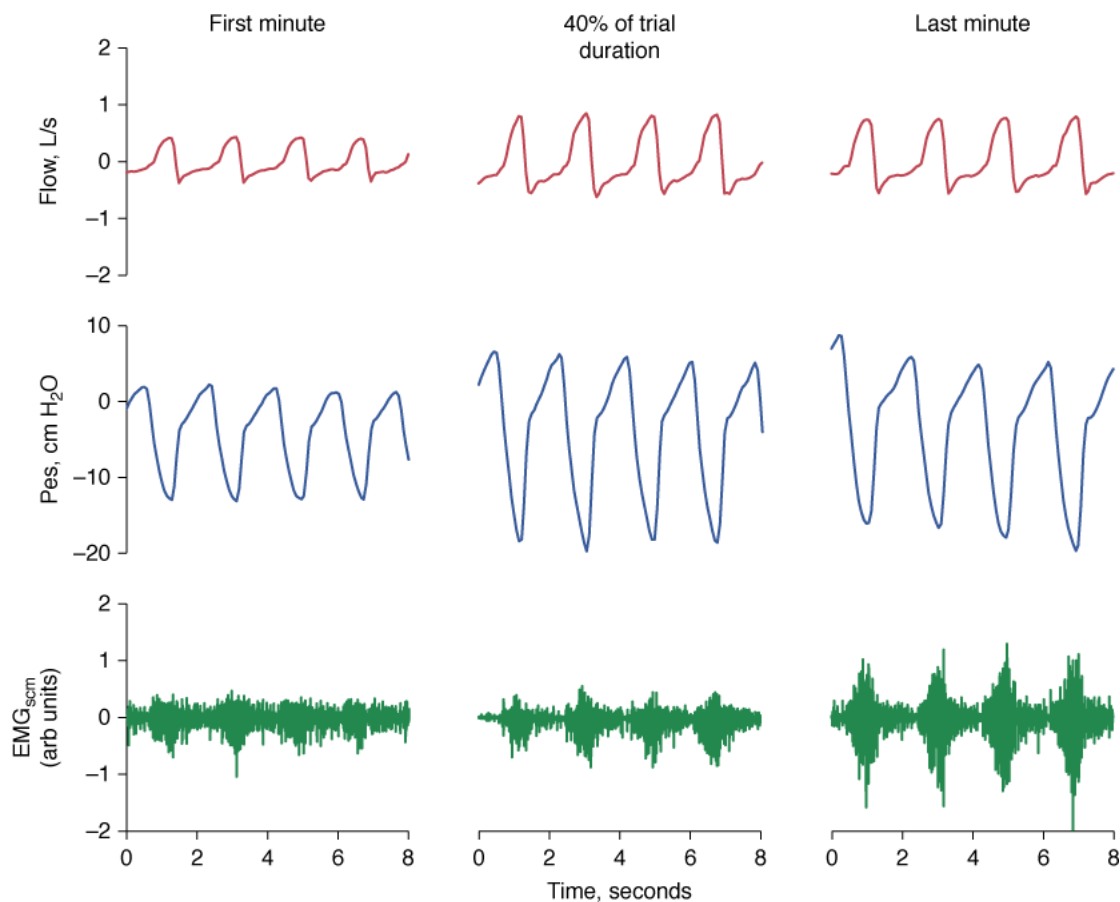
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Expiratory rise in gastric pressure (*Pga*) during the course of a weaning trial in failure (•) and success patients (•). Between the onset and the end of the trial, failure patients developed an increase in the expiratory rise in *Pga* ($p = 0.0005$) whereas success patients did not. Over the course of the trial, failure patients had higher values of expiratory rise in *Pga* ($p = 0.004$) than did success patients. Bars represent $\pm$ standard error (SE). (Modified, with permission, from Parthasarathy et al.¹⁸)

In the study of Parthasarathy et al.,¹⁸ sternomastoid EMG activity, measured with fine-wire electrodes, was evident in $83 \pm 9\%$ of all the breaths in the weaning-failure group and in $19 \pm 10\%$ of all breaths in the weaning-success group ($p = 0.002$) (Fig. 58-10). Sternomastoid activity became evident within the first minute of the trial in eight of the eleven failure patients and in one of the eight success patients. By the end of the trial, sternomastoid activity was noted in all failure patients but in only three of the success patients, and this activity was modest. The immediate increase in sternomastoid activity in the failure patients probably results from increased respiratory motor output in response to a combination of decreased capacity of the respiratory muscles to generate pressure²⁹ and an increase in mechanical load that occurs early during the weaning trial. In addition to increased sternomastoid activity, weaning-failure patients displayed greater inspiratory rib cage muscle contribution to tidal breathing throughout the trial than did the success patients (Fig. 58-11).

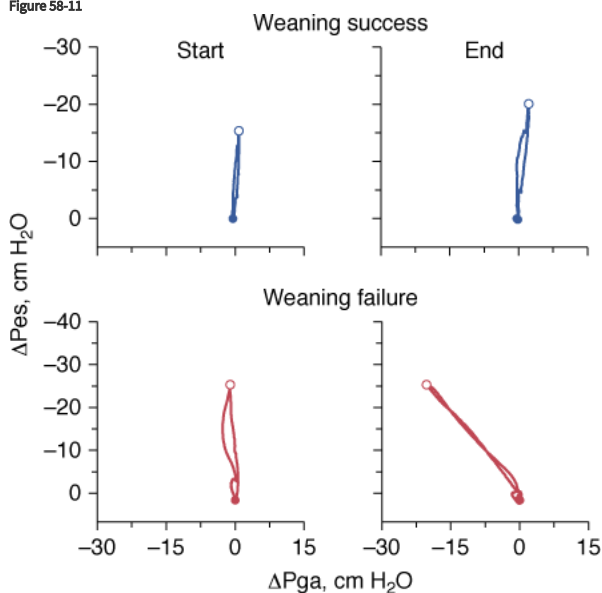
Figure 58-10



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Representative tracings of flow, esophageal pressure (Pes), and electromyogram of the sternomastoids (EMG_{scm}) in a weaning-failure patient. Recordings were obtained during the first minute of the weaning trial, 40% of trial duration, and last minute of the trial. Phasic inspiratory activity of the sternomastoid muscle was evident within the first minute of the trial, and it increased progressively over the course of the trial. Note that phasic activity of the sternomastoids persists into expiration. (Used, with permission, from Parthasarathy et al.¹⁸)

Figure 58-11



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Plots of tidal changes in esophageal pressure (ΔPes) against tidal changes in gastric pressure (ΔPga) in a weaning-success patient and a weaning-failure patient. At the start of a weaning trial, the success patient (top left panel) exhibited swings in esophageal pressure that became markedly more negative between the onset (closed symbol) and the end of inspiration (open symbol); in contrast, gastric pressure increased only slightly. Therefore, the slope of the Pes-Pga plot at the onset of inspiration was steeper than the slope recorded in healthy subjects during resting breathing, where the tidal change in gastric pressure is often

greater than the tidal change in esophageal pressure. The steep Pes–Pga plot (*top left panel*) indicates a greater-than-usual contribution of the rib cage muscles to tidal breathing than that of the diaphragm. Between the onset (*top left panel*) and the end of weaning (*top right panel*), the slope of the Pes–Pga plot changed very little, indicating a constant contribution of the diaphragm and rib cage muscles to tidal breathing over the course of the weaning trial. In the case of the weaning-failure patient, the inspiratory swings in esophageal pressure and gastric pressure had a similar pattern at the start of the trial to that in the success patient (*bottom left panel*). At the end of the trial, the failure patient exhibited a markedly negative slope in the Pes–Pga plot, signifying a further increase in inspiratory rib cage muscle recruitment that was out of proportion to diaphragmatic recruitment.

A striking feature of the weaning-failure patients is the timing at which different muscle groups become active.¹⁸ The sequence begins with activity of the diaphragm and with greater activity of inspiratory rib cage muscles than is the case in the success patients; recruitment of sternomastoids and rib cage muscles is near maximum within 4 minutes of trial commencement, whereas the expiratory muscles are not recruited until quite late in the trial (at 17 to 20 minutes).¹⁸ The existence of a hierarchy of respiratory muscle activation is supported by the known delayed activation of the sternomastoid muscles³⁸ and expiratory muscles in healthy volunteers^{39,40} and in ambulatory patients with COPD.⁴¹

Cardiovascular Performance

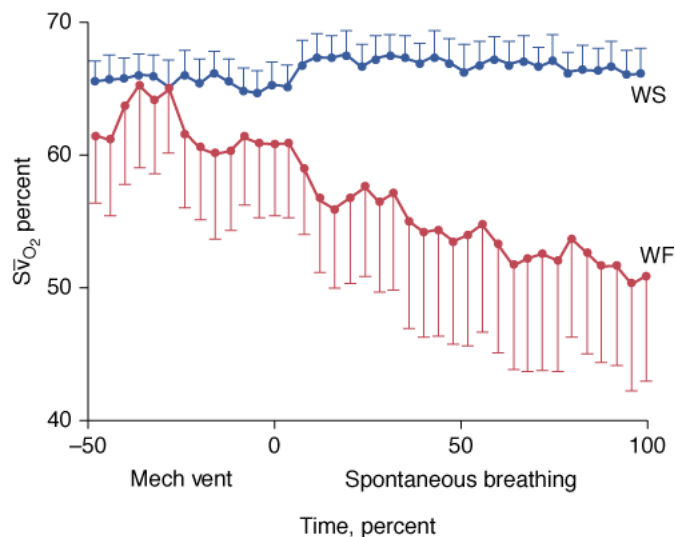
Although the respiratory muscles do not develop fatigue, they perform a huge workload. Thus, they depend on an efficient transport of oxygen by the cardiovascular system. Aware of this fact, several researchers have examined cardiovascular performance during weaning. Lemaire et al⁴² studied fifteen patients with COPD, seven of whom had documented ischemic heart disease. After 10 minutes of breathing (through the ventilator without PEEP), the patients developed increases in transmural pulmonary artery occlusion pressure (PAOP) (8 to 25 mm Hg), cardiac index (3.2 to 4.3 L/min/m²), left ventricular end-diastolic volume index (65 to 83 mL/m²), and right ventricular end-diastolic volume index (83 to 103 mL/m²). The investigators attributed the increase in left ventricular end-diastolic volume to augmentation of venous return (secondary to low pleural pressure during spontaneous breathing and central translocation of blood volume secondary to peripheral vasoconstriction) and increased left ventricular afterload (secondary to markedly negative pleural pressure swings and increased catecholamine release). Nine of the fifteen patients were weaned after 10 days of diuretic therapy, at which time PAOP had fallen to 9 mm Hg.

Although a particular PAOP has not been rigorously linked with the inevitable onset of cardiogenic pulmonary edema, an increase in PAOP above 18 mm Hg during a weaning trial is widely considered to indicate the onset of weaning-induced pulmonary edema.^{43,44} Several research groups have reported increases in PAOP in patients as they failed a weaning trial, although these researchers did not detect a concomitant decrease in cardiac output.^{42,45–47} As such, the cardiorespiratory stress posed by the resumption of spontaneous breathing after a period of mechanical ventilation resembles the challenges posed by whole-body exercise, which also results in increases in cardiac output and work of breathing.⁴⁸ Increases in adrenergic tone that accompany increased cardiorespiratory stress can produce increases in venous return, left-ventricular afterload, cardiac work, and myocardial oxygen demand—all of which may precipitate myocardial ischemia in predisposed patients.^{42,43}

Richard et al⁴⁹ studied a group of twelve patients with COPD who did not have documented coronary artery disease. All of the patients tolerated at least two 30-minute T-tube trials. (This finding suggests that they were weaning-success patients, although it is not stated that they tolerated extubation.) Ejection fraction, measured by technetium^{99m} radionuclide angiography, was 54.5 ± 12.4% during mechanical ventilation. Spontaneous breathing resulted in a fall in ejection fraction to 47 ± 13%. The fall was homogenous and not accompanied by regional wall abnormalities that occur with myocardial ischemia. Moreover, thallium imaging performed 15 minutes after the weaning trial revealed normal myocardial perfusion. The investigators attributed the decrease in ejection fraction to increased left ventricular afterload.

The increases in oxygen consumption during a weaning trial means that patients who are unable to achieve sufficient increases in cardiac output are at risk of experiencing a decrease in oxygen transport. To investigate these considerations, Jubran et al³⁹ continuously recorded mixed venous oxygen saturation ($\bar{Sv}_{O_2}$) in eight weaning-failure and eleven weaning-success patients over the course of T-tube trials that lasted about 40 minutes. Immediately before the trial, $\bar{Sv}_{O_2}$ was equivalent in the two groups. On discontinuation of the ventilator, $\bar{Sv}_{O_2}$ fell progressively in the failure patients (to 51.5 ± 7.9% at the end of the trial), whereas it did not change in the success patients (Fig. 58-12). Oxygen demand ($\dot{V}_{O_2}$) was similar in the two groups during the weaning trial, although it differed in the manner with which it was met.

Figure 58-12

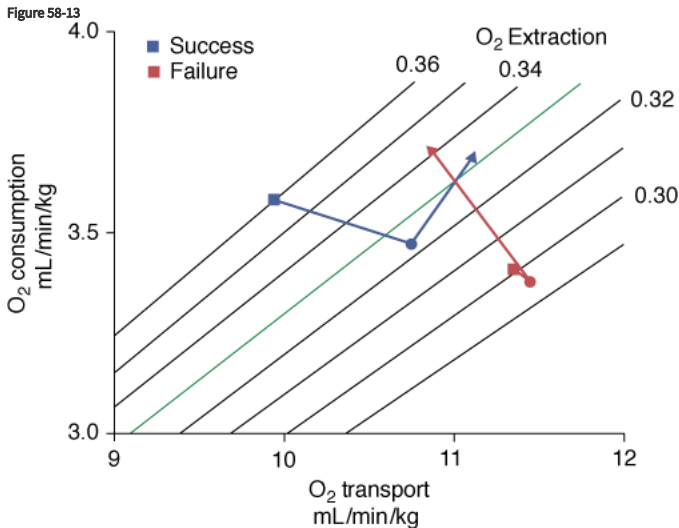


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Ensemble averages of the interpolated values of mixed venous oxygen saturation (SvO_2) during mechanical ventilation (Mech Vent) and a trial of spontaneous breathing in weaning-success patients (blue symbols) and weaning-failure patients (red symbols). During mechanical ventilation, SvO_2 was similar in the two groups ($p = 0.28$). Between the onset and the end of the trial, SvO_2 decreased in the failure patients ($p < 0.01$), whereas it did not change in the success patients ($p = 0.48$). Over the course of the trial, SvO_2 was lower in the failure patients than in the success patients ($p < 0.02$). (Bars represent standard error [SE].) (Used, with permission, from Jubran et al.⁴⁵)

The success patients demonstrated an increase in cardiac index between mechanical ventilation and the end of the trial, 3.07 to 3.51 L/m/m², which was accompanied by an increase in oxygen transport (Fig. 58-13). The failure group did not experience an increase in oxygen transport (partly because of elevations in right and left ventricular afterload); instead, they experienced an increase in oxygen extraction ratio, which, in turn, contributed to the fall in SvO_2 . The failure patients also had more impaired pulmonary gas exchange ($\dot{Q}_{VA}/\dot{Q}_T$ was 0.32 at the start of the trial). The combination of greater venous admixture and low SvO_2 led to rapid arterial desaturation and a relative decrease in oxygen being supplied to the tissues.



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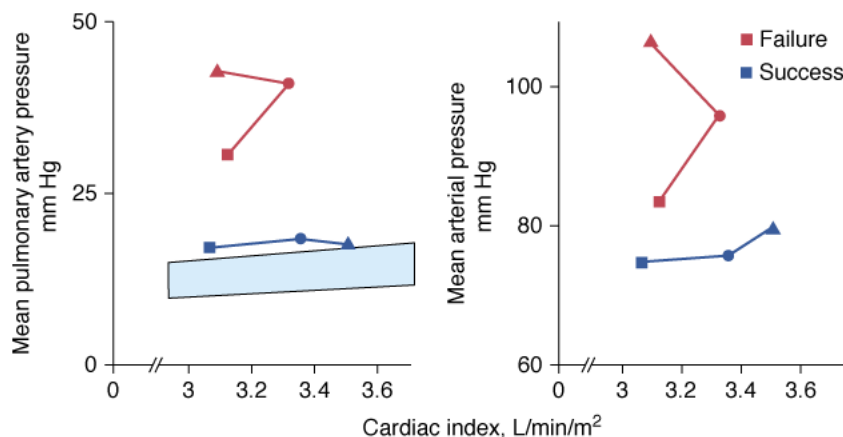
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Oxygen transport, oxygen consumption, and isopleths of oxygen extraction ratio in weaning-success (blue symbols) and weaning-failure patients (red symbols) during mechanical ventilation (squares) and at the onset (circles) and end (triangles) of a T-tube trial. (Used, with permission, from Jubran et al.⁴⁵)

Mean pulmonary artery pressure was higher in the failure patients than in the success patients during mechanical ventilation (Fig. 58-14). The pressure increased further over the course of the trial in the failure patients, whereas the success patients showed no change. Several factors account for the increase in pulmonary artery pressure. Hypoxemia and acidosis are potent vasoconstrictors. Pulmonary artery pressure can also be increased by alveolar vessel compression secondary to the increase in alveolar pressure that accompanies the dynamic hyperinflation and deterioration in pulmonary mechanics in weaning failure. During mechanical ventilation, the two groups had an equivalent mean arterial pressure, which increased in the failure patients by the end of the trial. The increase in mean arterial pressure combined with no change in cardiac index indicates an increase in left ventricular afterload.

Figure 58-14

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Mean pulmonary artery pressure and mean arterial pressure versus cardiac index during mechanical ventilation (*squares*) and at the onset (*circles*) and end (*triangles*) of a T-tube trial in weaning success (*blue symbols*) and weaning-failure patients (*red symbols*). The *blue shaded area* represents the normal range of increase in mean pulmonary artery pressure with cardiac index. In the success patients, cardiac index increased between mechanical ventilation and the end of the trial, mean pulmonary artery pressure remained slightly above the normal range, and mean arterial pressure did not change. Conversely, in the failure patients, cardiac index was similar during mechanical ventilation and at the end of the T-tube trial, but both mean pulmonary artery pressure and mean arterial pressure were higher by the end of the trial ($p < 0.025$ and $p < 0.05$, respectively). The increases in these vascular pressures, together with the lack of change in cardiac index, indicate increases in right and left ventricular afterload in the failure patients. (Used, with permission, from Jubran et al.⁴⁵)

In an investigation of similar design, Zakyntinos et al.³¹ studied twelve weaning-success and eighteen weaning-failure patients during a spontaneous breathing trial. Half of the failure patients increased their *oxygen* consumption and this increase was met mainly by an increase in *oxygen* extraction. The remaining half of the failure patients did not exhibit an increase in *oxygen* consumption; instead, increase in *oxygen* delivery was accompanied by a decrease in *oxygen* extraction. These studies^{31,42,45} demonstrate variability in circulatory and global tissue oxygenation responses during weaning failure.

Gas Exchange

A primary goal of mechanical ventilation is to improve gas exchange, and accordingly one expects some deterioration in gas exchange with the resumption of spontaneous breathing. The most detailed study of gas exchange during weaning is that conducted by Beydon et al.⁵⁰ They studied eight patients with COPD who were considered ventilator dependent (although the patients were able to sustain at least 1- to 3-hour periods of spontaneous breathing). When switched from controlled ventilation, patients developed an increase in frequency, fall in V_T (without change in $\dot{V}_E$), and an increase in P_{CO_2} (41 to 49 mm Hg). Using the multiple inert gas technique, the investigators found that the distribution of ventilation to regions of ventilation-perfusion ($\dot{V}_A/\dot{Q}$) relationships above 100 (i.e., V_D/V_T) increased from $39 \pm 8\%$ during controlled ventilation to $46 \pm 7\%$ during spontaneous breathing. Perfusion of low $\dot{V}_A/\dot{Q}$ regions was higher during spontaneous breathing than during controlled ventilation (15 ± 11 vs. $6 \pm 8\%$). The investigators also performed isotope scans, which revealed a decrease in $\dot{V}_A/\dot{Q}$ ratios between the apex and the base of the lungs. This observation indicated that the low $\dot{V}_A/\dot{Q}$ units identified by the inert gas technique were located at the lung bases. The major determinant of the $\dot{V}_A/\dot{Q}$ abnormalities was the size of V_T : It correlated with perfusion in the low $\dot{V}_A/\dot{Q}$ range, the decrease of $\dot{V}_A/\dot{Q}$ ratios in the bases, and widening of the isotopic craniocaudal gradient. The maldistribution $\dot{V}_A/\dot{Q}$ of ratios during spontaneous breathing were improved by controlled ventilation but not by pressure support of 10 cm H₂O.

Torres et al.⁵¹ used the multiple inert gas technique to study eight patients with COPD who were apparently successfully weaned. Measurements were first obtained during assist-control ventilation (V_T 700 mL, rate 12 breaths/min, and $FiO_2 \leq 0.40$). On discontinuation of the ventilator, the patients developed rapid shallow breathing (relative to ventilator settings) and acute respiratory acidosis (increase in P_{CO_2} from 49 to 59 mm Hg, decrease in pH from 7.42 to 7.36). Spontaneous breathing caused an overall worsening of ventilation-perfusion inequality: The fraction of cardiac output distributed to low $\dot{V}_A/\dot{Q}$ (< 0.1) areas increased from 9.4% to 19.6%, and the dispersion of ventilation distribution increased. Despite the deterioration in P_{CO_2} relationships, the expected fall in partial pressure of *oxygen* (P_{O_2}) was prevented by an increase in cardiac output (4.7 to 6.7 L/min) and increase in mixed venous P_{O_2} (37 to 42 mm Hg). A largely similar pattern of gas exchange was reported by Ferrer et al.⁵² who studied seven patients with COPD who were not yet ready to tolerate complete discontinuation of mechanical ventilation.

Predicting Outcomes

General Principles of Medical Decision Analysis

When talking about weaning indices or weaning predictors, many clinicians overlook the fact that these measurements constitute a form of diagnostic testing.

Accordingly, evaluation of the reliability of weaning-predictor tests must comply with the canons developed for evaluating diagnostic tests.^{53–55} To understand the

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the reader has to cope with a huge number of specialized terms, many of which have meanings that contradict the meaning

of the same words in everyday speech. Specificity, for example, measures the fraction of patients who are correctly identified as not having the disease under consideration. In regular everyday speech, specificity means that something that “possesses properties that characterize a species” or that is “clearly defined and definite.” These ideas are the opposite of the absence of a property (disease). The terms are also duplicative. Specificity is also known as true-negative rate and as 1 minus false-positive rate. All of the specialized expressions are simply different ways of combining true-positive, true-negative, false-positive, and false-negative results. Life would be easier if research on weaning predictors were solely communicated using these four test characteristics, and terms such as specificity were avoided. The reader, however, cannot adopt this minimalist approach and must instead grapple with the many terms because of their widespread use.

A second problem arises when a reader desires a more fundamental understanding and consults a textbook on medical decision analysis. Diagnostic test results are called “positive” (or abnormal) when they diagnose a “disease” (or undesirable condition), and “negative” when they indicate a normal or desirable condition.⁴⁵ When applying these concepts to weaning, the reader must think of “disease” as “weaning success” and a positive or abnormal test result as one that predicts weaning success. This orientation may seem counterintuitive. But too large a body of research already exists to make a fundamental change.

A third problem for the reader of a textbook on decision analysis is that the discussion is usually in generic terms, covering all possible disease states and all forms of diagnostic testing. This generic or abstract approach makes the discussion long-winded. The reader has to relate abstract concepts to the particular clinical situation that the reader is interested in. To ease discussion, we discuss the fundamental principles of decision analysis in terms of a single diagnostic test. We select a test used for predicting weaning outcome, the f/V_T , because it has been subjected to the most investigation. And we focus on a single outcome: a patient’s ability to tolerate a 30-minute T-tube trial without distress that is followed by extubation. This outcome is weaning success. Thus, a positive diagnostic test result is a recording that predicts actual successful weaning outcome without the need for reintubation. The development of distress during a T-tube trial that leads to the reinstitution of mechanical ventilation is weaning failure; extubation followed by reintubation is not a necessary requirement to satisfy the definition of weaning failure. Thus, a negative diagnostic test result is a reading that predicts either the development of distress during a T-tube trial (leading to the reinstitution of mechanical ventilation) or the need for reintubation after extubation. The reader can apply the same framework to other diagnostic tests (such as maximum inspiratory pressure) and to other outcomes (such as the prediction of reintubation in a patient who is extubated after a successful weaning trial).

The data generated by most diagnostic tests, including f/V_T , are reported as continuous variables. It is common, however, to focus on a threshold and view the data in dichotomous terms: An f/V_T value equal to or less than 100 (breaths/min/L) is referred to as a positive result (indicating a high likelihood of weaning success). All tests are inaccurate to varying degrees: No diagnostic test has a one-to-one correspondence with a disease state. The degree of error is assessed by comparing a test’s performance against a reference test. For weaning predictors, the reference standard is a patient’s ability to tolerate a weaning trial that leads to extubation. This reference standard gives rise to significant difficulties in interpreting research findings (see below). Its lack of concreteness (necessary if a disorder is to fit an ontologic model of disease) contrasts with more rigorous reference standards available for other diagnostic tests (histologic findings for evaluating a cancer marker).

The characteristics of test results are most easily displayed by a fourfold table, often referred to as a “2 × 2” table (Fig. 58-15). For f/V_T , a true-positive result is a reading of equal to or less than 100 (the test predicts weaning success) in a patient who actually tolerates a T-tube trial that leads to extubation (weaning success). A true-negative result is an f/V_T reading greater than 100 (the test predicts weaning failure) in a patient who actually develops distress during a T-tube trial and requires the reinstitution of mechanical ventilation (weaning failure). A false-positive result is an f/V_T reading equal to or less than 100 (the test predicts weaning success) in a patient who actually fails a weaning trial. A false-negative result is an f/V_T reading equal to or less than 100 (the test predicts weaning failure) in a patient who actually tolerates a weaning trial and is extubated.

Figure 58-15

		Gold standard	
		Success	Fail
Test (f/V_T)	Positive (≤ 100)	TP	FP
	Negative (> 100)	FN	TN

TP = Test predicts weaning success and patient actually succeeds

TN = Test predicts weaning failure and patient actually fails

FP = Test predicts weaning success and patient actually fails

FN = Test predicts weaning failure and patient actually succeeds

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \text{TPR} = [1 - \text{FNR}]$$

$$\text{Specificity} = \frac{TN}{TN + FP} = \text{TNR} = [1 - \text{FPR}]$$

$$\text{PPV} = \frac{TP}{TP + FP}$$

$$\text{NPV} = \frac{TN}{TN + FN}$$

$$\text{FN rate} = 1 - \text{sensitivity}$$

$$\text{FP rate} = 1 - \text{specificity}$$

$$\text{Likelihood ratio for a positive test} = \text{TPR} / \text{FPR} = \text{sensitivity} / (1 - \text{specificity})$$

$$\text{Likelihood ratio for a negative test} = \text{FNR} / \text{TNR} = (1 - \text{sensitivity}) / \text{specificity}$$

$$\text{Prevalence} = TP + FN / (TP + TN + FP + FN)$$

$$\text{Diagnostic accuracy} = [TP + TN] / [TP + TN + FP + FN]$$

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A 2 × 2 tabular display of the characteristics of diagnostic tests. The vertical columns represent the results of the reference standard test. The horizontal rows represent the results of the index test. Readings of f/V_T equal to or less than 100 are classified as positive test results and readings greater than 100 are classified as negative test results. The relationship of these binary results to the outcome of a T-tube weaning trial forms a decision matrix that has four possible combinations.

Each cell in Figure 58-15 represents one of the four unique characteristics of a test (true-positive, true-negative, false-positive, and false-negative results). When developing a new weaning predictor, researchers need to undertake the initial evaluation in roughly equal numbers of weaning-success and weaning-failure patients if the predictor is to prove reliable under future testing.⁴⁶ Once the four test characteristics have been determined in a broad spectrum of weaning-success and weaning-failure patients, the test characteristics are considered constant and the test can be applied to the evaluation of any given patient. There is, however, one major assumption, which is all too often ignored. The formulae assume that the prevalence of weaning success and weaning failure in the new groups in which the test is being applied is the same as in the sample from which the four test characteristics were originally developed.^{53,56,57} If researchers are mindful of this fundamental requirement, much confusion can be avoided.

Sensitivity (also known by the more intuitive term, true-positive rate) answers the question: “In a weaning-success patient, what is the likelihood that the predictor test will be positive ($f/V_T \leq 100$)?” Thus, sensitivity measures the proportion of weaning-success patients in whom the predictive test is positive ($f/V_T \leq 100$). When clinicians are primarily interested in screening, they employ diagnostic tests that have a high sensitivity.⁵³ The purpose of a screening test is to pick up as many cases of a disease as possible out of the population being tested; screening can also be viewed as an exercise in ruling out disease, and a test for this purpose should have a low number of false-negative results (thus, a high sensitivity).⁵⁸ (Mnemonics have been proposed to remember these relationships. SnNout: If a test has a sufficiently high Sensitivity, a Negative result rules out the target disorder.) In the weaning context, this step is equivalent to using a predictor that will identify as many patients as possible who will actually pass a T-tube trial. Screening tests are typically performed in situations in which the pretest probability of the disease in question is low.⁵⁵ Because results are most often negative, the test should be easy to perform. As such, a test that takes 30 minutes or more to conduct, such as a T-tube trial, is not a satisfactory screening test.

Specificity (also known as true-negative rate) answers: “In a weaning-failure patient, what is the likelihood that the predictor test will be negative ($f/V_T \leq 100$)?” Thus, specificity measures the proportion of weaning-failure patients in whom the predictive test is negative ($f/V_T > 100$). When clinicians are primarily interested in confirming (or ruling in) the presence of a disease, they employ diagnostic tests that have a low number of false-positive results and, thus, a high specificity.^{53,58} (A mnemonic to remember this relationship is SpPin: If a test has a sufficiently high Specificity, a Positive result rules in the target disorder.) Because f/V_T has a relatively low specificity (0.64 in the original study,⁵⁹ it alone is not sufficient to confirm the presence of weaning failure. Instead, clinicians should undertake additional diagnostic testing, such as with a T-tube trial.

What are the characteristics of an ideal weaning-predictor test? Consider the following three diagnostic tests which differ in sensitivity and specificity. Test A has a sensitivity of 0.96 and specificity of 0.70; test B has a sensitivity of 0.70 and specificity of 0.96; and test C has a sensitivity of 0.83 and specificity of 0.83. Which test is best? The question cannot be answered until you are informed for what purpose the diagnostic test will be used. Many clinicians look on diagnostic testing from a monolithic perspective—a test is a test is a test. In reality, diagnostic testing is expected to fulfill two very different demands.^{53–55} The first is screening, that is, to pick up cases of a condition at the earliest possible time. This demand requires a test with high sensitivity.^{53–55} The second is confirmation of a condition for which there is already a strong suspicion. This demand requires a test with high specificity.^{53–55} “A single test can seldom be excellent for the goals of both discovery [screening] and confirmation,” Alvan Feinstein emphasized. “With rare exceptions, the same procedure cannot be sensitive enough to find all cases of the disease while simultaneously being specific enough to avoid false-positive identifications.”⁵⁴ For example, chest radiography is reasonably sensitive (but nonspecific) in detecting lung cancer. Almost all patients with lung cancer will have an abnormal chest radiograph. (A normal chest radiograph is good in ruling out lung cancer.) But not everyone with an abnormal chest radiograph has lung cancer (high false-positive rate). Conversely, a positive histology result on bronchoscopic biopsy is a reasonably specific diagnostic method (false-positive results are uncommon). But it is insensitive (often failing to capture cancers at inaccessible sites). For these reasons, clinicians commonly use diagnostic tests in combination.

A weaning-predictor test is used to spot the earliest point in time that a patient might tolerate a weaning trial. It serves solely as a screening test. On its own a positive predictor-test result is not used as justification for extubation. Before that step, a confirmatory test, such as with a T-tube trial, is undertaken. The ideal time to undertake a screening test is when the pretest probability of weaning success is 20% or less.⁵³ In contrast, weaning trials are commonly performed when the pretest probability of success is 75% or more.

The development of a reliable screening test hinges on avoiding false-negative results (a test predicting failure, but the patient actually succeeds).^{53,54} Simultaneously the test needs to pick up every possible true-positive result—the mindset is to miss no patient who can breathe without the ventilator. To capture the maximum meaningful number of true-positive results, the threshold for defining a positive screening test may be set deliberately high.^{53,54} This necessarily increases the number of false-positive results, producing a proportional decrease in specificity. Sensitivity captures exactly the components that define the reliability of a screening test because it contains only true-positive and false-negative rate. Likewise, specificity captures exactly the constituents of a reliable confirmatory test: avoidance of false-positive results (a test predicting success, but the patient actually fails) and maximizing true-negative rate.^{53,54}

Sensitivity and specificity are often regarded as constant properties of a diagnostic test. The characteristics, however, for any diagnostic test are derived from data collected in a selected group of patients. Consequently, sensitivity and specificity of diagnostic tests vary across different parts of the clinical spectrum of the disease they are attempting to identify or exclude.^{57,60} Both sensitivity and specificity also perform differently in populations with different distributions of disease severity. For example, the sensitivity and specificity of electrocardiographic stress testing differs between patients with triple-vessel coronary artery disease and patients with mild single-vessel disease.⁶¹

False-positive rate answers: “What is the likelihood that a weaning-failure patient will have a positive test result ($f/V_T \leq 100$)?” Thus, the false-positive rate measures the proportion of positive test results ($f/V_T \leq 100$) in all weaning-failure patients. In a weaning-failure patient, test results are only true-negatives or false-positives. Thus, false-positive rate is the complement of true-negative rate (false-positive rate = 1 minus true-negative rate).

False-negative rate answers: “What is the likelihood that a weaning-success patient will have a negative test result ($f/V_T > 100$)?” Thus, false-negative rate measures the proportion of negative test results ($f/V_T > 100$) in all weaning-success patients. In a weaning-success patient, test results are only true-positives or false-negatives. Thus, false-negative rate is the complement of true-positive rate (false-negative rate = 1 minus true-positive rate).

Sensitivity and specificity are calculated in patients in whom a diagnosis is already known. Clinicians, however, are faced with positive and negative results in patients whose diagnosis is not yet established. When contemplating a diagnosis, a clinician is not oriented down the vertical columns of the 2×2 table, but across the horizontal rows (see Fig. 58-15). Thus, clinicians think more in terms of positive-predictive and negative-predictive values than in terms of sensitivity and specificity.⁵² Positive-predictive value answers: “What is the likelihood of weaning success in a patient who has an f/V_T equal to or less than 100?” Thus, positive-predictive value measures the fraction of patients with positive test results ($f/V_T \leq 100$) who are successfully weaned. Negative-predictive value answers: “What is the likelihood of weaning failure in a patient who has an f/V_T greater than 100?” Thus, negative-predictive value measures the fraction of patients with negative test results ($f/V_T > 100$) who fail a weaning trial. The positive-and negative-predictive values of a diagnostic test are particularly susceptible to variation in the prevalence of the condition under consideration (see below).

Likelihood ratio combines sensitivity and specificity into a single number. The likelihood ratio for a positive test relates the likelihood that a weaning-success patient will have a positive test result ($f/V_T \leq 100$) to the likelihood that a weaning-failure patient will have a positive test result. In other words, it is the probability of a

[Loading \[Contrib\]/a11y/accessibility-menu.js](#) hg-success patients divided by probability of the same test result in weaning-failure patients. It is calculated as true-positive

rate/false-positive rate (or: sensitivity/[1 minus specificity]). The likelihood ratio for a negative test relates the likelihood that a weaning-success patient will have a negative test result ($f/V_T > 100$) to the likelihood that a weaning-failure patient will have a negative test result. In other words, it is the probability of a negative test result ($f/V_T > 100$) in weaning-success patients divided by probability of the same test result in weaning-failure patients. It is calculated as false-negative rate/true-negative rate (or: [1 minus sensitivity]/specificity).

Evidence-Based Medicine Task Force on Weaning

An Evidence-Based Medicine (EBM) Task Force of the American College of Chest Physicians^{62,63} evaluated the usefulness of weaning-predictor tests. Employing a meta-analysis, the Task Force calculated pooled likelihood ratios for several predictors. They concluded that all weaning-predictor tests have low power, and recommended that clinicians should start the weaning process with a spontaneous breathing trial (a confirmatory test), and use the initial few minutes of the trial as a screening test.⁶² This reverses the logic axiomatic to diagnostic testing. It is analogous to saying that when you suspect diabetes, start with a glucose-tolerance test and then, as the test gets underway, ask the patient for a urine sample in order to do a dipstick.⁸

The Task Force's meta-analysis contained more than fifteen methodologic errors, including several examples of the three main types of systematic error—selection bias (test-referral bias, spectrum bias), misclassification bias (categorizing reintubation as weaning failure), and confounding (considering pressure support to represent a form of unassisted breathing)—as well as several errors of interpretation.^{64,65} Any one of the systematic errors would be sufficient to scupper the conclusions of the meta-analysis; yet when these flaws were subsequently identified and reported, the EBM Task Force did not contend a single one of them.^{65–68} Instead, the Task Force viewed the errors as side issues that did not detract from their recommendations.^{65,67} To ignore test-referral bias in the evaluation of a diagnostic test is analogous to a physiologist who claims that a partial pressure of arterial oxygen (Pa_{O_2}) of 80 mm Hg is always better than a Pa_{O_2} of 60 mm Hg, and the fact that the measurements were made at inspired oxygen concentrations of 50% and 21%, respectively, is an academic distraction best ignored.

Physicians recognize that writing an order for a T-tube trial constitutes a clinical decision. A corollary might be proffered that not writing an order means that no decision has been made. That interpretation would be wrong. Whenever a physician looks at a ventilated patient and does not order a T-tube trial (or weaning-predictor test), the physician is deciding that the patient is not weanable at this time (stage 2 of weaning; see [Seven Stages of Weaning](#)). This subtle distinction is missed by the recommendation of the EBM Task Force to dispense with screening and commence the weaning process with a confirmatory test (a spontaneous breathing trial). They are thus encouraging physicians not to test for weanability until a patient looks ready for a T-tube trial. Their recommendation betrays a failure to comprehend the very purpose of weaning predictors. The sole purpose of a screening test (weaning predictors) is to alert a physician to consider doing a T-tube trial sooner than is the physician's custom—for the trial to occur earlier than would otherwise happen.⁸ A positive result on a weaning-predictor test acts as a “physician alert,” and aids in the cognitive process known as diagnostic triggering.⁶⁹

The Bayes' Theorem

Before clinicians perform a diagnostic test, they formulate a pretest (or prior) probability of disease. In the context of weaning, clinicians form an initial gestalt of a patient's likelihood of passing a T-tube trial based on their previous experience of patients with similar clinical characteristics. After measuring a weaning predictor test, and knowing its test characteristics (sensitivity and specificity), the clinician formulates a new probability statement (of whether the patient is likely to pass the T-tube trial). The new statement is the posttest (or posterior) probability. The Bayes' theorem is an equation that describes the relationship between pretest probability and posttest probability. It is used to estimate how much the uncertainty of weaning outcome changes from before measurement of a predictor test (the pretest probability) to after obtaining the new information (the conditional probability). In particular, the Bayes' theorem is used to transform the information contained in sensitivity and specificity into a format that can be employed in diagnostic testing (calculation of posttest probability, in the format of positive-predictive and negative-predictive value).⁵⁴

Conditional probability refers to the probability that a particular event will occur given that some other condition has been met.⁵⁷ In the weaning context, it addresses, “What is the probability of a positive f/V_T result (< 100) conditional upon the patient's passing (or failing) a weaning trial?” A clinician can calculate the posttest probability of weaning success if the clinician has three pieces of information: (a) the pretest probability of weaning success (typically, the prevalence); (b) the probability of a positive f/V_T result (≤ 100) conditional upon the patient's passing a weaning trial (true-positive rate, or sensitivity); and (c) the probability of a positive f/V_T result (≤ 100) conditional upon the patient's failing a weaning trial (false-positive rate, or 1 minus specificity). A useful weaning predictor test has a high conditional probability (a high likelihood ratio), and thus markedly alters the posttest probability of weaning success. The Bayes' theorem is employed to convert the vertical indices in the 2×2 table (sensitivity and specificity) into the desired horizontal indices of disease prediction, which indicate posttest probability.⁵⁴ The posttest probability of weaning success after obtaining a positive test result ($f/V_T \leq 100$) is the positive-predictive value. The posttest probability of weaning failure after obtaining a negative test result ($f/V_T > 100$) is the negative-predictive value.

The Bayes' theorem operates on the assumption that the sensitivity and specificity of a test are constant irrespective of the pretest probability of disease. But “[i]n the few instances in which this assumption has been checked,” notes Feinstein,⁵⁴ “it was found to be erroneous.” In a study of exercise electrocardiography for the diagnosis of coronary artery disease, Hlatky et al⁷⁰ found sensitivity was 0.80 in patients with typical angina and 0.53 in patients with atypical chest pain; sensitivity was 0.85 in patients with triple-vessel disease and 0.48 in patients with single-vessel disease. Specificity was 0.85 when left-ventricular ejection fraction was equal to or greater than 50% and 0.73 when ejection fraction was 30% to 49%. In a study of patients being evaluated for urinary tract infection, Lachs et al⁶⁰ found that sensitivity of the dipstick test was 0.92 in patients with a high (> 0.50) pretest probability of infection (patients with dysuria, urgency, and hematuria), and sensitivity was 0.42 in patients with a low (≤ 0.50) pretest probability of infection. Conversely, specificity was 0.42 in patients with a high pretest probability for infection, and 0.78 in patients

with a low pretest probability. Indeed, for any condition, history and physical examination is likely to be more abnormal when a disease is extensive, and thus, the clinician will assign a high pretest probability.⁷¹ In many studies of weaning predictors, researchers have assumed that reported values for sensitivity and specificity apply in all circumstances. This assumption can give rise to errors when Bayes' theorem is applied.^{54,57,72}

Pretest Probability of Successful Outcome

In evaluating the performance of weaning predictors, probably no aspect is ignored more often—and the source of greater confusion—than recognizing the fundamental importance of pretest probability (the anticipated prevalence of a disorder). The pretest probability is the fraction of weaning- success patients out of an entire weaning population (both success and failure patients). This pretest probability markedly affects the interpretation of new test information. In their book on medical decision analysis, Sox et al⁵⁵ state, “Perhaps the most important idea in this book is the following: The interpretation of a test result depends on the pretest probability of disease.” That is, a clinician's interpretation of new diagnostic information depends on what the clinician believed before doing the test. As a clinician's estimate of pretest probability of weaning success increases, so also does the posttest probability of weaning success.

The pretest probability has an enormous influence over the ability of the results of any diagnostic test to alter the posttest probability of disease. When the pretest probability of weaning success is high, a positive test result ($f/V_T \leq 100$) has little effect. (The probability cannot increase much after a positive test because of a ceiling effect.) A negative test result ($f/V_T > 100$) will drop the probability considerably, but only into the large middle range of probability that is diagnostically inconclusive.⁷² Conversely, when the pretest probability of weaning success is low, a negative test result ($f/V_T > 100$) has little effect. (The probability does not have anywhere to drop after a negative test. A positive test result [$f/V_T \leq 100$] has a large effect, but it only brings the clinician into the large middle range of probability.) Thus, when the pretest probability is already close to diagnostic certainty (high or low), a test that does not confirm the diagnostic suspicion can produce substantial changes in posttest probability. The above description is not unique to weaning predictors, but applies to all diagnostic tests in medicine. The only exception to this relationship between pretest probability and postprobability is an imaginary test with 100% sensitivity and 100% specificity.

A clinician gains the maximum increase in posttest probability for a positive test when pretest probability is 40%, and gains the maximum increase for a negative test when the pretest probability is 60%.⁵³ That is, physicians have most to gain from diagnostic testing when the pretest probability of a condition is close to 50%.⁵³ For a weaning predictor test to influence decision making, it needs to be measured early in a patient's course (expediting the transition between weaning stage 2 and stage 3). If a clinician observes that half or more of the clinician's measurements of weaning predictors generate positive results (indicating that a patient is ready for a weaning trial), that clinician is not measuring the predictor test early enough in his or her patient's course. A clinician who is measuring predictor tests early in a patient's course will obtain negative test results (indicating that a patient is not ready for a weaning trial) at least as often as positive results. Measuring weaning predictors in patients who have already passed a weaning trial is a futile undertaking. Unfortunately, most studies of weaning predictors have been conducted in patients who had a pretest probability of weaning success of 75% or higher (Table 58-1). Such a high pretest probability markedly decreases the apparent reliability of a weaning-predictor test (Fig. 58-16).

Table 58-1: Accuracy of F/V_T in Predicting Weaning or Extubation Outcome

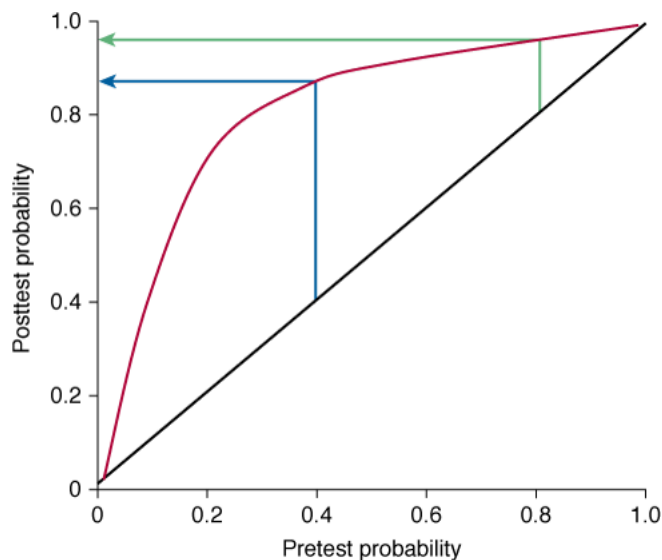
Author	No. Patients	Outcome End Point	Threshold	Sensitivity	Specificity	PPV	NPV	Positive Likelihood Ratio	Negative Likelihood Ratio	Pretest Prob of Success	Data Available to Primary	Location
1. Yang & Tobin ⁵⁹	64	WF or EF	≤105	0.97	0.64	0.78	0.95	2.69	0.05	0.56	No	MICU
2. Gandia & Blanco ¹³	40	WF or EF	<96	0.89	0.83	0.93	0.77	5.24	0.13	0.7	No	NS
3. Sassoon & Mahutte ¹⁴	45	WF or EF	≤100	0.97	0.40	0.85	0.80	1.62	0.08	0.78	No	NS
4. Yang ¹²⁸	31	WF or EF	≤100	0.94	0.73	0.79	0.92	3.48	0.08	0.1352	No	MICU
5. Mohsenifar ¹¹⁷	29	WF or EF	≤105 (PS 7 to 8)	1.00	0.27	0.69	1.00	1.37	0.00	0.62	Not clear	RICU
6. Lee ¹³³	52	EF only	≤105 (PS ?)	0.72	0.11	0.79	0.08	0.81	2.55	0.83	Yes	MICU
7. Capdevila ¹⁵	67	EF only	60	0.73	0.75	0.92	0.36	2.92	0.36	0.82	Yes	Multidisc ICU
8. Epstein ¹	94	EF only	<100	0.92	0.22	0.83	0.40	1.18	0.36	0.81	Yes	MICU
9a. Chatila ¹⁰⁸	100	WF or EF	≤100	0.89	0.41	0.72	0.68	1.51	0.27	0.63	Yes	MICU, CCU
9b. Chatila ¹⁰⁸	100	WF or EF	≤100	0.98	0.59	0.83	0.94	2.39	0.03	0.63	Yes	MICU, CCU
10. Dojat ¹²⁹	38	WF or EF	<100	0.94	0.81	0.80	0.94	4.95	0.07	0.45	No	MICU, SICU
11. Leitch ⁷⁵	163	EF only	≤100 (PS 7)	0.96	0.00	0.98	0.00	0.96	>2.55	0.982	Yes	M-SICU
12. Mergoni ¹⁸¹	75	WF or EF	<105	0.65	0.58	0.60	0.63	1.54	0.61	0.49	Yes	M-SICU
13. Bouachour ¹²⁰	15	WF only	≤105	1.00	0.40	0.77	1.00	1.67	0.00	0.67	Not clear	NS
14. Baumeister ¹⁸²	47 Ped	EF only	≤11 breaths/min/mL/kg	0.79	0.78	0.94	0.47	3.59	0.27	0.81	No	PICU
15. Голоторский ¹⁸³	127	Not Defined	Not stated	0.84	0.83	0.80	0.86	—	—	—	Not clear	SICU
16a. Jacob ¹⁰⁹	183	WF or EF	100	0.97	0.33	0.94	0.50	1.45	0.09	0.92	Yes	SICU
16b. Jacob ¹⁰⁹	183	WF or EF	100	0.96	0.31	0.94	0.40	1.39	0.13	0.92	Yes	SICU
17a. Kreiger ¹³⁰	49	WF	≤105	0.74	0.73	0.90	0.44	2.74	0.36	0.78	Yes	MICU
17b. Kreiger ¹³⁰	49	WF	≤130 at 3 hours	0.93	0.89	0.97	0.80	8.45	0.08	0.78	Yes	MICU
18a. Rivera & Weissman ¹³²	40	WF only	65 (PS 5)	0.90	0.80	0.90	0.70	4.50	0.13	0.7	Yes	SICU

Author	No. Patients	Outcome End Point	Threshold	Sensitivity	Specificity	PPV	NPV	Positive Likelihood Ratio	Negative Likelihood Ratio	Pretest Prob of Success	Data Available to Primary	Location
18b. Rivera & Weissman ¹³²	40	WF only	65 (PS + IMV)	1.00	0.82	0.84	1.00	5.56	0.00	0.7	Yes	SICU
19. Farias ¹⁸⁴	84 Ped	WF	≤11 breaths/min/mL/kg	0.48	0.86	0.53	0.83	3.43	0.60	0.75	No	PICU
20. Vallverdu ⁷⁷	217	WF or EF	≤100	0.90	0.36	0.66	0.73	1.41	0.28	0.58	Yes	M-SICU
21. Thiagarajan ¹⁸⁵	227 Ped	EF only	≤8 breaths/min/mL/kg	0.74	0.74	0.97	0.22	2.85	0.35	0.89	No	PICU
22. Zeggwagh ¹²⁷	101	EF only	<88	0.77	0.79	0.68	0.86	3.67	0.29	0.63	No	MICU
23. Maldonado ¹¹⁸	27	WF only	≤105	0.93	0.75	0.83	0.89	3.72	0.09	0.56	Yes	RICU
24. Uusaro ⁷⁸	68	EF only	<100 (PS 5)	0.96	0.18	0.78	0.60	1.17	0.22	0.75	Yes	M-SICU
25. Khamiees ⁷⁹	100	EF only	≤105	0.84	0.17	0.82	0.19	1.01	0.94	0.82	Yes	MICU, CCU
26. Smina ⁸¹	115	EF only	<100	0.90	0.42	0.92	0.36	1.55	0.24	0.89	Yes	MICU, CCU
27. Conti ¹³¹	51	WF or EF	≤100	0.81	0.14	0.71	0.22	0.94	1.36	0.73	No	MICU, SICU
28. Fernandez ⁸²	57	WF; EF	<50	0.35	0.56	0.81	0.14	0.80	1.16	0.84	Yes	MICU, SICU
29. Jiang ¹⁸⁶	55	EF only	≤105	0.81	0.57	NR	NR	1.88	0.33	0.58	Yes	MICU

Abbreviations: CCU, cardiac care unit; EF, extubation failure; f/V_T , frequency-to-tidal-volume ration; IMV, intermittent mandatory ventilation; MICU, medical intensive care unit (ICU); M-SICU, medical-surgical ICU; Multidisc ICU, multidisciplinary ICU; NPV, negative-predictive value; NS, nonstated location; PICU, pediatric ICU; PPV, positive-predictive value; PS, pressure support; RICU, respiratory ICU; SICU, surgical ICU; WF, weaning failure.

The listed studies are those that reported data on the accuracy of f/V_T as a predictor of weaning outcome. Four groups of investigators, Chatila et al, Jacob et al, Krieger et al, and Rivera and Weissman (studies 9, 16, 17, and 18 in the table), report data under two different conditions in their articles; both sets of data are presented. Pretest probability of success in a study is the fraction of patients with a successful outcome out of the total population (both success and failure patients) included in the study.

Figure 58-16



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Relationship between pretest probability and posttest probability for a good weaning-predictor test, sensitivity of 0.9 and specificity of 0.9, is characterized by the red curve. If pretest probability of weaning success is 0.40, the Bayes' theorem dictates that a positive result on the weaning-predictor test will yield a posttest probability of 0.86. If pretest probability is 0.80, posttest probability will be 0.97. The increase between pretest and posttest probability in the second instance (21%, 0.17/0.80) is only a fraction of that in the first instance (115%, 0.46/0.40) despite the sensitivity and specificity being identical. Thus, a high pretest probability markedly decreases the apparent reliability of a weaning-predictor test.

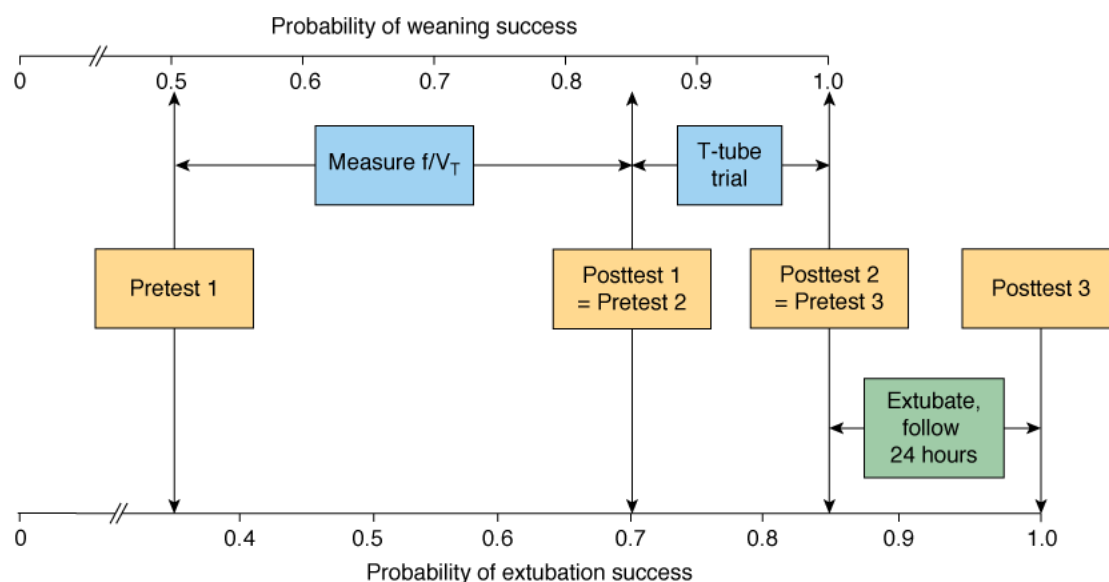
Sequential Diagnostic Testing

The implications of pretest probability are greater for weaning than for many clinical situations because weaning involves a sequence of three diagnostic tests: measurement of predictors, followed by a weaning trial, followed by an extubation trial. The Bayes' theorem assumes that the conditional probability of a test result is independent of pretest (prior) information—the assumption of conditional independence. The idea of independence is one of the most important concepts in probability theory.⁷³

Two events are judged independent if the knowledge that one event has occurred tells you nothing about whether the second event will occur. The events “the patient has a skull fracture” and “the patient has cholecystitis” are independent. That is, conditioning on one event does not change the probability of the other event. In contrast, the events “the patient has cholecystitis” and “the patient has abdominal pain” are not independent. The conditioning probability of abdominal pain, given cholecystitis, is much higher than the nonconditional probability of abdominal pain (the proportion of all patients in the world with abdominal pain). When both an exercise electrocardiogram and a radionuclide myocardial scan are performed before coronary arteriography, the false-positive rate of each test depends on the results of the other test.⁵⁵ That is, the assumption of conditional independence of the two tests is invalid. Likewise, the conditioning probability that a T-tube trial will be successful and lead to extubation is much higher given a positive f/V_T test result (≤ 100). The two events, a positive f/V_T test result (≤ 100) and a successful T-tube trial, are not conditionally independent.

Because the assumption of conditional independence has not been verified for most combinations of two or more tests, experts recommend that the posttest probability of the first test be used as the pretest probability of the second test.⁵⁵ Figure 58-17 shows typical changes in pretest probability of successful extubation and posttest probability of successful extubation with the sequential performance of f/V_T measurement, a T-tube trial, and a trial of extubation.

Figure 58-17



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Interpreting the sequence of diagnostic testing in a patient who is weaned from mechanical ventilation and then extubated, with successful outcome defined as the ability to breathe spontaneously without ventilator assistance for 24 hours after extubation. Pretest 1 is the pretest probability of extubation success before measurement of f/V_T . The f/V_T reading constitutes the posttest results of the first diagnostic test, and also pretest probability of extubation success for the second diagnostic test, a T-tube trial. The outcome of the T-tube trial constitutes the posttest result of the second diagnostic test, and also the pretest probability of extubation success for the third diagnostic test, a trial of extubation. The outcome of the trial of extubation constitutes the posttest result of the third diagnostic test. In this hypothetical example, pretest probability of weaning success for the first diagnostic test is set arbitrarily at 0.50; the extent of the change for each subsequent step is based on average changes reported in published studies.

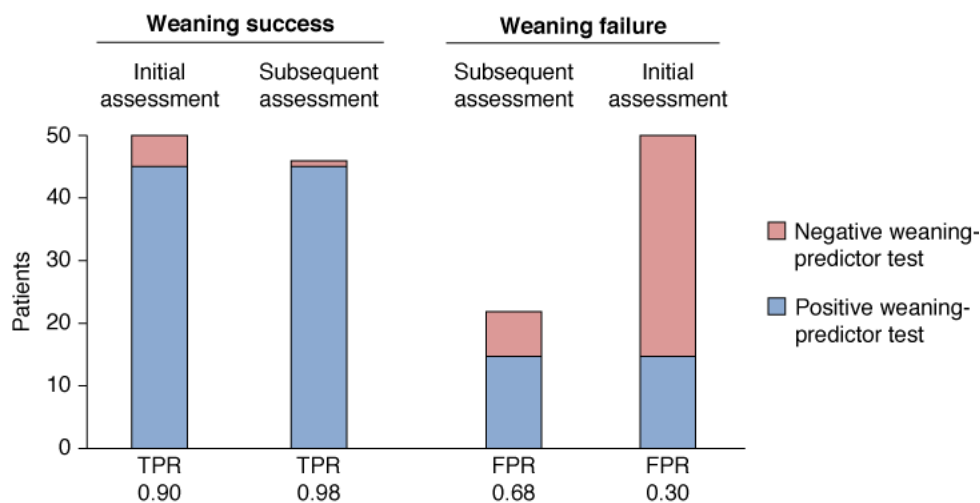
Spectrum Bias and Test-Referral Bias

The term *spectrum* denotes the range of disease presentation and severity found in patients used to challenge the sensitivity and specificity of a diagnostic test.⁵⁸ Spectrum bias occurs when a diagnostic test performs differently in different groups of patients, as occurs when the new study population contains more sick (or less sick) patients than the population in which the diagnostic test was originally developed.⁶⁰ Another form of bias, test-referral bias, occurs when the results of a test under evaluation (f/V_T) are used to select patients for the reference-standard test (T-tube trial).

To illustrate spectrum bias, we discuss the introduction of a hypothetical new weaning-predictor test. To make our account easy to understand, we use f/V_T as the previously unevaluated weaning-predictor test. To determine if the new test is reliable in predicting extubation outcome, the investigators study 100 patients. The average value of f/V_T in these patients is 111. Because the investigators do not know whether f/V_T is reliable or not, all patients proceed to a T-tube trial; and if they pass that, the patients continue to a trial of extubation. The investigators report that an f/V_T value of less than 100 is reliable in predicting a successful outcome. Subsequently, another group of investigators decide to reevaluate the usefulness of the weaning-predictor test. They also study 100 patients, and the average value of f/V_T is again 111. Of the study population, seventy patients have positive f/V_T results (values <100), and thirty have negative f/V_T results (values >100). The seventy patients with positive f/V_T results proceed to a T-tube trial. Because the initial study revealed that negative f/V_T results were reliable in predicting weaning failure, the clinicians refuse to undertake a T-tube trial in the thirty patients with negative f/V_T results. These are thus excluded in the second study. The exclusion of patients with negative results skews the study population toward less severely ill patients, an effect termed *spectrum bias*. The patients in the first study, who had an average f/V_T of 111, were more severely ill than the patients in the second study who had an average f/V_T of ninety. The selection of less severely ill patients in the second study produces an increase in the pretest probability of success; it also alters the sensitivity and specificity of the test.^{54,55}

A second form of bias, test-referral bias, occurs when the results of a test under evaluation are used to select patients for the reference-standard test.^{54,55} Consider a weaning-predictor test where its reliability is evaluated in terms of its ability to predict the successful toleration of extubation. If patients are required to pass a weaning trial before extubation, this study-entry requirement necessarily excludes all patients who fail. This step has three effects on the study population: first, fewer patients with negative results (of the weaning-predictor test) are included; second, relatively more patients with positive results are included; third, pretest probability of success is increased.^{54,55} The first consequence produces a decrease in the specificity of the weaning-predictor test in this population compared with the population in which the test was originally developed. The second consequence increases the sensitivity of the test. Figure 58-18 provides a hypothetical example of how test-referral bias leads to changes in pretest probability, sensitivity, and specificity. The risk for the occurrence of test-referral bias is enormous when three diagnostic tests are undertaken in a sequential manner, as is the case with weaning and extubation.

Figure 58-18



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Test-referral bias. An initial study population (*outer two columns*) for the development of a new weaning predictor consists of fifty weaning-success patients and fifty weaning-failure patients. Of the fifty success patients, forty-five have a positive-test result (true-positive rate [TPR] or sensitivity of 0.90) (*first column*). Of the fifty failure patients, thirty-five have a negative-test result (false-positive rate [FPR] of 0.30 or specificity of 0.70) (*fourth column*). In a subsequent study (*second and third columns*), clinicians decide not to refer 80% of patients with negative test results (that predict weaning failure) for a T-tube trial. The 80% decrease in negative-test results in the success arm causes a decrease in false-negative results (true-positive results do not change); thus, the true-positive rate (sensitivity) increases to 0.98 (second column). The 80% decrease in negative-test results (twenty-eight fewer patients with negative-test results) in the failure arm causes a decrease in true-negative results (false-positive results do not change); thus, the false-positive rate increases to 0.68 (specificity decreases to 0.32) (*third column*).

Spectrum bias and test-referral bias accounts for much confusion in the interpretation of studies of weaning predictor tests. A skilled clinician who measures f/V_T will obtain on average at least as many (if not more) negative test results as positive test results (if the clinician is trying to move the transition between stage 2 and stage 3 of weaning [see Fig. 58-1] to an earlier time in a patient's course). Thus, pretest probability of weaning success is 50% or lower. Clinicians commonly use the f/V_T result to decide which patients will progress that day to a weaning trial. They subsequently use the results of the weaning trial to decide which patients will undergo a trial of extubation. If tolerance of extubation is used as the reference gold standard for evaluating the accuracy of f/V_T , the requirement to pass a weaning trial before extubation (as a criterion for entry into a study) will lead to fewer patients with negative test results ($f/V_T > 100$) and relatively more patients with positive test results ($f/V_T \leq 100$) in the study population (test-referral bias). Thus, the patients in the new study population will have a lower f/V_T on average than the population in which the weaning predictor was originally developed (spectrum bias). Test-referral bias leads to an increase in both true-positive rate and false-positive rate.

Several studies of weaning predictors suffer from the flaws of spectrum bias and test-referral bias.^{15,74-82} The misinformation that has resulted from failure to take into account test-referral and other forms of bias brings to mind the comment of the late Alvan Feinstein, the founding father of clinical epidemiology:⁵⁶ "The most important issues in biostatistics are not expressed with statistical procedures. The issues are inherently scientific, rather than purely statistical, and relate to the architectural design of research, not to the numbers with which the data are cited and interpreted."

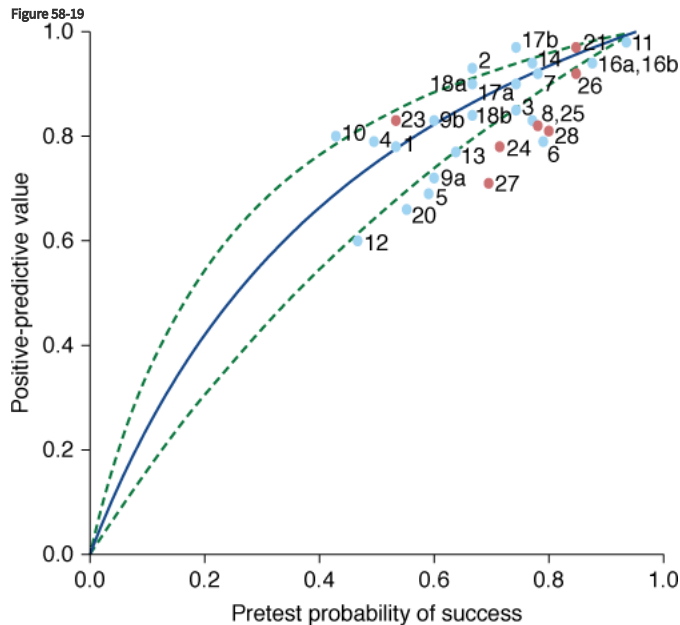
The extreme case of test-referral bias is the situation in which investigators require a positive f/V_T result (value ≤ 100) before subjecting a patient to a T-tube trial. By design, no patient with a negative f/V_T result (value > 100) would undergo a T-tube trial. In that instance, all failed T-tube trials would be in patients who had positive f/V_T results. Thus, the false-positive rate of f/V_T equal to or less than 100 would tend toward 1.0, and specificity toward 0. Investigators who have evaluated the accuracy of f/V_T have not required an f/V_T equal to or less than 100 per se before referring patients for a weaning trial (the reference gold standard test). Vallverdu et al⁷⁷ and Fernandez et al,⁸² however, did require patients to have a frequency equal to or less than 35 breaths/min and V_T equal to or greater than 5 mL/kg; for an average 70-kg patient, these criteria result in f/V_T less than 100. Khamiees et al⁷⁹ required f/V_T less than 125 for entry. Not surprisingly, these three groups reported very low specificities for f/V_T (see Table 58-1). (Specificity in the study of Fernandez et al⁸² is inflated through use of an f/V_T threshold of 50.)

Bayesian Analysis of Reported Performance of f/V_T

Motivated by the meta-analysis undertaken by the EBM Task Force^{62,63} and cognizant of the group's failure to employ Bayesian principles in their pooled calculation, Tobin and Jubran³² examined whether variation in reported reliability of f/V_T in predicting weaning success might be explained by spectrum and test-referral bias, as reflected by variation in pretest probability of success. They took data from the twenty studies included in the EBM Task Force's meta-analysis and nine additional studies published after the meta-analysis. Table 58-1 lists the evaluated studies.

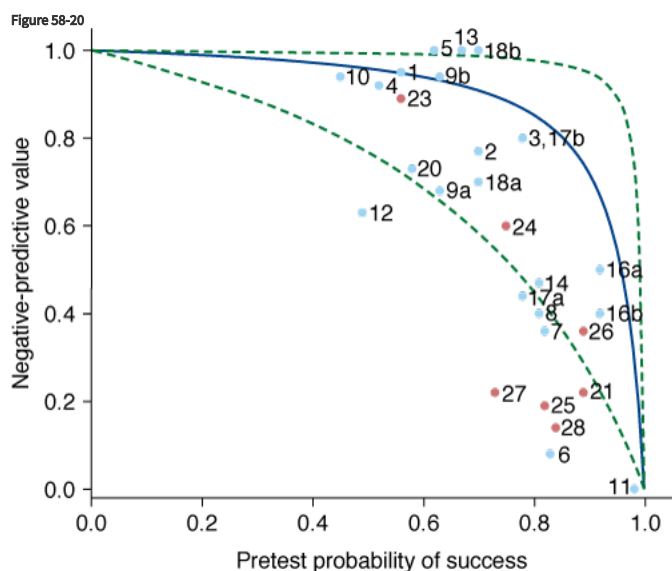
Pretest probability of successful outcome varied from 0.45 to 0.98; reported sensitivity ranged from 0.35 to 1.00, with a mean of 0.87 ± 0.14 ; and specificity ranged from 0.00 to 0.89 (Table 58-1). The data were entered into a Bayesian model with pretest probability (prevalence of success) as the operating point. A weighted Pearson correlation analysis, which adjusts for the number of patients contained in a study, was used to compare the relationship between pretest probability and the reported values of positive- and negative-predictive value.

Figures 58-19 and 58-20 show that most of the positive-predictive and negative-predictive values in the studies fall close to or above the lower 95% confidence interval (CI) of the values predicted by the Bayes' theorem for pretest probability, using the sensitivity and specificity originally reported by Yang and Tobin.⁵⁹ Figure 58-21 shows that reported positive-predictive values were significantly correlated with the values of positive-predictive values predicted on the basis of the sensitivity and specificity originally reported by Yang and Tobin: $r = 0.86$, $p < 0.0001$. The correlation between reported and predicted negative-predictive values was $r = 0.82$, $p < 0.0001$. These correlations provide de facto confirmation of the sensitivity and specificity reported in the original 1991 study.



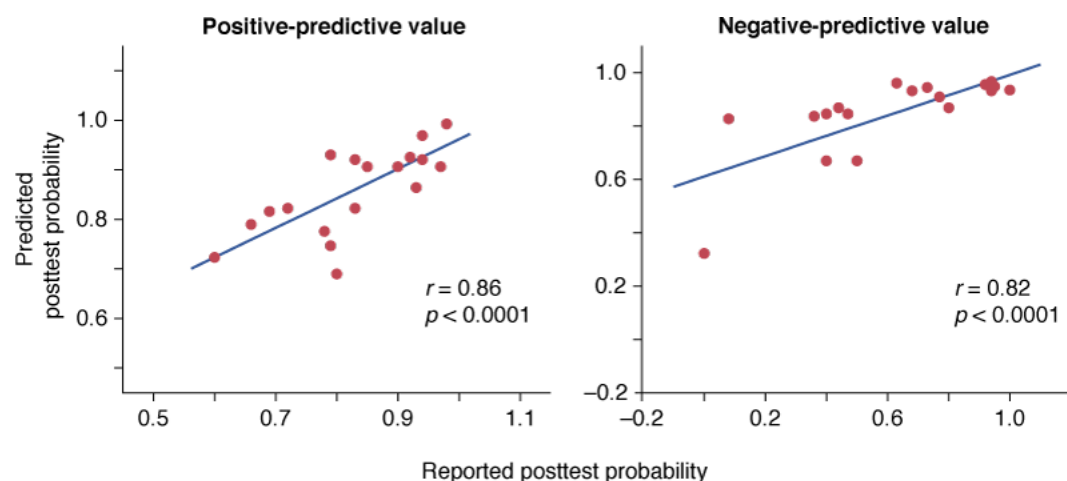
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Positive-predictive value (posttest probability of successful outcome) for f/V_T plotted against pretest probability of successful outcome. Studies included in EBM Task Force meta-analysis are indicated by *blue symbols*; additional studies are indicated by *red symbols*. The curve is based on the sensitivity and specificity originally reported by Yang and Tobin⁵⁹ and the Bayes' formula for 0.01-unit increments in pretest probability between 0.00 and 1.00.⁵⁵ The lines represent the upper and lower 95% CIs for the predicted relationship of the positive-predictive values against pretest probability. The observed positive-predictive value in a study is plotted against the pretest probability of weaning success. The numbering of a study corresponds to its numbering in Table 58-1, and not to the numbering of the references. (Used, with permission, from Tobin and Jubran.³²)



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Negative-predictive value (posttest probability of unsuccessful outcome) for f/V_T . Studies included in EBM Task Force metaanalysis are indicated by *blue symbols*; additional studies are indicated by *red symbols*. The curve, its 95% CIs, and placement of a study on the plot are described in the legend to Figure 58-19. The observed negative-predictive value in a study is plotted against the pretest probability of weaning success (prevalence of successful outcome). The numbering of a study corresponds to its numbering in Table 58-1, and not to the numbering of the references. (Used, with permission, from Tobin and Jubran.³²)



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The relationship between the reported posttest probabilities for f/V_T (among the studies included in the EBM Task Force's meta-analysis) and the values predicted by observed pretest probability together with the sensitivity and specificity originally reported by Yang and Tobin.⁵⁹ The weighted Pearson correlation for positive-predictive value was $r = 0.86$, $p < 0.0001$ (left panel), and for negative-predictive value the correlation was $r = 0.82$, $p < 0.0001$ (right panel). (Used, with permission, from Tobin and Jubran.³²)

Also evident in Table 58-1 is the high rate of successful outcome, 75% or higher in more than half the studies. The high success rate reflects occurrence of spectrum bias, and the lower value of f/V_T in subsequent studies than in the original report (77.4 vs. 89.1) provides further evidence of spectrum bias. The high occurrence of test-referral bias is suggested by the high success rate together with the lower specificity of f/V_T in subsequent studies than in the original report (0.52 vs. 0.64). These analyses indicate that the low values of likelihood ratio for f/V_T reported by the Task Force are largely explained by their failure to correct for the occurrence of spectrum bias and test-referral bias.

A key requirement in establishing a scientific fact is to test a hypothesis under various settings and see if a finding is reproducible. The data from the twenty-nine studies in Table 58-1 are aggregated data from thousands of patients, in various settings, collected by investigator groups who varied widely in research skills. When these data were entered into a Bayesian model with pretest probability as the operating point, the observed posttest probabilities were closely correlated with the values predicted by the original study on f/V_T ($p < 0.0001$).

A further limitation in the Task Force's analysis was their failure to focus on the goal of a weaning-predictor test: to screen for weanability. Their use of likelihood ratio represents conflation of screening and confirmation. Likelihood ratio is not precisely suited to the evaluation of a screening test because it includes test components vital for screening (true-positive and false-negative rates) but also components not directly focused on screening (true-negative and false-positive rates); the latter cloud the contribution of the vital components.⁵⁴ In contrast, sensitivity captures exactly the components that define the reliability of a screening test because it contains only true-positive and false-negative rate. The average reported sensitivity of f/V_T was 0.87 (Table 58-1). Thus, contrary to the conclusion reached by the EBM Task Force, the facts included in the aggregated studies show that f/V_T is a reliable predictor of weaning success.

Tailoring a Predictor to Patient Needs

The threshold value of a test (the single value that separates positive from negative results) is determined by conducting research in a large number of patients. When applying the test to an individual patient, however, clinicians may decide to raise or lower the threshold value, taking into account unique aspects of a particular patient's condition.

Consider a severely immunocompromised patient in whom the physician believes that the risk of complications with continued ventilation far outweigh the risk of weaning failure or reintubation. With that mindset, the physician wants a weaning predictor test that has a very low rate of false-negative results. (A false-negative result is a test that predicts failure, $f/V_T > 100$, but the patient succeeds.) False-negative results are reflected in sensitivity (for f/V_T , 0.97)⁵⁹ and negative-predictive value (for f/V_T , 0.95).⁵⁹ To minimize the likelihood of a false-negative result in this patient, the physician may decide to undertake a T-tube trial even when f/V_T is higher than the usual threshold of 100 (say, 115 or 125). Use of a higher threshold moves borderline patients away from being kept on the ventilator and pushes them toward a weaning attempt. The clinician must recognize that a lower false-negative likelihood can only be achieved by also accepting a higher false-positive likelihood for the test result.

The second example relates to a patient with a neck deformity who the physician is leaning toward keeping on the ventilator for an additional day or longer rather than taking a risk of weaning failure and reintubation. The physician wants a weaning predictor test that has a very low rate of false-positive results. (A false-positive is a test that predicts success, $f/V_T < 100$, but the patient will fail a weaning trial.) False-positive results are reflected in specificity (for f/V_T , 0.64)⁵⁹ and positive-predictive value (for f/V_T , 0.78).⁵⁹ To minimize the likelihood of a false-positive result in this patient, the physician may decide to undertake a T-tube trial only if the

f/V_T threshold is lower than the usual 100 (say, 90 or 80). But, again, the lower false-positive likelihood can be achieved only by accepting a higher false-negative likelihood for the test result.

If the particular group of patients that a clinician manages usually have a high likelihood of weaning success (high prevalence, or pretest probability, of success), the clinician should use an f/V_T threshold higher than 105, and thus accept a higher false-positive rate so as to maintain a high true-positive rate.⁸³ This new threshold will result in a higher sensitivity and lower specificity.

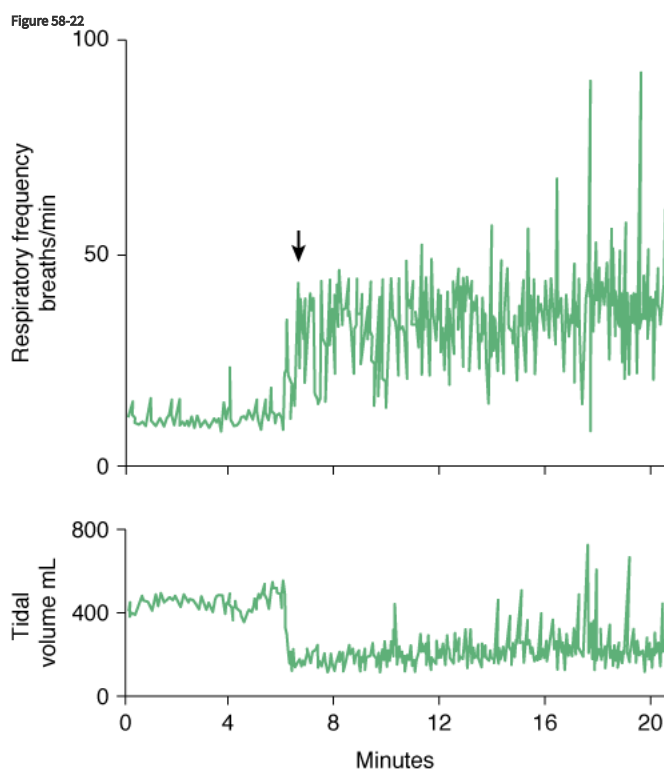
How much to adjust thresholds in an attempt to minimize false-negative results and false-positive results is an area of active decision-analysis research. In population studies, the threshold of a test is set such that the burden of false-positive results multiplied by the number of false-positive results balances the burden of false-negative results multiplied by the number of false-negative results. When adjusting the threshold value for an individual patient, the clinician needs to do an analogous computation. Reducing the occurrence of one undesirable test result can only be achieved by simultaneously increasing the occurrence of the other undesirable result. In the language of game theory, this balancing action is a zero-sum game (one goal is achieved at the expense of another).

Prediction of Weaning Outcome

Frequency-to-Tidal-Volume Ratio

Technical Details

The introduction of f/V_T as a weaning predictor stemmed from research into the pathophysiology of weaning failure. In 1986, Tobin et al¹⁰ showed that patients who failed a T-tube trial exhibited an increase in frequency and a fall in V_T in the first few minutes after removal of ventilator support (Fig. 58-22).¹⁰ In 1991, Yang and Tobin⁵⁹ reported that f/V_T was superior to nine other weaning predictors. In that study, sensitivity was 0.97, specificity 0.64, positive-predictive value 0.78, and negative-predictive value 0.95. Since the initial study, f/V_T has been employed in hundreds of studies, not all of them related to weaning. The manner of making the measurement and the use of f/V_T in decision making has varied considerably. The key steps that should be followed in order to obtain a reliable measurement of f/V_T are discussed in the following paragraphs.



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A time-series, breath-by-breath plot of respiratory frequency and tidal volume in a patient who failed a weaning trial. The arrow indicates the point of resuming spontaneous breathing. Rapid, shallow breathing developed almost immediately after discontinuation of the ventilator. (Used, with permission, from Tobin et al.¹⁰)

The measurement should be performed relatively early in the patient's course. In the original study, Yang and Tobin⁵⁹ pointed out that the measurement was aimed at detecting "the earliest time that a patient might resume spontaneous breathing." That is, f/V_T functions solely as a screening test. The intent is to move the transition between stage 2 and stage 3 of weaning (see Fig. 58-1) to an earlier time in a patient's course. It makes no sense to make the measurement after the

Loading [Contrib]/a11y/accessibility-menu.js reathing, as many researchers have done.^{15,84}

In the original study, the measurement of f/V_T was made with a handheld spirometer after disconnecting the patient's endotracheal tube from the ventilator circuit. That is, the measurement was made while the patient breathed room air, as that was the custom when obtaining weaning predictors in the 1980s. This is the approach we still employ (Fig. 58-23). (Some patients exhibit oxygen desaturation during the measurement, which indicates that they are not ready to proceed to a weaning trial.)

Figure 58-23



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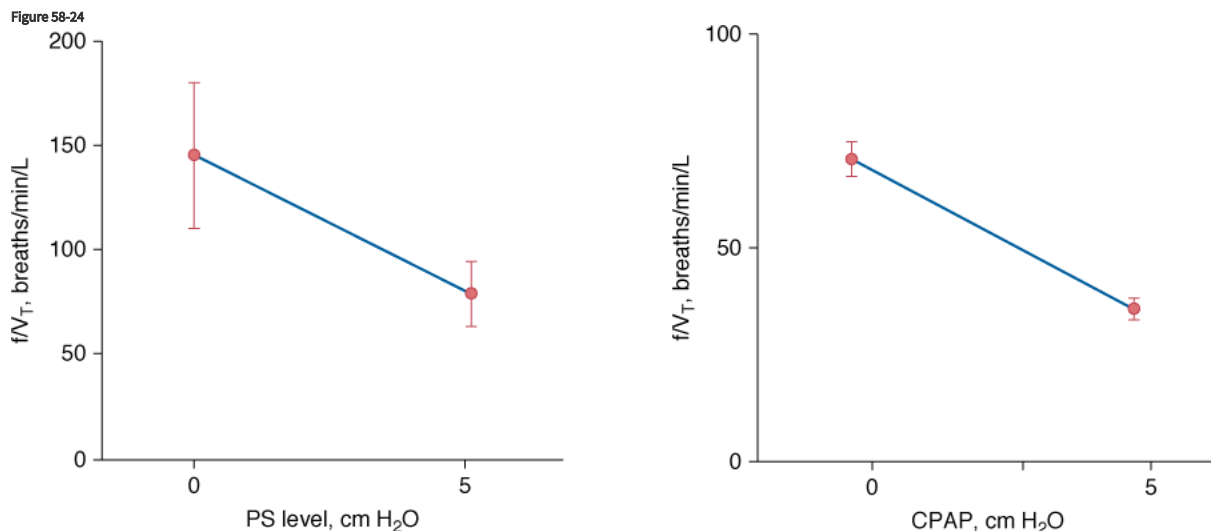
Measurement of f/V_T . The patient's endotracheal tube is disconnected from the ventilator circuit and from supplemental oxygen, and the patient breathes room air. A handheld spirometer (with a filter) is attached to the end of the endotracheal tube. After the patient has established a regular respiratory rhythm (which may take about 1 minute), cumulative ventilation is measured for exactly 1 minute while respiratory frequency is simultaneously counted. The average tidal volume is calculated as minute ventilation divided by respiratory frequency. f/V_T is calculated as respiratory frequency divided by the average tidal volume.

To measure f/V_T , some researchers use the pneumotachograph within the ventilator, taking care to place the patient in the “flow-by” mode and ensure that pressure support (PS) and continuous positive airway pressure (CPAP) are both set at zero. Patel et al⁸⁵ recently reported that such use of the pneumotachograph within an Evita ventilator (Evita 4 or Evita XL, Dräger, Telford, PA) yielded equivalent values of f/V_T to those obtained manually with a Wright spirometer.⁸⁵

The measurement should be obtained over a full minute of unassisted breathing. Some clinicians employ a shorter duration, but this is unreliable because of variability in breathing. It is also important that the patient's breathing pattern achieves equilibrium before starting the measurement. Mechanical ventilation can depress respiratory motor output through several mechanisms,⁸⁶ and shallow breaths and apneic pauses are common during the first minute after disconnecting a patient from the ventilator. We wait until we see that the patient has established a regular respiratory rhythm, which generally occurs by the end of the first minute of unassisted breathing. We then make the measurement of f/V_T over the subsequent minute (see Fig. 58-23).

A common mistake is to measure f/V_T while the patient is receiving ventilator assistance, either in the form of PS or CPAP. The goal of recording a patient's respiratory frequency and tidal volume is to forecast the values of these variables following removal of the endotracheal tube—when a patient will be breathing without PS or CPAP. Thus, it is irrational to measure f/V_T while the patient is receiving ventilator assistance. Moreover, the numbers themselves are drastically different. Values of f/V_T during unassisted breathing are 23% to 52% higher than during PS of 5 cm H₂O; values during unassisted breathing are 46% to 82% higher than during PS of 10 cm H₂O (Fig. 58-24).^{87–90} Some clinicians assert that a PS of 5 to 10 cm H₂O simply overcomes the resistance posed by an endotracheal tube. This claim ignores the Loading [Contrib]/a11y/accessibility-menu.js in the upper airways after an endotracheal tube has been in place for some time. After the endotracheal tube has been

removed, the mucosal swelling produces an increase in upper airway resistance. Straus et al⁹¹ demonstrated experimentally that the respiratory work dissipated against the supraglottic airway after extubation is almost identical to the work dissipated against the endotracheal tube before extubation (see [Techniques of Weaning](#)).



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Effect of pressure support (PS) and continuous positive airway pressure (CPAP) on frequency-to-tidal volume ratio (f/V_T). *Left panel:* In twelve patients with COPD, f/V_T was 145 ± 121 (standard deviation [SD]) while patients received no ventilator assistance; the addition of PS 5 cm H₂O caused f/V_T to decrease to 79 ± 53 (45% decrease).^{90,138,139} *Right panel:* In thirty-three postoperative patients, f/V_T was 71 ± 23 (SD) during unassisted breathing; the addition of CPAP 5 cm H₂O caused f/V_T to decrease to 36 ± 14 (49% decrease).⁹³

The measurement of f/V_T while a patient is receiving CPAP also suffers from flawed thinking. In healthy subjects, CPAP 5 cm H₂O caused f/V_T to decrease by 38% as compared with unassisted breathing.⁹² In thirty-three patients recovering from coronary bypass surgery, f/V_T was 49% lower when measured in the presence of CPAP 5 cm H₂O than during unassisted breathing: f/V_T 36 ± 14 and 71 ± 23 (standard deviation [SD]), respectively (see [Fig. 58-24](#)).⁹³ In sixty ventilated patients, f/V_T was 21% lower when measured in the presence of CPAP 5 cm H₂O (delivered through the ventilator circuit) than when patients breathed through a T-piece circuit without CPAP: median values of 71 (IQR: 52 to 88) and 90 (IQR: 59 to 137), respectively ($p < 0.001$).⁸⁵ In the latter study, 89% of patients satisfied the threshold f/V_T for weaning success, equal to or less than 105, while receiving CPAP 5 cm H₂O, but only 65% satisfied it while breathing through a T-piece circuit. The ability of CPAP to alter the distribution of patients on each side of the threshold is sufficient to account for much of the variation in performance of f/V_T in research studies.

Some authors claim that use of CPAP 5 cm H₂O simply replaces the loss of “physiologic PEEP,” which results from the presence of an endotracheal tube.⁶³

Physiologic PEEP, however, is a myth. Static recoil pressure of the respiratory system is zero at end-expiration. The addition of external PEEP while a patient is still intubated confounds clinicians in their ability to assess the challenges the patient will have to face following extubation.

In the original study on f/V_T , the value that best discriminated weaning-success patients from weaning-failure patients was equal to or less than 105.⁵⁹ This threshold is commonly rounded to 100 because sensitivity was the same for both f/V_T equal to or less than 105 and f/V_T equal to or less than 100. Although it is crucial for investigators to adhere meticulously to a threshold of 105 (or 100) when conducting a research study, clinicians should not feel tethered to such a threshold in every patient (see “[Tailoring a Predictor to Patient Needs](#)”). Dichotomizing the readings of a diagnostic test on the basis of a single cutoff value inevitably means that much information is discarded; an f/V_T reading of 106 does not carry the same clinical significance as an f/V_T of 136 (despite being handled identically in a research study). Physicians should use their judgment in making decisions based on the actual reading of f/V_T and the characteristics of the patients rather than feeling constrained by a single cutoff value.

Comparison of Frequency-to-Tidal-Volume Ratio with Other Diagnostic Tests

A detailed evaluation of a diagnostic test, when confined to that test alone, leaves a reader with an incomplete understanding of its usefulness. It is helpful for a reader to compare the performance of the test against other tests in common use. [Table 58-2](#) provides a listing of true-positive rate (sensitivity), false-positive rate (1 minus specificity), and likelihood ratios for f/V_T as a weaning predictor and a number of other tests in common use, as reported by Sox et al.⁵⁵

Table 58-2: Test Characteristics of Some Common Diagnostic Tests

Test	TPR	FPR	Positive LR	Negative LR
$f/V_T \leq 105$	0.97	0.36	2.72	0.04
$f/V_T 100$	0.97	0.32	3.03	0.04
$f/V_T 80$	0.81	0.11	7.52	0.22
Acid phosphatase for prostatic metastases	0.83	0.01–.003	42	0.17
Angiography in abdominal aortic aneurysm	0.70	0.05	14	0.32
Chest X-ray in lung cancer	0.60	0.04	15	0.42
CPK-MB in acute myocardial infarction	0.94	0.10	9.4	0.07
Ventilation–perfusion scan (segmental/larger) in PE	0.76	0.20	3.8	0.30
$Pa_{O_2} < 90$ mm Hg in pulmonary embolism	0.95	0.50	1.9	0.10
$FEV_1 < 80\%$ for COPD				
Young, minimal symptoms, nonsmoker	0.07	0.05	1.4	0.98
Middle age, moderate symptoms, smoker	0.27	0.10	2.7	0.81
Stress ECG for coronary artery disease	0.60	0.17	3.5	0.45
Theophylline > 20 mcg/mL for toxicity	0.60	0.10	6.0	0.44
Ultrasound for endocarditis	0.37	0.04	9.3	0.66
WBC indium scan for abdominal abscess	0.86	0.05	17.2	0.15

Abbreviations: COPD, chronic obstructive pulmonary disease; CPK-MB, creatine phosphokinase-myocardial band; ECG, electrocardiogram; FEV_1 , forced expiratory volume in 1 second; FPR, false-positive rate (1 – specificity); f/V_T , frequency-to-tidal-volume ratio; LR, likelihood ratio; Pa_{O_2} , partial arterial oxygen pressure; PE, pulmonary embolism; TPR, true-positive rate; WBC, white blood cell count.

LR > 10 or < 0.1 = generates large and often conclusive changes in pretest to posttest probability;

LR 5 to 10 or 0.1 to 0.2 = generates moderate and usually useful shift in pretest to posttest probability;

LR 2 to 5 or 0.5 to 0.2 = generates a small and sometimes important change in pretest probability;

LR 1 to 2 or 0.5 to 1.0 = generates a small and rarely important change in pretest probability.

Source: Modified, with permission, from Sox et al.⁵⁵

Randomized, Controlled Trial

Tanios et al.⁹⁴ undertook a randomized, controlled trial (RCT) to determine whether the inclusion of f/V_T in a weaning protocol influenced weaning time. They studied 304 patients: 153 in the f/V_T -protocol group and 151 in the nonprotocol group. Patients in the f/V_T -protocol group could proceed to a weaning trial (CPAP 5 cm H₂O and PS 7 cm H₂O or less) if—and only if—they had an f/V_T of less than 106. In the second study arm, clinicians did not follow the protocolized approach to weaning. The rate of extubation was the same in both groups: 82%. The duration of weaning was longer in the f/V_T -protocol group than in the nonprotocol group: median 3 (IQR: 1, 5) versus 2 (2, 6) days ($p = 0.04$). The authors concluded that including a weaning predictor (f/V_T) in a protocol prolonged weaning time.

The study has three major problems. The first arises with clinical management of patients in the control group. It is assumed that physicians managing patients in this group did not calculate f/V_T . Physicians are highly aware that respiratory frequency and tidal volume are key variables in deciding whether a patient will tolerate weaning and extubation. Once this knowledge has crept into a physician's brain, it cannot be surgically extirpated at the point of commencing an RCT. To ensure that physicians did not employ breathing pattern in their decision making, Tanios et al.⁹⁴ would have had to have taken steps to hide or occlude the display of frequency and tidal volume on the bedside monitor and ventilator screen. Alternatively, Tanios et al.⁹⁴ would have had to compare their usual care group with matched

patients weaned in their unit before the late 1980s—before research studies began emphasizing the importance of respiratory frequency and tidal volume in clinical decision making on weaning.^{10,59,95}

A second problem is the pretest probability of weaning/extubation success, 82%, in the patients of Tanios et al.⁹⁴ At a pretest probability of weaning success of 82%, the posttest probability for a positive f/V_T result, as calculated by the Bayes' theorem, is 93%. The benefit that accrues from such a small increment (from 82% to 93%) is extremely difficult to detect by means of an RCT.

The third problem is that f/V_T was one of several components in a weaning protocol. It is important to make a distinction between the use of a protocol in conducting a research study and its use in everyday clinical practice. In the research protocol of Tanios et al.,⁹⁴ patients who had an f/V_T of 105 or less progressed to a weaning trial whereas patients with an f/V_T of 106 or higher were returned to the ventilator. When conducting research, this is exactly how a protocol must be specified and followed. No flexibility was permitted. A competent clinician, however, would think it daft to slavishly comply with a protocol that decided an entire day of ventilator management on a one-unit difference in a single measurement of f/V_T . Rather, intelligent physicians customize knowledge to the particulars of each patient and are expected to outperform the inflexible application of a protocol—as has been shown in numerous studies of weaning protocols (see [Weaning by Protocol versus Usual Care](#)).

Tanios et al.⁹⁴ concluded that the inclusion of f/V_T in a protocol results in prolonged weaning time, and also that “ f/V_T should not be used routinely in weaning decision making.” The latter, however, is a non sequitur. The authors are conflating the use of f/V_T as one component of an inflexible protocol and its use in clinical decision making. The authors' experimental design did not test the use of f/V_T in decision making per se; rather, the design tested whether the use of a protocol that keeps a patient ventilated for an additional day when f/V_T is 106 as opposed to 105 increases the time required for weaning.

Arterial Blood-Gas Values

Other than in the course of withdrawing life support, mechanical ventilation is virtually never discontinued in a patient who has severe hypoxemia, such as a Pa_{O_2} less than 55 mm Hg with fractional inspired oxygen concentration (Fi_{O_2}) greater than 0.40. Computations such as the Pa_{O_2}/Fi_{O_2} ratio or alveolar-arterial P_{O_2} gradient are often used to quantify gas-exchange function. The relationship of these variables to more direct measurements of ventilation-perfusion relationships and shunt is based on many assumptions, and the relationship is variably affected by changes in the concentration of inhaled oxygen.⁹⁶ Few of these measurements have been rigorously evaluated as predictors of weaning outcome.

Krieger et al.⁹⁷ found that a Pa_{O_2}/Fi_{O_2} ratio of 238 (equivalent to a Pa_{O_2} of 50 with an Fi_{O_2} of 0.21) had a positive-predictive value of 90% and a negative-predictive value of 10%. Yang and Tobin⁵⁹ found that a Pa_{O_2} -to-alveolar partial pressure of oxygen (PA_{O_2}) ratio of 0.35 provided the best separation between weaning-success and weaning-failure patients in an initial training data set. When that threshold was evaluated in a subsequent validation data set, the positive-predictive value was 0.59 and negative-predictive value was 0.53. These poor test performances do not mean that measurement of gas exchange is of no value in predicting weaning outcome. All such studies suffer from test-referral bias and patients with severe hypoxia were excluded from the study population. Although threshold values of the efficiency of gas exchange cannot be recommended for weaning prediction, weaning attempts are not recommended in patients with borderline hypoxemia.

Minute Ventilation

Normal minute ventilation ($\dot{V}_E$) is approximately 6 L/min.^{98,99} Based on an initial report of Sahn and Lakshminarayan,¹⁰⁰ $\dot{V}_E$ of less than 10 L/min became one of the standard weaning predictors. [Table 58-3](#) summarizes the subsequent studies with accessible data on the accuracy of $\dot{V}_E$ as a predictor of weaning outcome. When interpreting these data, it is essential to recognize the influence of test-referral bias, because clinicians are reluctant to initiate weaning attempts in patients with a high $\dot{V}_E$.

Table 58-3: Accuracy of Minute Ventilation in Predicting Weaning Outcome

Threshold L/min	Sensitivity	Specificity	Positive-Predictive Value	Negative-Predictive Value	Number of Patients	Probability of Weaning Success	Authors
<9.9	1.00	0.32	0.46	1.00	101	0.63	Zeggwagh et al ¹²⁷
10	0.45	0.78	0.89	0.25	47	0.81	Tahvanainen et al ¹⁸⁷
<10	0.79	0.32	0.67	0.48	100	0.63	Chatila et al ¹⁰⁸
	0.50	0.67	0.98	0.20	163	0.988	Leitch et al ⁷⁵
	0.31	0.61	0.50	0.40	64	0.56	Yang and Tobin ⁵⁹
≤10	0.76	0.07	0.92	0.14	183	0.92	Jacob et al ¹⁰⁹
	0.96	0.47	0.90	0.73	100	0.83	Sahn and Lakshminarayan ¹⁰⁰
<12	0.40	0.50	0.74	0.19	45	0.78	Sassoon and Mahutte ¹⁴
≤12	0.86	0.14	0.72	0.31	92	0.73	Conti et al ¹³¹
>12.5	0.75	0.64	0.45	0.86	40	0.70	Gandia and Blanco ¹³
<15	0.97	0.11	0.65	0.67	100	0.63	Chatila et al ¹⁰⁸
	0.78	0.18	0.55	0.38	64	0.56	Yang and Tobin ⁵⁹
≤15	0.81	0.20	0.52	0.50	31	0.52	Yang ¹²⁸

Maximum Inspiratory Pressure

The airway pressure during a maximum inspiratory effort, maximum inspiratory pressure ($P_{I\max}$), provides a global measure of inspiratory muscle strength. The measurement is usually made at the opening of the endotracheal tube with an aneroid manometer while the patient makes a maximum effort against a closed airway.¹⁰¹ The use of $P_{I\max}$ as a weaning predictor stems from a study by Sahn and Lakshminarayan.¹⁰⁰ They found that all patients with a $P_{I\max}$ value more negative than -30 cm H_2O were successfully weaned, whereas all patients with a $P_{I\max}$ less negative than -20 cm H_2O failed a weaning trial. Table 58-4 summarizes subsequent studies with accessible data on the accuracy of $P_{I\max}$ as a predictor of weaning outcome.

Table 58-4: Accuracy of Maximal Inspiratory Pressure in Predicting Weaning Outcome

P _{imax} Threshold cm H ₂ O	Sensitivity	Specificity	Positive-Predictive Value	Negative-Predictive Value	Number of Patients	Probability of Weaning Success	Authors
≤-30	0.68	0	0.74	0	42	0.81	Tahvanainen et al ¹⁸⁷
≤-30	0.57	0	0.44	0	12	0.58	Sassoon et al ¹¹
	1	0	0.67	0	17	0.65	Fiaastro et al ³⁰
	NR	NR	0.92	0.21	269	0.9	Krieger et al ¹³⁰
	0.67	0.69	0.78	0.55	100	0.63	Chatila et al ¹⁰⁸
	0.86	0.21	0.58	0.55	100	0.56	Yang and Tobin ⁵⁹
≤-25	0.59	0.75	0.59	0.79	101	0.63	Zeggwagh et al ¹²⁷
≤-20	NR	NR	0.91	0.22	269	0.93	Krieger et al ⁹⁷
	1	0.14	0.6	1	100	0.56	Yang and Tobin ⁵⁹
	0.91	0.3	0.82	0.55	45	0.78	Sassoon and Mahutte ¹⁴
	0.9	0.26	0.67	0.6	100	0.63	Chatila et al ¹⁰⁸
	0.96	0.07	0.92	0.14	183	0.92	Jacob et al ¹⁰⁹
<-16	0.92	0.07	0.72	0.36	92	0.73	Conti et al ¹³¹
<-15	1	0.11	0.59	1	100	0.56	Yang and Tobin ⁵⁹

Abbreviation: NR, not reported.

Vital Capacity

The normal vital capacity is usually between 65 and 75 mL/kg, and a value of 10 mL/kg or more has been suggested to predict a successful weaning outcome. Many investigators have found that vital capacity is often unreliable. For example, Milbern et al¹⁰² found vital capacity of 15 mL/kg to be falsely positive in 15% and falsely negative in 63% of patients. In a study of ten patients with Guillain-Barré syndrome, Chevrolet and Deleamont¹⁰³ reported that vital capacity was helpful in guiding the weaning process. Patients with a vital capacity of less than 7 mL/kg were unable to tolerate as few as 15 minutes of spontaneous breathing. As vital capacity increased to greater than 15 mL/kg with recovery from the illness, patients were safely extubated. Apart from unique circumstances, such as patients with Guillain-Barré syndrome, vital capacity is rarely used as a weaning predictor.

Airway Occlusion Pressure

The pressure generated during the first 0.1 second of an airway occlusion (P_{0.1}) is widely used as an index of respiratory center motor output.^{104,105} During resting breathing in normal subjects, P_{0.1} is approximately 0.5 to 1.5 cm H₂O. Several investigators have evaluated the usefulness of P_{0.1} as a predictor of weaning outcome (Table 58-5). In these studies, the threshold value of P_{0.1} discriminating between weaning success and weaning failure ranged from 3.4 to 6.0 cm H₂O.

Table 58-5: Accuracy of Occlusion Pressure in Predicting Weaning Outcome

Threshold cm H ₂ O	Sensitivity	Specificity	Positive-Predictive Value	Negative-Predictive Value	Number of Patients	Probability of Weaning Success	Authors
>2.8	0.67	0.52	0.21	0.89	130	0.88	Fernandez et al ⁸²
≥3.4	0.75	0.61	0.45	0.85	30	0.7	Gandia and Blanco ¹³
≤4.0	0.83	0.90	0.93	0.78	20	0.6	Fernandez et al ¹⁸⁸
≤4.0	0.94	0.07	0.73	0.33	92	0.73	Conti et al ¹³¹
≤4.2	0.78	1.00	1.00	0.89	20	0.35	Herrera et al ¹⁸⁹
	0.71	0.43	0.56	0.6	11	0.5	Montgomery et al ¹²
≤4.5	1.00	1.00	1.00	1.00	13	0.46	Conti et al ¹⁹⁰
	0.75	0.55	0.7	0.62	217	0.58	Vallverdu et al ⁷⁷
5.0	0.87	0.91	0.96	0.65	75	0.82	Capdevila et al ¹⁵
<5.5	0.91	0.36	0.56	0.82	68	0.75	Uusaro et al ⁷⁸
≤5.5	09.7	0.40	0.85	0.80	45	0.78	Sassoon and Mahutte ¹⁴
≤6.0	0.86	0.29	0.55	0.67	11	0.5	Montgomery et al ¹²
<6.0	1.00	1.00	1.00	1.00	12	0.58	Sassoon et al ¹¹

Reproducibility of Standard Weaning Predictors

As with any diagnostic test, it is important to have precise details of the instrumentation used for the measurement, the conditions of the measurement, key steps in making the measurement, and reproducibility of the values. Reproducibility should be assessed in terms of variation in measurements made by one individual on several occasions, and variations in measurements made by several individuals at one point in time. The reproducibility of weaning predictor tests has been assessed to varying extents.

Before worrying about reproducibility, the first requirement for measuring a physiologic variable is to ensure that the patient has achieved a steady state (before the recording is commenced). This requirement can be difficult to satisfy when measuring weaning predictors: The switch between assisted ventilation and autonomous breathing means that there will be some intervening period when a steady state will not be present. For example, if measurement of frequency and V_T commences immediately after disconnecting a patient from the ventilator (say, within 10 seconds of the last ventilator breath and continued over the next 60 seconds), the patient may initially experience apneas (consequent to ventilator-induced neuromechanical inhibition).^{106,107} The recorded values of frequency and V_T will be different than if the investigators had waited a minute or so to first ensure that the patient had begun to breathe in a steady manner. Chatila et al¹⁰⁸ reported that measurements of f/V_T at 30 minutes into a weaning trial more accurately predict outcome than measurements in the first minute.¹⁰⁹ Although it is true that including data of the first 30 seconds or so may be unrepresentative, this does not mean that it takes 30 minutes to establish a steady state.

Jubran et al²² studied the time required for f/V_T to reach a point of equilibration in thirty-five weaning-failure and twenty-five weaning-success patients. The median time to reach $\pm 10\%$ of the final value of f/V_T was 2 (IQR: 1 to 2) minutes in both the weaning-success and weaning-failure patients (see Fig. 58-6). Within 2 minutes of the onset of the T-tube trial, 77% of the failure patients and 73% of the success patients had reached $\pm 10\%$ of the final value of f/V_T . The rapid equilibration of f/V_T contrasts with the slower equilibration of swings in esophageal pressure. The median time to reach $\pm 10\%$ of the final value of swings in esophageal pressure was 7.5 (4.2 to 14.75) minutes in the failure patients and 5 (2 to 8.5) minutes in the success patients. Based on these data, it is reasonable to usually commence measurement of f/V_T at 60 seconds after removal of the ventilator and then continue the measurement for another 60 seconds.

Yang and Tobin⁵⁹ investigated the methods used for measuring $\dot{V}_E$ and its subsets. In a survey of twenty-five hospitals, the most common condition for the measurement was unassisted breathing while receiving room air (thirteen hospitals). Ten hospitals made the measurement $\dot{V}_E$ during assisted breathing while receiving oxygen supplied by three different methods. Two hospitals measured during ventilator support. The latter error seemed laughable in 1991, but today it is even more widely made.

Yang and Tobin¹¹⁰ measured $\dot{V}_E$ and its subsets in thirty-three patients, who were clinically stable and considered ready to undergo a trial of weaning. $\dot{V}_E$ was 11.9 ± 0.8 L/min while patients spontaneously breathed oxygen through the ventilator circuit (the same FI_{O_2} as patients had received during ventilator support; achieved Pa_{O_2} 89 ± 4 mm Hg). When the patients were switched to room air, $\dot{V}_E$ increased to 13.5 ± 1.1 L/min; frequency also increased from 30.1 ± 11.8 to 33.9 ± 2.2 breaths/min, whereas V_T did not change. Fifteen patients satisfied the $\dot{V}_E$ threshold to progress to a weaning trial ($\dot{V}_E < 10$ L/min) while they breathed oxygen, but seven of them (47%) no longer satisfied this threshold when switched to room air. The study illustrates a wide range of conditions under which $\dot{V}_E$ is measured, and emphasizes the need to control the conditions of the measurement.

Yang¹¹¹ examined the reproducibility of predictor measurements in thirty patients about to undergo a weaning trial. Repeat measurements were obtained by a single individual on three trials over 15 minutes. The coefficients of variation for the repeated measurements (during 1 minute of room air breathing) were $10.8 \pm 2.1\%$ for $\dot{V}_E$, $6.7 \pm 1.2\%$ for frequency, $7.6 \pm 1.2\%$ for V_T , and $9.5 \pm 1.1\%$ for f/V_T . The coefficient of variation for measurements of $P_{I\max}$ (measured with a one-way valve, ensuring a lung volume below functional residual capacity) was $10.6 \pm 1.5\%$. Repeat measurements of vital capacity were the least reproducible, with a coefficient of variation of $19.6 \pm 2.8\%$.

Multz et al¹¹² undertook a more detailed study of the reproducibility of $P_{I\max}$ (measured with an aneroid manometer) in fourteen ventilator-dependent patients. Triplicate measurements were obtained by five experienced investigators who encouraged the patients to make vigorous inspiratory efforts. Measurements of $P_{I\max}$ at a single sitting by a single investigator showed good reproducibility: coefficient of variation of $12 \pm 1\%$. Much greater variation was observed when $P_{I\max}$ was measured in the same patient on the same day by different investigators: coefficient of variation $32 \pm 4\%$.

Measurements of $P_{0.1}$ exhibit the greatest variability, perhaps because the pressures recorded are very small (normal: 0.5 to 1.5 cm H₂O) and the timing (0.1 second) must be precise. The coefficient of variation within individual subjects is approximately 50%.^{113,114} The interindividual coefficient of variation is approximately 20% to 33% during a single set of measurements,^{114,115} but on repeated measurements (during CO₂ rebreathing trials) it is approximately 60%.¹¹⁶

Gastric Tonometry

The reliability of gastric tonometry in predicting weaning outcome was discussed in considerable depth in the second edition of this book. The rationale for its use is that the gastrointestinal mucosa becomes ischemic early with the development of either hemodynamic compromise or a redistribution of blood flow.^{78,117–121} New studies supporting the use of this technique for weaning prediction have not been published during the interval between the second edition and this edition. We are not aware of any research or clinical group that advocates the use of gastric tonometry as a weaning predictor and the topic will not be discussed.

B-Type Natriuretic Peptides

Pro-brain natriuretic peptide (BNP), a prohormone, is synthesized by cardiomyocytes in response to mechanical stretch as occurs with increased atrial or ventricular pressure, volume overload or both.¹²² Once secreted, pro-BNP is immediately cleaved into the biologically active 32-amino acid carboxyl-terminal peptide (BNP) and its inactive 76-amino acid amino terminal (N-terminal) fragment (NT-proBNP).¹²³ Both systolic and diastolic dysfunction of the left ventricle can result in high levels of BNP and NT-proBNP in the bloodstream, and these biomarkers are widely used as screening tools in patients with suspected cardiac disorders. Several investigators have evaluated the accuracy of BNP and NT-proBNP in identifying the likelihood and presence of cardiac causes of weaning failure.

In 102 patients about to undergo a 1-hour weaning trial, Mekontso-Dessap et al¹²⁴ measured plasma BNP and repeated the measurement in the first sixty patients at the end of the trial. Forty-two patients (41.2%) were classified as outcome failures: thirty-seven patients failed the weaning trial and five patients failed extubation. Plasma BNP was higher in patients with an unsuccessful outcome. The area under the receiver operating characteristic (ROC) curve for plasma BNP that predicted an unsuccessful outcome was 0.89 ± 0.04 ; a cutoff value of 275 pg/mL provided the greatest diagnostic accuracy (86%). The higher value of BNP, however, before the weaning trial in the outcome failures does not necessarily mean that cardiac dysfunction was responsible for the failed weaning attempt. Plasma BNP can be elevated for reasons other than left-heart dysfunction, such as advanced age, female gender, renal dysfunction, sepsis, pulmonary hypertension, vigorous fluid management, and extensive use of agents such as diuretics, angiotensin-converting enzyme inhibitors, and β -blockers.^{43,125} Moreover, the level of BNP did not significantly change between the beginning and the end of the weaning attempt. Nine patients who failed the initial weaning attempt were successfully weaned after diuretic therapy, which was associated with a significant decrease in BNP from 517 to 226 pg/mL ($p = 0.01$).¹²⁴

Grasso et al¹²⁵ measured plasma levels of NT-proBNP in nineteen patients with COPD before and at the end of a weaning trial. Eight patients (42%) were classified as having acute cardiac dysfunction at the end of the trial. Baseline NT-proBNP levels were significantly higher (median: 5000; IQR: 4218 pg/mL) in these patients than in patients without evidence of acute cardiac dysfunction (median: 1705; IQR: 3491 pg/mL). Plasma levels of NT-proBNP increased significantly at the end of the weaning trial only in patients with acute cardiac dysfunction (median: 12,733; IQR: 16,456 pg/mL; $p < 0.05$). The increase in NT-proBNP over the course of the weaning trial, but not baseline NT-proBNP, provided a good prediction of weaning-induced cardiac dysfunction: area under ROC curve, 0.909 (95% CI, 0.69 to 0.98; $p < 0.0001$,

cutoff = 184.7 pg/mL). The response pattern is the opposite of that reported by Mekontso-Dessap et al,¹²⁴ who found that the baseline value of plasma BNP, but not the change in BNP over the course of a weaning trial, predicted weaning failure.

The relative accuracy of BNP and NT-proBNP for predicting and identifying cardiac dysfunction as the cause of weaning failure was investigated by Zapata et al¹²⁶ in a group of 100 patients who had received mechanical ventilation for more than 48 hours. Thirty-two patients failed the weaning trial, and twelve of these were classified as resulting from cardiac failure, diagnosed on the basis of PAOP greater than 18 mm Hg or echocardiographic features of elevated filling pressures. Before the weaning trial, BNP and NT-proBNP were higher in patients who went on to fail because of cardiac causes than in the remaining patients: Threshold values based on ROC curve analyses were 263 ng/L for BNP ($p < 0.001$) and 1343 ng/L for NT-proBNP ($p = 0.08$). The threshold value of BNP that most accurately predicted weaning-induced cardiac failure was lower than the value reported by Mekontso-Dessap et al,¹²⁴ plasma BNP of 263 and 275 ng/L, respectively, and the level of diagnostic accuracy was also lower, 68% and 85%, respectively. Both BNP and NT-proBNP increased significantly over the course of the weaning trial in patients who developed cardiac failure. BNP was superior to NT-proBNP both in predicting and making a diagnosis of cardiac failure. Zapata et al¹²⁶ attributed the superior performance of BNP to its shorter circulating half-life (20 minutes). The longer half-life of NT-proBNP (120 minutes) promotes increased accumulation of the peptide after secretion, and thus produces a wider range of plasma concentrations.

At this juncture, there is considerable divergence among the results of studies evaluating the ability of natriuretic peptides to predict weaning outcome. Some investigators report that baseline values have the greatest predictive power,¹²⁴ whereas others report that the change in a peptide over the course of a weaning trial is more reliable,¹²⁵ some investigators report that NT-proBNP is more reliable than BNP,¹²⁵ whereas others report the opposite,¹²⁶ and the threshold value that discriminate between weaning-success and weaning-failure patients differs considerably among the studies.

Techniques of Weaning

Researchers who developed weaning predictors never recommended that patients be immediately extubated if a patient passed the predictor test. A one-step approach would be reasonable if the risks associated with the development of postextubation distress and the need for reintubation were trivial. Instead, clinicians decide whether to extubate a patient after performing two diagnostic tests in sequence (measurement of predictors followed by a weaning trial). The importance of this two-step approach is illustrated by the study of Zeggwagh et al.¹²⁷ They studied 101 patients, ventilated for more than 48 hours (mean: 10 ± 10 days; range: 2 to 60 days), who were considered ready for weaning by the unit team. The unit team then extubated the patients without first undertaking any formal weaning trial. (The extubation decision was based on relatively crude variables, and not the results of more sophisticated weaning predictor tests.) Of the 101 patients, thirty-seven required reintubation. This rate is double (or more) the rate reported in most other studies. These data of Zeggwagh et al¹²⁷ emphasize the need for a formal weaning trial before extubation.

The two-step strategy long employed by clinicians is consistent with the theoretical reasoning of the ideal approach to diagnostic testing. Clinicians first contemplate the possibility of a condition (typically a disease, but here it is the possibility of weaning success). At this stage they need a test with very high sensitivity. When the test results are negative (normal), the condition can be confidently ruled out.⁵³ If the test results are positive (abnormal), the clinician ideally follows the now stronger suspicion (based on the initial very sensitive test) with a second diagnostic test that is very specific. When the result of a very specific test is positive (abnormal), it essentially rules in a disease.⁴⁷ Studies of f/V_T have shown it to have a much higher sensitivity than specificity; several investigators have reported sensitivities of 0.90 or higher^{1,77,81,118,128–131} or 0.97 or higher.^{14,59,108,109,132,133}

For the second diagnostic test, a T-tube or some other weaning trial, clinicians operate on the assumption that it has a high specificity, although this assumption has never been formally tested. A test with high specificity has a low rate of false-positive results (patients who pass a T-tube trial, are extubated, but require reintubation). The usual rate of reintubation after passing a T-tube trial is 15% to 20%, although some investigators have reported reintubation rates as high as 29%.^{78,80,118,129,134} The true-negative rate is also needed for calculation of specificity. Determination of the true-negative rate for a T-tube trial would require the extubation of all patients who fail the trial and counting how many require reintubation, an experiment that is unethical.

A second rationale for the use of weaning techniques is that they somehow improve the patient's likelihood of tolerating extubation. That is, weaning techniques have some therapeutic action beyond their diagnostic role. This rationale is generally couched in terms of improved reconditioning of the respiratory muscles.

A weaning task force^{62,63} has suggested that measuring predictors may inhibit expeditious weaning, and they advocated the bypassing of this step and starting weaning evaluation with a trial of spontaneous breathing. Yet in a study used to support this viewpoint, Ely et al¹³⁵ found that physicians refused to order a spontaneous breathing trial in 64% to 89% of patients who had already demonstrated satisfactory respiratory function on their weaning predictors. Respiratory therapists had demonstrated greater than 95% compliance in measuring f/V_T . It is difficult to understand how the demonstration of good physiologic performance on weaning predictors could have resulted in the failure to undertake a trial of spontaneous breathing in up to 90% of patients.

The major techniques that are used include T-tube trial, PS, intermittent mandatory ventilation (IMV), or some combination of these three.

Intermittent Mandatory Ventilation

Chapter 7 discusses IMV in detail. With IMV, the patient receives periodic positive-pressure breaths from the ventilator at a preset volume and rate. In addition, the patient can take spontaneous breaths between these mandatory breaths.

When IMV is used for weaning, the mandatory rate from the ventilator is reduced in steps of 1 to 3 breaths/min, and an arterial blood gas is obtained about 30 minutes after each change. Unfortunately, adjusting the number of ventilator breaths in accordance with the results of blood gases can lead to a false sense of security. As little as 2 to 3 positive-pressure breaths per minute can achieve acceptable blood gases, but these values provide no information about a patient's work of breathing, which may be excessive.

IMV was originally seen as the ideal mode for weaning: The ventilator breaths were expected to provide respiratory muscle rest, and the intervening spontaneous breaths to facilitate reconditioning. Accordingly, it was thought that respiratory muscle rest would be proportional to the number of positive-pressure breaths.

Subsequent studies revealed that patients have difficulty in adapting to the intermittent nature of the assistance.^{136,137} Studies revealed that inspiratory effort is equivalent for the assisted and unassisted breaths. At IMV rates of 14 breaths/min or less, tension-time index for both the assisted and unassisted breaths is above the threshold associated with respiratory muscle fatigue. At a moderate level of ventilator assistance (where the ventilator accounts for 20% to 50% of the total ventilation), electromyographic recordings reveal that activity of the diaphragm and sternomastoid muscles during assisted breaths is no lower than that during unassisted breaths.¹³⁷ These findings suggest that respiratory center output is preprogrammed and unable to adapt to breath-by-breath changes in load as occur with IMV. Consequently, IMV may contribute to the development of respiratory muscle fatigue or prevent recovery from it.

Pressure Support

Chapter 8 discusses pressure support (PS) in detail. Like IMV, PS provides graded assistance, but differs from IMV in that the clinician sets the level of pressure (rather than the volume) to augment every spontaneous respiratory effort. The level of pressure is usually adjusted in accordance with the patient's frequency. The frequency, however, that signals a satisfactory level of respiratory muscle rest has never been well defined, and recommendations range from 16 to 30 breaths/min.^{90,138,139}

When PS is used for weaning, the level of pressure is reduced gradually, in decrements of 3 to 6 cm H₂O, titrated to the patient's frequency. PS is commonly used to counteract the work imposed by breathing through the endotracheal tube and ventilator circuit. Consequently, the notion arose that if a patient was able to sustain ventilation at this "compensatory level" of PS, the patient should be able to breathe without difficulty after extubation. Investigators have estimated this compensatory level of PS to range from 3 to 13 cm H₂O. No method has ever been shown to reliably estimate the compensatory level of PS required by an individual patient.

Individuals who recommended the addition of PS for overcoming resistance of the endotracheal tube failed to recognize that the upper airways become inflamed and edematous after an endotracheal tube has been in place for some time. When the endotracheal tube is removed, the resistance of the upper airway is higher than normal. This subject was investigated rigorously by Straus et al.⁹¹ They studied fourteen patients who tolerated a 2-hour T-tube trial and were then extubated. Work of breathing did not change between the start and end of the trial, 20.0 ± 9.1 and 22.1 ± 10.6 J/min, respectively. Importantly, work did not decrease after extubation, 22.6 ± 9.7 J/min. The work dissipated against the supraglottic airway after extubation was almost identical to the work dissipated against the endotracheal tube before extubation (approximately 11% of total work of breathing). These data indicate that the addition of any level of PS will lead to underestimation of a patient's work of breathing after extubation—the very goal that is being attempted by a weaning trial that uses a low level of PS. Mehta et al.¹⁴⁰ measured work of breathing in twenty-two patients before and 15 minutes after extubation. The work of breathing at PS 5 cm H₂O before extubation underestimated the work of breathing performed after extubation by 36%; work of breathing at CPAP 5 cm H₂O underestimated the work of breathing performed after extubation by 23%.

Most investigators have focused on the effect of PS on inspiratory muscle effort. The switch between inspiration and expiration can give rise to problems, particularly in patients with COPD. With PS, ventilator assistance ceases when the patient's inspiratory flow falls to a preset amount (such as 25% of the peak flow). Air flow changes more slowly in patients with COPD, and these patients often begin to exhale while the ventilator is still pumping gas into their chests. In a study of patients with COPD who were receiving PS 20 cm H₂O, almost half recruited their expiratory muscles before the ventilator had completed mechanical inflation.⁹⁰

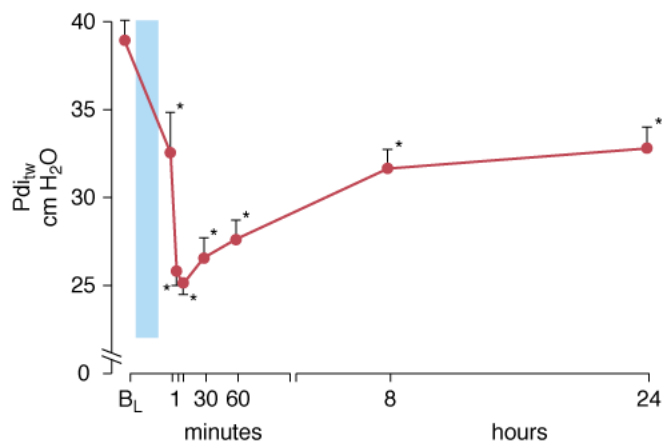
T-Tube Trials

The oldest weaning technique is to undertake trials of spontaneous breathing through a T-tube circuit. In the past, these trials started with a duration of approximately 3 to 5 minutes, and were repeated every 30 minutes.¹⁴¹ The duration of spontaneous breathing was progressively increased according to patient tolerance, as decided by physical examination and arterial blood-gas measurements. Patients were not extubated until they tolerated the T-tube trial for several hours: up to 8 hours in the classic study of Sahn and Lakshminarayan,¹⁰⁰ up to 16 hours in a study from the late 1980s,⁹⁷ and 12 hours in a study from the mid-1990s.¹⁴² The need for frequent changes to the ventilator circuit (every hour or more often) placed enormous demands on the intensive care staff, and the approach became extremely unpopular. By the late 1980s, T-tube trials had been largely supplanted by IMV.¹⁴³

Today, it is usual to limit a T-tube trial to 2 hours or less. We typically extubate a patient who does not develop distress during a 30-minute T-tube trial. If a patient fails a T-tube trial, we wait 24 hours before we undertake a subsequent trial. Our reasoning is based on the knowledge that most patients who fail a T-tube trial experience considerable stress on the respiratory muscles secondary to marked increases in their work of breathing. The respiratory muscles take 24 hours or longer to recover from this stress (Fig. 58-25).¹⁴⁴ Accordingly, we reinstitute full assistance with assist-control ventilation for at least 24 hours before reassessing the patient for another T-tube trial.

Figure 58-25

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Induction of diaphragmatic fatigue (*stippled bar*) produced a significant fall in transdiaphragmatic pressure (Pdi) elicited by twitch stimulation of both phrenic nerves. Significant recovery of twitch pressure was noted in the first 8 hours after completion of the fatigue protocol; no further change was observed between 8 and 24 hours, and the 24-hour value was significantly lower than baseline. The delay in reaching the nadir of Pdi_{TW} probably results from twitch potentiation, induced by repeated contractions, which was present at the end of the protocol. Values are mean $\pm$ standard error (SE). *, significant difference compared with baseline value, $p < 0.01$. (Used, with permission, from Laghi et al.¹⁴⁴)

Optimal plumbing of the T-tube circuit is necessary to avoid imposing respiratory work. Humidified gas is commonly provided in the form of a heated or cool aerosol of water from a large-volume nebulizer. This system can provoke bronchospasm in patients with reactive airways disease. Such patients can be managed with a non-aerosol-generating system, such as a heated passover humidifier.

A T-tube trial serves as a diagnostic test. It is primarily a means of evaluating a patient's ability to sustain spontaneous ventilation. Patients are judged to have failed a T-tube trial when they develop severe tachypnea, increased accessory muscle activity, diaphoresis, facial signs of respiratory distress, oxygen desaturation, tachycardia, arrhythmias, or hypotension. The degree of change in these variables, however, varies from report to report.

A standardized approach to patient monitoring during a T-tube trial does not exist. Indeed, there is no agreement as to whether the monitoring of any variable helps in deciding whether to continue a T-tube trial for an initially planned duration, prolong it, or curtail it. We monitor all such patients with a pulse oximeter. In a report on more than 1000 spontaneous breathing trials, Ely et al¹³⁵ observed only one transient episode of desaturation—a rate so staggeringly low as to beggar belief. In a study of seventeen patients with COPD who failed a T-tube trial, Jubran and Tobin¹⁷ found that severe hypoxemia ($P_{O_2} \leq 46$ mm Hg) was the primary cause of failure in two patients. In a study of 100 spontaneous breathing trials in eighty-three patients, Salam et al¹⁴⁵ observed seven instances in which the results of arterial blood gases caused physicians to defer extubation in patients judged otherwise to have passed the trial. Recent data indicate that patients who fail a T-tube trial do not develop low-frequency fatigue.²⁹ These data, however, do not exclude the possibility of high-frequency fatigue.²⁷ Moreover, weaning-failure patients (on average) experience more than four times the normal level of inspiratory effort (at the end of a T-tube trial), and some patients experience more than six times the normal value.¹⁷ Given the magnitude of these changes, we do not recommend the undertaking of a T-tube trial in a cavalier manner. Such large increases in respiratory effort cause severe dyspnea, and, apart from humanitarian considerations, we do not know what imprint severe dyspnea leaves on a patient's psyche.

Jubran et al²² investigated whether repeated measurements of esophageal pressure throughout a trial of spontaneous breathing might provide additional guidance over a single measurement obtained during the first minute of the trial. They quantified the change in esophageal pressure over the first 9 minutes of the trial using a multivariate adaptive regression spline procedure. In a study of sixty patients, an esophageal pressure–trend index had a sensitivity of 0.91 and specificity of 0.89. Specifically, an esophageal pressure–trend index reading of equal to or less than 0.44 was 8.2 times more likely to occur in weaning failure than in weaning-success patients. These data suggest that the continuous monitoring of esophageal pressure swings during a T-tube trial may provide additional guidance in patient management over tests used for deciding when to initiate weaning.

An implied goal of techniques that involve a gradual reduction in ventilator assistance—whether with IMV, PS, or T-tube trials of increasing duration—is to recondition the respiratory muscles, which may have been weakened during the period of mechanical ventilation. Theoretically, a once-daily trial of spontaneous breathing and a prolonged period of rest may be the most effective method of eliciting adaptive changes.^{146,147} That approach meets the three principal requirements of a conditioning program: overload, particularity, and reversibility.¹⁴⁶ During a T-tube trial, patients breathe against an elevated intrinsic load, thus satisfying the overload requirement. Particularity is also satisfied, in that the trial is an endurance stimulus and the desired objective is enhanced endurance. Finally, the use of a daily trial prevents regression of the adaptive changes. This reasoning, however, is solely based on indirect evidence. The effect of different weaning techniques on respiratory muscle reconditioning has not been investigated.

Spontaneous Breathing Trials

During a T-tube trial, a patient is not connected to the ventilator. To convey the lack of assistance, T-tube trials were also described as trials of spontaneous breathing, and used interchangeably. Subsequently, the term “spontaneous breathing trial” was also used to describe

patients who were breathing through the ventilator but not receiving any positive-pressure inspiratory assistance as occurs with use of “flow-by.” Flow-by avoided the substantial respiratory work imposed by the demand valves in older ventilators, and became a popular alternative to a T-tube circuit for conducting a trial of spontaneous breathing.

Authors, unfortunately, now use the term “spontaneous breathing trial” to describe weaning trials conducted with a fixed level of PS, ranging from 5 to 10 cm H₂O. It is an oxymoron to label PS as spontaneous breathing. The word “support” is an antonym of “unassisted.” The problem is not simply semantic. It is unscientific to regard positive-pressure assistance (of 5 to 10 cm H₂O) with every inspiratory effort as unassisted breathing. This unscientific thinking has led authors to believe that measurements of breathing pattern at PS of 5 to 10 cm H₂O is no different than measuring breathing pattern while a patient is disconnected from the ventilator. The argument that PS of 5 to 10 cm H₂O does no more than compensate for the work imposed by an endotracheal tube is now known to be invalid (see [Pressure Support](#) above).

The weaning of a patient from the ventilator boils down to making clinical judgments as to whether a patient will be able to sustain ventilation on his or her own, and if not, how much ventilator assistance is required. It is unfortunate that the same term—“spontaneous breathing”—is now used to describe both breathing without any ventilator assistance and breathing with the delivery of positive-pressure assistance with every inspiratory effort. An expression that formerly facilitated meaningful communication has been corrupted such that it misleads not only readers but also the authors themselves.

The sloppiness of communication has reached its apogee in the trials of noninvasive ventilation for management of postextubation distress. Before extubation, patients undergo a trial of “spontaneous breathing” at PS 5 to 8 cm H₂O. After extubation, the major intervention is the delivery of PS (perhaps at a lower level) by face mask. The nonchalant representation of PS 7 cm H₂O as “spontaneous breathing” signifies a fundamental misunderstanding of physiology.

Addition of Positive End-Expiratory Pressure to Replace “Physiologic Positive End-Expiratory Pressure”

Some clinicians recommend that all patients with an endotracheal tube should receive PEEP of 5 cm H₂O, claiming that this level of PEEP is replacing a loss of physiologic PEEP produced by intermittent narrowing of the vocal cords. These claims do not square with long-standing knowledge of pulmonary physiology. Lung volume at end-expiration generally approximates the relaxation volume of the respiratory system (i.e., the lung volume determined by the static balance between the opposing elastic recoil of the lung and chest wall).^{148,149} The static recoil pressure of the respiratory system is thus zero at end-expiration in a healthy person.

The addition of PEEP of 5 cm H₂O is not without consequence. In ten ventilated patients who had PEEPi of 6.2 ± 1.0 cm H₂O, Smith and Marini¹⁵⁰ found that the addition of external PEEP of 5 cm H₂O decreased work of inspiration by 19%. The addition of PEEP can also substantially increase cardiac output in patients with left-ventricular failure. In either circumstance, the rapid removal of PEEP may lead to rapid decompensation at the time of extubation. Thus, the generic order of “5 of PEEP” may cause physicians to underestimate the likely challenges that a patient will face after extubation.

Head-to-Head Comparisons

The first RCT of the three major weaning techniques was conducted by Brochard et al.³ They randomized 109 difficult-to-wean patients (defined as failure to tolerate their first 2-hour T-tube trial) to three study arms. At 21 days, there were fewer weaning-failure patients among the PS group (2/26, 8%) than among the patients weaned by T-tube trials (10/30, 33%) or IMV (16/41, 39%). Also, less time was taken to wean with PS (5.7 ± 3.7 days) than with T-tube trials (8.5 ± 8.3 days) or IMV (9.9 ± 8.2 days). Statistically, however, the only difference was a shorter weaning time for PS patients (5.7 ± 3.7 days) than for the combined T-tube and IMV patients (9.3 ± 8.2 days; $p < 0.05$).

Esteban et al.⁴ undertook an RCT that had a similar design to that of Brochard et al. They randomized 130 difficult-to-wean patients (again defined as failure to tolerate their first 2-hour T-tube trial) to four study arms: IMV, PS, once daily T-tube trials, and intermittent spontaneous breathing trials (this terminology was used because six patients breathed through a circuit providing CPAP 5 cm H₂O; the remaining twenty-seven had T-tube trials; the trials were of gradually increasing duration, at least twice daily, with ≥ 1 hour of assist-control ventilation between each trial). At 14 days, mechanical ventilation was still required by 17% of IMV patients, 11% of PS patients, 3% of once-daily T-tube patients, and 3% of patients weaned by intermittent T-tube or CPAP trials. The time taken to wean was a median of 5 days for IMV (IQR: 3 to 11), 4 days (IQR: 2 to 12) for PS, 3 days (IQR: 1 to 6) for once-daily T-tube trials, and 3 days (IQR: 2 to 6) for intermittent T-tube/CPAP trials. The rate of successful weaning with once-daily T-tube trials was about three times faster than with IMV and two times faster than with PS; there was no difference between the speed of weaning with once-daily T-tube trials versus intermittent T-tube/CPAP trials.

These two studies concur in finding that IMV was the least-effective method for weaning a difficult patient. The steps taken in setting the IMV rate were largely the same in the two studies: The initial IMV rate was half the respiratory frequency seen during assist-control ventilation (IMV of about 10 breaths/min in both studies). Then, IMV rate was decreased by 2 to 4 breaths/min twice a day if the patient tolerated it. The point for extubation differed between the studies. In the study of Brochard et al, patients were extubated when they tolerated an IMV of 4 breaths/min for 24 hours. In the study of Esteban et al, patients were extubated when they tolerated an IMV of 5 breaths/min for 2 hours. This difference probably accounts for the shorter weaning time with IMV in the study of Esteban et al⁴ versus that of Brochard et al³: 5 days versus 10 days.

The data appear to differ in the relative efficacy of PS and T-tube trials. The reason for the difference can again be understood by examining the two protocols. The initial level of PS was largely similar in the two studies: pressure titrated to respiratory frequency (20 to 30 breaths/min³; < 25 breaths/min⁴). This resulted in a PS of 18 cm H₂O in both studies. PS was decreased by 2 to 4 cm H₂O twice a day in the study of Brochard et al, and by that amount or more in the study of Esteban et al.

[†] Loading [Contrib]/a11y/accessibility-menu.js In the studies. In the study of Brochard et al, patients were extubated when they tolerated PS of 8 cm H₂O for 24 hours. In the

study of Esteban et al, patients were extubated when they tolerated PS of 5 cm H₂O for 2 hours. This difference probably accounts for the shorter weaning time with PS in the study of Esteban et al than in the study of Brochard et al: 4 days versus 6 days. The approach to T-tube trials also differed. Brochard et al performed trials of increasing duration twice a day. Esteban et al used two approaches to T-tube trials: One arm consisted of T-tube trials once a day, and the second arm consisted of intermittent spontaneous breathing trials. The point for extubation differed between the studies. In the study of Brochard et al, extubation was contemplated when patients tolerated a 2-hour T-tube trial, but there could be as many as three separate 2-hour trials. In the study of Esteban et al, patients were extubated when they first tolerated a 2-hour trial.

The investigators of the two preceding studies^{3,4} set the duration of T-tube trials at 2 hours. In a subsequent study, conducted in 526 patients who had received more than 48 hours of ventilation, Esteban et al¹⁵¹ found that the rate of weaning failure was equivalent for trials lasting 30 and 120 minutes: 12% (33/270) and 16% (40/256), respectively. The rates of reintubation were virtually identical (13%); approximately 74% of patients in the two arms were successfully extubated. This study consisted of patients undergoing a first weaning attempt, unlike the two earlier trials that were limited to patients with demonstrable difficulty in being weaned. The findings, however, suggest that most patients can be weaned after a 30-minute trial, which simplifies management by freeing staff time for other patient care tasks. With a similar goal, Perren et al¹⁵² found that the rate of weaning failure was equivalent when ninety-eight patients were randomized to breathe with PS of 7 cm H₂O for 30 or 120 minutes: 6.5% (3/46) and 11.5% (6/52). The rates of reintubation did not differ significantly (7% and 4%, respectively).

Koh et al¹⁵³ investigated whether adding a 1-hour T-tube trial would alter outcome in patients who had already undergone progressive decreases in PS (3 to 5 cm H₂O every hour) to a level estimated to offset imposed respiratory work (by the endotracheal tube and ventilator circuit: mean: 7.6 ± 0.4 cm H₂O [SE]; range: 4 to 13 cm H₂O). The rate of weaning failure was not significantly different in patients who were extubated after the additional T-tube trial and those extubated directly, 45% (10/22) and 30% (6/20), respectively, nor did the rates of reintubation differ, 18% (4/22) and 20% (4/20), respectively.

Matic and Majeric-Kogler¹⁵⁴ studied 260 patients who had received mechanical ventilation for more than 48 hours and who had who satisfied several criteria of readiness for weaning (such as frequency <35 breaths/min, $V_T > 5$ mL/kg, $f/V_T < 100$, and $P_{O_2} > 60$ mm Hg on $FI_{O_2} < 0.40$). Patients were randomized to a 2-hour T-tube trial or a 2-hour trial of PS 8 cm H₂O. Of the 150 patients in the PS trial group, 120 (80.0%) passed the trial and were extubated. Of the 110 patients in the T-tube trial group, eighty (72.7%) passed the trial and were extubated ($p = 0.06$). The remaining patients who failed the initial weaning trial were returned to assist-control ventilation, and the previously assigned weaning technique was again attempted after 24 hours (or when the patient's condition permitted). The rate of weaning success was higher in the PS group than in the T-tube trial group: twenty-six of thirty (86.7%) versus twenty-one of thirty (70%). In addition, the time for weaning was shorter for the PS group, 54 (IQR: 47 to 88) versus 94 (IQR: 79 to 132) hours, as was total time for mechanical ventilation, 215 hours (IQR: 187 to 259) versus 262 hours (IQR: 216 to 328). The rate of reintubation was equivalent for the PS and T-tube trial groups: one (3.5%) versus two (6.7%) patients.

Vitacca et al¹⁵⁵ studied seventy-five patients with COPD who had been transferred to a long-term weaning unit. The patients had required more than 15 days of mechanical ventilation, and all had a tracheostomy. Of the seventy-five patients, twenty-three (31%) tolerated a T-tube trial for 48 hours and were studied no further. The remaining fifty-two patients failed the T-tube trial after 290 ± 452 minutes, and were randomly assigned to weaning by PS or T-tube trials. Weaning was deemed successful when patients tolerated spontaneous breathing for 48 hours or longer. Clinical outcomes were equivalent in the PS and T-tube trial groups: weaning success rate 73% (19/26) versus 77% (20/26); mortality 11.5% (3/26) versus 7.6% (2/26); ventilator duration 181 ± 161 versus 130 ± 106 hours; weaning unit stay 33 ± 12 versus 35 ± 15 days; and hospital stay 49 ± 27 versus 50 ± 32 days. The investigators also retrospectively compared fifty-five patients managed by the protocols of this study with sixty-two control patients managed in the same institution during the 2 years preceding the study. The study patients exhibited better outcomes than the historical control group: 30-day weaning success rate 87% versus 70%; ventilator duration 103 ± 144 versus 170 ± 127 hours; weaning unit stay 27 ± 12 versus 38 ± 18 days; and hospital stay 38 ± 17 versus 47 ± 18 days.

Weaning by Protocol versus Usual Care

Several investigators have examined whether formalizing weaning steps into a systematic protocol might alter weaning outcome. Many of these studies were motivated by the results of a 1996 study by Ely et al,¹⁵⁶ who combined two steps from previous studies—measurement of f/V_T (and other predictors) and a 2-hour T-tube trial—into an algorithm. They randomized 300 patients into an intervention arm and a usual-care arm. Patients in the intervention arm were screened in the early morning. To pass the screen, patients had to meet all of the following five criteria: f/V_T less than 105; Pa_{O_2}/FI_{O_2} greater than 200; PEEP equal to or less than 5 cm H₂O; adequate cough during suctioning; and no infusions of vasopressors or sedatives (with the exception of dopamine ≤ 5 mcg/kg/min and intermittent sedative boluses). Patients who passed the screen automatically underwent a trial of spontaneous breathing (a T-tube trial or flow-by and CPAP 5 cm H₂O) later that morning (without requiring an order from the primary physician). If the patient tolerated the trial, the primary physician was informed both orally and by a note in the chart.

The time taken to wean (from the point of satisfying the predictor criteria until the discontinuation of mechanical ventilation) was shorter in the intervention group than in the usual-care group: median 1 day (IQR: 0 to 2) versus 3 days (IQR: 2 to 7). The duration of ventilation was also shorter: 4.5 days (IQR: 2 to 9) versus 6 days (IQR: 3 to 11). Reintubation was less common in the intervention group than in the usual-care group: 4% versus 10%. This study demonstrates that a two-step approach—systematic measurement of predictors followed by a single daily T-tube (or flow-by/CPAP) trial—was superior to usual care.

A closer inspection of the data illustrates some important points about weaning. Among the 149 patients in the intervention group, 113 (76%) met the predictor criteria. Of these 113 patients, eighty-eight (78%) passed the spontaneous breathing trial. Yet, extubation was not attempted that day in 45% (40/88) of the patients who passed the trial. In a subsequent analysis of the data, Ely et al¹⁵⁷ noted that eighty-four of the 300 patients never satisfied the weaning predictor thresholds. Of the remaining twenty-five patients (25/300; 8.3% of the overall study group) were successfully

extubated despite never satisfying the predictor thresholds. Of these twenty-five patients, fifteen (15/300) had $\text{PaO}_2/\text{FiO}_2$ ratios less than 200 (5% of the overall study group) and ten (10/300) (3.3% of the overall study group) had f/V_T readings higher than 105. These data highlight that (a) the major source of delay in weaning of patients in the intervention group resulted from the failure of physicians to take action in patients who passed a spontaneous breathing trial (extubation was delayed in 45% of these patients), (b) almost 80% of patients who meet predictor thresholds will tolerate an immediate spontaneous breathing trial, and (c) f/V_T (105) falsely identifies approximately 3% of patients who can be extubated.

Ely and members of the Evidence-Based Medicine Task Force on weaning^{62,158} subsequently argued that the data from the 1996 study provided evidence in support of the use of formalized protocols in weaning management. To claim that use of a protocol per se improves patient outcome, the weaning approach must be the same in both the protocol arm and nonprotocol arm of a research study. In the 1996 study,¹⁵⁶ however, 76% of patients in the usual-care arm were managed with IMV, whereas no patient in the intervention arm was weaned using IMV. Several studies had already shown that IMV impeded weaning. Such a study design lacks internal validity and cannot be used as the basis for claims that use of a weaning protocol per se expedites weaning.¹⁵⁹

The next study to test the benefit of protocolized management of weaning was undertaken by Kollef et al.² They randomized 179 patients to a protocol group and 178 patients to usual care. The study was conducted in four intensive care units (ICUs), and weaning techniques were a mixture of PS, CPAP, and IMV. Overall, the duration of mechanical ventilation was shorter for the protocol group than for the usual-care group: median 35 hours (IQR: 15 to 114) versus 44 hours (IQR: 21 to 209). The difference, however, was significant in only one of the four ICUs. Moreover, interpretation of the data for this ICU is not straightforward. Patients in the usual-care group were significantly sicker than were the patients in the protocol group: Acute Physiology, and Chronic Health Evaluation (APACHE) II scores were 15.4 versus 13.6. In separate analysis, the investigators found that APACHE II score was at least as important as protocol management as a determinant of the duration of mechanical ventilation. In the other three ICUs, there was no difference in duration of mechanical ventilation between the protocol and usual-care groups; in one ICU, the trend was in the opposite direction.

Marelich et al.¹⁶⁰ undertook a randomized comparison of weaning by protocol versus usual care in 335 patients, about half of whom were in the medical ICU and half in the surgical ICU. They randomized 186 patients to a protocol arm and 189 patients to a usual-care arm. The duration of mechanical ventilation was shorter for the protocol arm than for the usual-care arm: 68 hours (IQR: 33 to 164) versus 124 hours (IQR: 54 to 334) ($p=0.001$).¹⁶⁰ Benefit, however, was confined to the medical ICU. Protocol management achieved no advantage in the surgical ICU.

Namen and members of the Ely research group¹⁶¹ investigated the benefit of protocolized weaning in a neurosurgical ICU. There were forty-nine patients in the protocol arm and fifty-one patients in the usual-care arm. Implementation of protocolized weaning had no beneficial effects on any patient outcome, such as number of days of mechanical ventilation, costs, morbidity, or mortality.

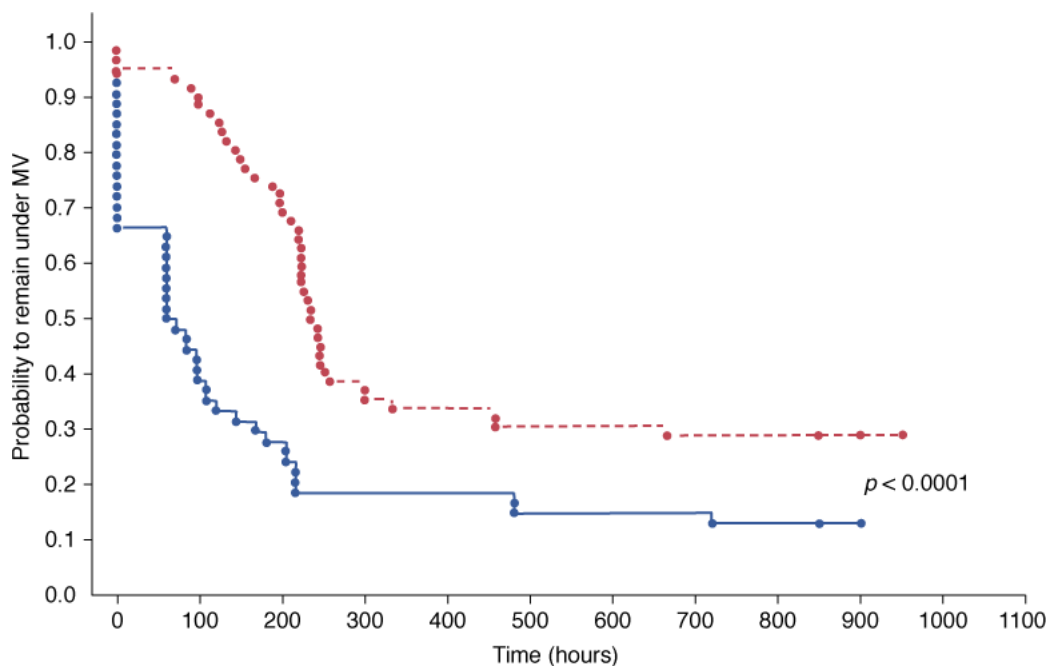
Randolph et al.¹⁶² randomized 182 children who had failed to satisfy criteria for extubation (and had been ventilated for >24 hours) to three arms: PS protocol, volume-support ventilation protocol, usual care. The rate of weaning failure was equivalent for PS, volume-support ventilation, and usual care, 15%, 24%, and 17%, respectively, as was the median duration of weaning, 1.6, 1.8, and 2.0 days, respectively. The authors concluded that protocols did not shorten the duration of weaning.

Krishnan et al.¹⁶³ did an RCT in 154 patients assigned to weaning by protocol and 145 patients weaned according to usual care. The duration of mechanical ventilation was equivalent for protocol weaning and usual care, 60.4 versus 68.0 hours, as was the rate of successful weaning, 74.7% versus 75.2%.

Of the six RCTs that have tested the benefits of weaning protocols, three research groups have reported that protocolized weaning is superior to usual care. Two of these studies, those by Ely et al.¹⁵⁶ and Kollef et al.,² have internal-validity problems of such magnitude that the data cannot be accepted as valid evidence on which to base a claim that protocols per se expedite weaning. The third study, that by Marelich et al.,¹⁶⁰ revealed no benefit in one of the two ICUs in the study. The trials conducted by Namen et al.,¹⁶¹ Randolph et al.,¹⁶² and Krishnan et al.¹⁶³ revealed no benefit for protocolized weaning. Thus, only half of a study (out of six studies) revealed valid support for protocolized weaning, with the remainder providing no evidence of benefit.

In addition to these studies of weaning protocols in ICU patients, Vitacca et al.¹⁵⁵ investigated the benefits of protocolized management of patients admitted to a long-term weaning facility. The investigators concluded that protocolized weaning significantly decreased the duration mechanical ventilation as compared with usual care: 103 ± 144 versus 170 ± 127 hours (Fig. 58-26). Inspection of the Kaplan-Meier curve, however, reveals that the plots differ only at time zero; thereafter, the two plots superimpose on one another. On first evaluation, more than 30% of patients classified as “difficult-to-wean” were immediately weaned. That is, benefit was not derived from use of a protocol, but from the presence of a “thinking” doctor at the bedside.

Figure 58-26



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Kaplan-Meier curves of probability of continued mechanical ventilation (MV) in patients managed by a weaning protocol versus usual care. The time of continued mechanical ventilation was significantly shorter with protocolized weaning as compared with usual care: 103 ± 144 versus 170 ± 127 hours ($p < 0.0001$). (Used, with permission, from Vitacca et al.¹⁵⁵)

There is a deeper reason why protocolized management does not lead to improvements in weaning outcome. This relates to a fallacy in historiography (the methodology employed by historians) known as the Whig-interpretation-of-history fallacy: the tendency among historians to write their account on the side of the winners, to praise military strategists and revolutionaries provided they have been successful.¹⁶⁴ The fallacy can be best appreciated by first considering an example, and then reflecting on the implications for clinicians and researchers.

Numerous computer versions of the Battle of Gettysburg exist on the market. These reproduce with great accuracy the terrain, corps involved, and serial chronology of events in the battle. A moderately good player taking the Confederate side can beat the Union side.¹⁶⁵ Yet, Robert E. Lee, universally regarded as the best strategist and tactician on either side, lost. How could a moderately accomplished player outperform Lee? The player today knows from the outset what Lee did not know. Lee did not know that the battle was going to take place at the specific location of Gettysburg. Lee did not know precisely which units would be involved and where, the time scale of preliminary stages of the battle, the constraints imposed by the terrain, and all the trivial contingencies that powerfully influenced the final outcome of the battle. The determinate shape of the battle was conferred only retrospectively by its outcome. There are no conclusive reasons that the series of events turned out the way they did—each could have turned out otherwise. It is impossible to incorporate all possible contingencies into a battle plan. Simply, the computer version does not reproduce the situation that Lee faced; instead, it is what historians call a Whig illusion.

The error committed by Whig historians is that endings seem inevitable—but only after the fact. Nothing, however, could ever happen if reality did not kill all the other potentialities originally inherent in any given situation.¹⁶⁶ From a historical perspective, every sequence of events appears as though it could not have happened otherwise. This produces an existential illusion. Weaning researchers who expect protocols to improve clinical outcome are committing the Whig fallacy. At the outset of a weaning attempt, there is not a set of determinate, enumerable factors, the totality of which comprises the situation—like the set of steps used to provide the proof for a theorem in Euclidean geometry. To suppose otherwise is to confuse an a posteriori (retrospective) perspective with an a priori (prospective) perspective. This is the fallacy committed by protocol advocates, Whig historians, and malpractice attorneys.

Computerized Approaches to Weaning

Investigators have attempted to apply various forms of artificial intelligence to mechanical ventilation.^{129,167,168} These approaches have been most fruitful in the management of weaning. A specially constructed ventilator based on these efforts was compared against usual care in a multicenter RCT conducted by Lellouche et al.¹⁶⁹ The computerized system was embedded in an Evita 4 ventilator (Dräger, Lübeck, Germany). The modified ventilator continuously monitored respiratory frequency, tidal volume, and end-tidal CO₂ tension, and repeatedly altered the level of delivered PS based on iterative changes in these three variables. Depending on a patient's response, the computerized system periodically and automatically decreased the level of PS by 2 to 4 cm H₂O. Importantly, the algorithm accepted transient instabilities, such as a brief increase in respiratory frequency, without making changes in a patient's classification. The pace of pressure reduction was tailored according to the needs and performance of each individual patient. Once a predesignated minimal level of PS was reached, the ventilator automatically subjected the patient to a weaning trial, conducted at a low level of PS. If the patient performed satisfactorily, a message was displayed on the screen recommending the removal of the ventilator. Lellouche et al.¹⁶⁹ enrolled 144 patients from five medical-surgical ICUs into the study. The patients had received at least 24 hours of

Loading [Contrib]/a11y/accessibility-menu.js at an early stage of their course, at a point when weaning predictor tests did not signal readiness to wean. In four of the five

ICUs, usual care involved the use of paper-based weaning protocols. Compared with usual care, the computerized system decreased weaning duration from a median of 5 to 3 days ($p = 0.01$) and decreased total duration of mechanical ventilation from 12 to 7.5 days ($p = 0.003$).

This study provides support for the use of computerized ventilators as a method to expedite weaning. The study also carries lessons for the better management of weaning using conventional ventilators. Of the seven stages of weaning, stage 2, the period of diagnostic triggering—the time when a physician begins to think that the patient just might be ready come off the ventilator—is the most critical (see Fig. 58-1). RCTs on weaning techniques have revealed that 60% to 80% of patients ventilated for a week or more were successfully disconnected from the ventilator successfully on the first day they were evaluated for weaning.^{3,4} If weaning-predictor tests had been performed in these patients at an earlier time, it is highly likely that many of them could have been removed from their ventilator a day, or several days, sooner than was the case.

With the artificial-intelligence-operated ventilator,¹⁶⁹ screening tests were performed repeatedly and automatically, and thus the delay inherent in diagnostic triggering (stage 2) was bypassed. Moreover, once a predesignated readiness threshold was reached, a confirmatory test—a weaning trial that advises on the appropriateness of extubation—was automatically performed, without requiring any action (or decision making) on the part of the staff. For patients in the usual-care arm of the trial, Lellouche et al¹⁶⁹ calculated that the number of T-tube trials performed according to patient readiness was only 51%. The delay in human decision making and action was further emphasized by the lapse of more than 14 hours (0.6 ± 2.65 days) between the time that a message first indicated that a patient had passed a weaning trial until removal of the endotracheal tube. Indeed, only 42% of the patients were extubated on the day that the message was first displayed.

Rose et al¹⁷⁰ undertook an RCT of the ventilator employed by Lellouche et al,¹⁶⁹ which now goes by the name SmartCare. A total of 102 patients in an Australian ICU were equally divided between the computerized ventilator and usual care. The median time from the first identified point of suitability for weaning commencement until the display of a message indicating that a patient had passed a weaning trial was 20 hours (IQR: 2 to 40) for the computerized ventilator versus 8 hours (IQR: 2 to 43) for usual care ($p = 0.3$). The adjusted probability of reaching separation potential was estimated to be 31% lower with the computerized ventilator as compared to usual care (95% CI, 51% lower to 9% greater). At least two factors are likely to have contributed to the markedly different outcome in the two studies. First, 68% of enrollees in the study of Lellouche et al¹⁶⁹ were medical patients, whereas 78% of enrollees in the Australian study¹⁷⁰ were surgical patients, which are typically less challenging to wean. Second, the nurse-to-patient ratio in the Australian ICU was 1:1.

A third RCT of computerized weaning versus usual care was undertaken by Taniguchi et al.¹⁷¹ These Brazilian researchers employed a different computerized system, termed *mandatory rate ventilation*, from that used in the studies of Lellouche et al¹⁶⁹ and Rose et al.¹⁷⁰ An algorithm, embedded in a Taema-Horus ventilator, automatically adjusted the level of PS according to a target respiratory rate. If the patient's respiratory rate (averaged over four respiratory cycles) exceeded the target by more than 3 breaths/min, the level of PS was increased by 1 cm H₂O, and vice versa. Taniguchi et al¹⁷¹ found no difference in weaning duration with the two approaches. The level of PS was lower with usual care than with the computerized ventilator (over at least the first three hours of the study). Similar to the study of Rose et al,¹⁷⁰ enrollees in the study of Taniguchi et al¹⁷¹ consisted solely of postoperative patients. Another important factor that contrasts with the research team of Lellouche et al¹⁶⁹ is that both the Brazilian and Australian clinicians knew they were competing against a machine that they did not help to create.

Respiratory Muscle Training

Over the past 25 years, researchers have emphasized that respiratory muscle fatigue is likely to be more important than respiratory muscle weakness in causing weaning failure (see [Pathophysiology of Weaning Failure: Respiratory Muscles](#)). There is growing recognition that this attribution is misplaced. Recordings of twitch pressures with the use of phrenic-nerve stimulation have revealed that many weaning-failure patients have very severe respiratory muscle weakness.^{172,173} A renewed focus on respiratory muscle strength has gathered impetus from a recent RCT, which revealed that deliberate measures aimed at improving respiratory muscle strength resulted in a significant increase in the weanability of patients.

Martin et al¹⁷⁴ enrolled sixty-nine patients in a single-blind RCT designed to determine whether inspiratory muscle-strength training would improve weaning outcome. The patients had received mechanical ventilation for approximately 40 to 50 days before entry into the study. Thirty-five patients were randomized to strength training and thirty-four patients to sham therapy.

Strength training was achieved with the use of a commercial threshold inspiratory muscle trainer (Threshold PEP; Respironics Inc., Murrysville, PA), which required patients to generate inspiratory pressures of -4 to -20 cm H₂O at the onset of an inspiration. During a training session, a patient was removed from the ventilator and the threshold device connected to the tracheostomy tube (keeping the cuff inflated). The patient was then required to perform four sets of six to ten threshold-loaded inspirations; between each set, the patient was returned to the ventilator for 2 minutes. The setting on the threshold device was adjusted to the highest pressure setting that the patient could consistently open during each inspiration (airway pressures at the tracheostomy tube were monitored). Over the course of the study, the pressure setting on the device was increased according to patient tolerance. Patients in the sham arm inspired through an inspiratory muscle training device that contained a large hole, with the result that little pressure was required to generate airflow. Sessions were performed 5 days a week (Monday to Friday).

In addition to the inspiratory muscle-strength training sessions, all patients were subjected to weaning trials of progressively longer duration. These trials involved the use of automatic tube compensation, CPAP, or low levels of PS. Weaning trials were conducted 7 days a week according to a predesignated scheme.

Upon completion of the study, twenty-five of the thirty-five (71%) patients in the strength-training arm were weaned as compared with sixteen of thirty-four (47%)

er the course of the study, the inspiratory pressure setting on the threshold device became more negative: -9.54 ± 3.70 (SD)

versus -14.52 ± 4.59 cm H₂O ($p = 0.0004$). Corresponding pressures in the sham arm were -3.10 ± 1.54 and -3.36 ± 2.08 cm H₂O ($p = 0.86$). Over the course of the trial, patients in the strength-training arm exhibited an increase in P_Imax from 44.4 ± 18.4 to 54.1 ± 17.8 cm H₂O ($p < 0.0001$); corresponding values of P_Imax in the sham arm were 43.5 ± 17.8 and 45.1 ± 19.5 cm H₂O.

The results of this study suggest that use of a simple device (a threshold loader) can markedly increase the weanability of patients. The challenge now is to reproduce these findings and find out whether further adjustments to the training regimen might achieve an even greater level of success. The positive outcome highlights the importance of respiratory muscle strength as a determinant of weanability. The results also suggest that other methods of increasing respiratory muscle strength, such as the use of transcutaneous neuromuscular electrical stimulation or novel biologic or pharmacologic compounds, may also expedite weaning.

Physical and Occupational Therapy to Aid Weaning

The use of techniques aimed at whole-body rehabilitation, including formal physical and occupational therapy, shortly after the initiation of mechanical ventilation, holds considerable promise for improving long-term functional performance after discharge from the ICU, but it may also expedite weaning. Preliminary research in this area had shown promise,¹⁷⁵ but interest was heightened by a recent RCT.¹⁷⁶

Schweickert et al¹⁷⁶ sought to determine whether early implementation of a program of formal physical and occupational therapy would lead to improvements in functional outcomes in ventilated patients. At two university hospitals, investigators enrolled patients who had received less than 72 hours of mechanical ventilation, which was expected to continue for at least another 24 hours; enrollees were also required to have exhibited functional independence 2 weeks before admission. From an initial screening of 1161 patients, 104 were enrolled: Forty-nine were randomized to early exercise and mobilization (physical and occupational therapy), starting on the day of enrollment, and fifty-five to usual care. (Usual care at the two study sites did not include routine physical therapy for ventilated patients.)

Each day, infusions of sedative agents were interrupted in both groups; by the end of the study, no differences in sedation and analgesia practices were observed in the two groups. In the intervention arm, patients also underwent passive range of motion exercises for all limbs, delivered by a physical and occupational therapist, every morning. Once patient interaction was achieved, sessions began with active assisted exercises in the supine position. If these exercises were tolerated, treatment was advanced to bed mobility activities, including transferring to upright sitting, followed by participation in activities of daily living. Such therapy took place on 87% of days of the study in the intervention group. During mechanical ventilation, the median duration of physical and occupational therapy differed between the groups: median 0.32 hours a day (IQR: 0.17 to 0.48) in the intervention arm and 0 hours a day in the control arm. Therapy commenced at a median of 1.5 days (IQR: 1.0 to 2.1) after intubation in the intervention group versus 7.4 days (IQR: 6.0 to 10.9) in the control group.

The primary end point, the number of patients returning to independent functional status at hospital discharge, was reached by twenty-nine (59%) patients in the intervention group and nineteen (35%) patients in the control group ($p = 0.02$): odds ratio 2.7 (95% CI 1.2 to 6.1). Over the first 28 days of the study, patients in the intervention arm had a higher number of ventilator-free days, 23.5 (IQR: 7.4 to 25.6), than did patients in the control group, 21.1 days (IQR: 0.0 to 23.8; $p = 0.05$); overall duration of mechanical ventilation was also shorter in the intervention group, 3.4 days (IQR: 2.3 to 7.3) versus 6.1 days (IQR: 4.0 to 9.6; $p = 0.02$). Use of formal physical and occupational therapy resulted in a higher number of independent activities of daily living and greater unassisted walking distance at hospital discharge than achieved in the control group.

Although efforts to mobilize ventilator-supported patients, including attempts to help them walk while receiving ventilator assistance, have been employed for decades as an aid to weaning,¹⁴¹ the impressive results achieved by Schweickert et al¹⁷⁶ crystallizes the components of physical and occupational therapy that are crucial to achieve success.

Role of Sedation in Weaning

Pharmacologic agents that produce sedation are almost invariably used in patients receiving mechanical ventilation with the goals of decreasing anxiety and agitation, enhancing patient-ventilator synchronization, and facilitating patient care. Commonly, sedative drugs are administered by means of continuous intravenous infusions. Sedative agents accumulate in the body far beyond the treatment period, especially in critically ill patients, as a result of impaired renal or hepatic function. Sedative agents also hinder the assessment of mental function, leading physicians to suspect organic brain disorders when none exist. To determine whether the accumulation of sedative agents hinders ventilator management, Kress et al¹⁷⁷ undertook an RCT in 128 adult patients receiving mechanical ventilation in a medical ICU. In the intervention arm, an investigator interrupted the infusion of midazolam, propofol, or morphine on a daily basis until the patients were awake and could follow instructions or until they became uncomfortable or agitated. A patient was considered “awake” if the patient was able to perform at least three of the following four actions: open the eyes in response to a voice, use the eyes to follow the investigator, squeeze a hand, and stick out the tongue on request. In the control group, sedative infusions were interrupted only at the discretion of the intensivist.

Daily awakenings resulted in a shorter duration of mechanical ventilation, median of 4.9 days versus 7.3 days in the control group ($p = 0.004$), and shorter ICU stay, 6.4 days versus 9.9 days ($p = 0.02$). Interruption of sedation also reduced the number of diagnostic tests (brain scan, lumbar puncture) ordered to assess changes in mental status, six versus sixteen in the control group ($p = 0.02$).

Girard et al¹⁷⁸ extended these findings by combining daily interruption of sedation with spontaneous breathing trials. The investigators randomized 336 ventilated patients to an intervention arm, consisting of paired use of sedative interruption and systematic use of spontaneous breathing trials versus a control arm, consisting of spontaneous breathing trials and the administration sedative agents at the discretion of the intensivist. The paired intervention resulted in more days breathing without ventilator assistance (3.1 days in the 28-day study, 95% CI 0.7 to 5.6), and earlier discharge from both the ICU and hospital. In an accompanying editorial, Brochard¹⁷⁹ pointed out several problems with the experimental design of the study—the absence of a requirement to stop sedatives in the control group before a

spontaneous breathing trial, the timing of weaning onset, and markedly increased rate of failed weaning trials in the control arm—that render the interpretation of this study highly problematic.

Strøm et al.¹⁸⁰ went a step further than previous investigators and sought to determine whether total withholding of sedative agents would prove superior to daily interruption of sedation. Patients in the control arm received propofol for the first 48 hours and midazolam thereafter, titrated to reach a Ramsay score of 3 to 4; sedation was interrupted every day until patients were awake. Patients in the intervention arm received bolus doses of morphine (2.5 or 5 mg), as did patients in the control arm, but did not receive intravenous infusions of sedatives or analgesics. No sedation was associated with significantly more days without ventilator assistance (mean: 4.2 days, 95% CI 0.3 to 8.1 days), and earlier discharge from both the ICU and hospital. Agitated delirium was more frequent in the intervention group than in the control group (20% vs. 7%; $p = 0.04$), and the use of haloperidol was also more frequent (35% vs. 14%). Moreover, an extra person was needed at the bedside for 2.5 days to comfort and reassure patients more often in the intervention group than in the control group (20% vs. 5%). Follow-up data have not been published on these patients to determine whether they are more or less prone to posttraumatic stress disorder or other psychological problems.

Conclusion

The hazards of mechanical ventilation make it imperative to wean patients at the earliest possible time in their clinical course. Premature weaning attempts, however, cause considerable respiratory distress that may set back a patient's course. Premature extubation is also hazardous. To minimize both delayed weaning and premature extubation, a two-step diagnostic strategy is recommended: measurement of weaning predictors followed by a weaning trial. Because each step constitutes a diagnostic test, clinicians must be mindful of the scientific principles of clinical decision making when interpreting the information generated by each step.

The critical step is for the physician to contemplate the possibility that a patient just might be able to tolerate weaning. Such diagnostic triggering is facilitated through use of a screening test, which is the rationale for measurement of weaning predictor tests. It is important not to postpone this first step by waiting for a more complex diagnostic test, such as a T-tube trial. A positive result on a screening test (weaning predictor test) is followed by a confirmatory test (weaning trial), to increase the likelihood that a patient will tolerate extubation.

Many complex facets of pulmonary pathophysiology impinge on weaning management. Thus, weaning requires individualized care at a high level of sophistication. Few other activities undertaken by physicians in the ICU require a greater intellectual input and carry greater potential for improving patient outcome than the application of physiologic principles in the weaning of patients.

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